Imbalanced semiprime representations of even integers

A revised modular proof with paired sieve weights and finite certificates

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Abstract

We study representations of sufficiently large even integers in the form $N = p + rq$, where p, q are primes and r is either 1 or a prime, with a restriction on the relative sizes of r and q. Write $H_N = C_N N/\log^2 N$ with the Chen singular factor C_N defined below. The construction developed here gives, for each fixed $1/2 < b < 1$,

$$\liminf_{\substack{N\to\infty \\ 2|N}} \frac{D_b(N)}{H_N} \geq 1.96176252 - 6.7617368\log(1/b),$$

where D_b counts distinct prime summands with $r \leq q^b$. In particular the resulting eventual lower bounds are $H_N/100000$ at $b = 0.74816964$ and $H_N/1000000$ at $b = 0.74816865$. The proof combines a positive upper sieve for the excluded ordered semiprimes, a full eleven-term Chen weight with a paired geometric deletion, and explicitly selected finite numerical inequalities. A further estimate on the same deleted cells uses the elementary bound $\omega(s) \geq 11/20$ for $s \geq 7/3$ and a sharper payment for the discarded saved lower estimators. A restricted moving-residue distribution argument retains the oscillatory phase and is matched to the actual compiled coefficients. We state its classical inputs and its scope, and include the complete analytic modules and a finite-certificate specification. The present version replaces the earlier distribution interface; it does not invoke the previously used external coefficient 1.9728. The proof is a research manuscript with computational certificates, not a formally verified theorem or a report of external mathematical acceptance.

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1 Statement and organization

Throughout, $N > 2$ tends to infinity through the even integers. All logarithms are natural. Put

$$C_2 = \prod_{\substack{\ell>2\\ \ell \text{ prime}}} \left(1 - \frac{1}{(\ell-1)^2}\right), \qquad C_N = C_2 \prod_{\substack{\ell \mid N\\ \ell>2}} \frac{\ell-1}{\ell-2}, \qquad H_N = C_N \frac{N}{\log^2 N}. \tag{1}$$

The letter ℓ in a prime product always denotes a prime. We write $\Omega(n)$ for the number of prime factors of n, with multiplicity, and use $P^-(1) = +\infty$.

Define counts of distinct primes p, rather than von Mangoldt weighted sums, by

$$D_{1,2}(N) = \#\{p < N : p \text{ prime},\ N - p > 1,\ \Omega(N-p) \le 2\}, \tag{2}$$

$$D_b(N) = \#\{p < N : p \text{ prime},\ N - p = rq,\ q \text{ prime},\ r = 1 \text{ or prime},\ r \le q^b\}, \tag{3}$$

$$B_b(N) = \#\{p < N : p \text{ prime},\ N - p = rq,\ r, q \text{ prime},\ q^b < r \le q\}. \tag{4}$$

Each prime p is counted once. When $b < 1$, an admissible composite remainder has $r < q$ and its ordered factorization is unique. A prime remainder is represented with $r = 1$. A square remainder is in B_b, not in D_b.

Theorem 1.1 (Current fixed-exponent bound)**.** *Set*

$$K = \frac{8452171}{1250000} = 6.7617368, \qquad c = \frac{49044063}{25000000} = 1.96176252. \tag{5}$$

For every fixed b with $1/2 < b < 1$,

$$\liminf_{\substack{N\to\infty\\ 2\mid N}} \frac{D_b(N)}{H_N} \ge c - K\log(1/b). \tag{6}$$

Consequently every fixed number strictly below the right-hand side is an eventual lower coefficient for all even N.

Corollary 1.2. *There is an N_0 such that every even $N \ge N_0$ simultaneously satisfies*

$$D_{0.74816964}(N) \ge \frac{H_N}{100000}, \tag{7}$$

$$D_{0.74816865}(N) \ge \frac{H_N}{1000000}, \tag{8}$$

$$D_{0.9}(N) \ge 1.24889 H_N. \tag{9}$$

In particular the first two counts are positive for all sufficiently large even N.

No effective value of N_0 is supplied. The two small exponents have different stated count coefficients; they are not interchangeable versions of one numerical assertion. The coefficient 1.24889 at $b = .9$ is retained as a conservative corollary; the stronger refined lower endpoint is not needed for that comparison.

The proof has the following order. A local convolution estimate is derived from stated classical analytic inputs. The coefficient and prime-product modules match that estimate to the actual finite sieve coefficients. A single upper coefficient then supplies both a pointwise majorant and its density, giving K. Only after K is established is it inserted in the positive switched costs of the finite lower construction, which yields c. Finally the ordered bad-semiprime mass is $\log(1/b)$. Thus the upper estimate does not depend on the lower H/h table or on the final Chen count it helps establish.

The appendices contain the complete analytic completion modules used in this version. The main text gives their interfaces, the full pointwise Chen argument, the finite-realization and

accounting steps, and the final transfer. Exact result records and the enclosure and partition proofs they require accompany the source distribution. Appendices A–J specify the baseline mechanisms and their then-selected numerical endpoints. Appendices K–L supply the adopted same-mask increment used in the present statement. Older c and b values are preserved in the baseline module texts for provenance and are explicitly superseded by (5) and Corollary 1.2. No baseline analytic input is replaced by this numerical refinement. Names and checksums specify which finite objects enter the proof; a checksum or a successful program exit is not itself a mathematical inequality.

Change from earlier versions. The active argument below replaces the former unrestricted moving-residue distribution interface by the restricted derivation in Appendices A–C. It retains the large phase and the full last range in the Chen weight. In particular, the earlier corollary based on an external Chen coefficient 1.9728 is withdrawn from the asserted results of this version. No result here depends on that coefficient, and this work does not claim to repair an unrestricted distribution theorem or to reinstate the earlier external corollary. Historical numerical and proof records are retained with their original scopes; they must not be read as the current statement.

2 Restricted distribution and the actual coefficient class

2.1 A local estimate with a moving residue

We use U, V for the local convolution lengths in this subsection, to distinguish them from the global even target N. Write $x = UV$ and let $\mathfrak{a}$ denote the local residue. The local theorem in Appendix A is formulated as follows.

Proposition 2.1 (Restricted medium convolution)**.** *Fix $\vartheta = 7/32$, $0 < \sigma < 1/100$, $c_0 > 0$, a divisor order B, and $0 \le \kappa \le \sigma/100$. Let $U, V, Q_1, Q_2, Q_3 \ge 1$. Suppose*

$$\begin{aligned} c_0 x \le |\mathfrak{a}| \le x^{1+\kappa}, \qquad & Q_1 < U x^{-\sigma}, \\ U^2 Q_2 Q_3^2 < x^{1-8\sigma}, \qquad & |\mathfrak{a}|^{\vartheta} U Q_1 Q_2^5 Q_3^2 < x^{2-15\sigma}, \\ U^{1+2\vartheta} Q_1 Q_2^{5-2\vartheta} Q_3^{2+2\vartheta} < x^{2-15\sigma}. & \end{aligned} \tag{10}$$

Let α be one-bounded, supported on an individual interval in $[U, 2U]$ and on $P^-(n) > x^{1/(\log\log x)^3}$. Assume its auxiliary-coprimality Siegel–Walfisz estimate: for every $A > 0$, every $k, q \ge 1$ and $(b_0, q) = 1$,

$$\left| \sum_{\substack{n \equiv b_0 \pmod q \\ (n,k)=1}} \alpha_n - \frac{1}{\varphi(q)} \sum_{(n,kq)=1} \alpha_n \right| \le C_A U \tau(k)^{C_0} (\log x)^{-A}, \tag{11}$$

with fixed C_0 and uniform family constants. Let $|\beta_m| \le \tau(m)^B$, $m \in [V, 2V]$, and let each γ_i be a separate one-bounded sequence on $q_i \asymp Q_i$, supported on $(q_i, \mathfrak{a}) = 1$. For

$$\Delta_l = \sum_{n,m} \alpha_n \beta_m \left(\mathbf{1}_{nm \equiv \mathfrak{a} \pmod l} - \frac{\mathbf{1}_{(nm,l)=1}}{\varphi(l)} \right),$$

the stated classical inputs of Appendix E give

$$\sum_{q_1,q_2,q_3} \gamma_1(q_1)\gamma_2(q_2)\gamma_3(q_3)\Delta_{q_1q_2q_3} = O_A\big(x(\log x)^{-A}\big). \tag{12}$$

The constant is uniform in the residue and families satisfying these conditions. Fixed endpoint multiples are absorbed by the strict power margins.

This is a signed estimate for factorized modulus coefficients. It is not an absolute discrepancy sum over arbitrary moduli. In its actual use, α is a chart-rough 1 or Möbius slot, with a permitted Mellin twist; the proof derives (11) for those slots. Arbitrary rough coefficients are not declared to satisfy that hypothesis.

We record the parts of the local proof that determine its scope. In dispersion, take a fixed nonnegative Gevrey majorant ψ equal to one on $[1,2]$, supported in $[1/2,5/2]$, with

$$\|\psi^{(j)}\|_\infty \le C^j (j!)^2, \qquad |\widehat{\psi}(t)| \ll e^{-c\sqrt{|t|}}.$$

One construction convolves uniform measures on intervals of lengths c_1/j^2, then convolves the resulting density with an interval indicator. The derivative bound follows by differentiating the first j box measures. This gives a power-small Poisson tail beyond $H \asymp (l/V)\log^5 x$. An unspecified smooth function would not by itself justify that tail at this cutoff.

The exact products $nm = \mathfrak{a}$ are removed before Cauchy and restored in the prescribed order, with separate power bounds. The positive principal Gram variance is retained and estimated. In the second dispersion term the clean coprimalities give the exact identity below. Here $e(t) = \exp(2\pi i t)$, and a bar denotes the multiplicative inverse modulo the denominator of the corresponding fraction.

$$\begin{aligned} &e\left(\frac{\mathfrak{a}h\overline{n_1w_2}}{qdw_1} + \frac{\mathfrak{a}h\overline{n_2qdw_1}}{w_2}\right) \\ &\quad = e\left(\frac{\mathfrak{a}h(n_1-n_2)\overline{n_2qdw_1}}{n_1w_2}\right) e\left(\frac{\mathfrak{a}h}{qdw_1w_2n_1}\right). \end{aligned} \tag{13}$$

The last phase is kept. Partial summation in its five normalized coordinates costs $x^{5\kappa+o(1)}$, once. After the coprime partition and polarization, the spectral return uses a common completed-variable interval and a common additive phase across levels.

For clarity, with the local length U, the rough lengths R, S and $Q = Q_1$, the returned bare bound has the form

$$|\mathcal{W}| \ll \frac{U^2}{Q} x^\epsilon \left\{ D_0^{1/2} + \left[(1+|\mathfrak{a}|/x)D_1 + D_2\right]^{1/4} \right\}, \tag{14}$$

$$D_0 = \frac{U^2R^2S}{x}, \qquad D_1 = \frac{|\mathfrak{a}|^\vartheta UQR^2S^5}{x^2}, \qquad D_2 = \frac{U^{1+2\vartheta}QR^{2+2\vartheta}S^{5-2\vartheta}}{x^2}.$$

Here R comes from Q_3 and S from Q_2. The four conditions in (10) imply $D_0 \ll x^{-8\sigma}$ and $D_1, D_2 \ll x^{-15\sigma}$. Including the retained phase, the two powers to be paid are

$$x^{-4\sigma+5\kappa+\epsilon}, \qquad x^{-15\sigma/4+21\kappa/4+\epsilon}.$$

They have a fixed saving for the stated κ. The preliminary logarithmic errors and coefficient truncation are handled separately; we use the arbitrary-logarithmic conclusion (12), not a stronger unstated uniform power estimate for the whole argument.

The spectral inputs are normalized Kuznetsov, the regular cumulative large sieve, the coprime Assing–Blomer–Li packet and continuous quadratic-form identities, the Kim–Sarnak parameter 7/32, and the exceptional estimates in Pascadi's Theorems A and J. In the ABL normalization $R(n) = \sqrt{|n|}\rho(n)$, the coprime packet removes $\mathfrak{a}$ with cost $(|\mathfrak{a}|\nu)^{\vartheta/2}$ and leaves the coefficient

$$\mathbf{1}_{(n,\mathfrak{a})=1}\, \beta(\nu n, z) e(-n/z).$$

It is independent of the spectral height. The level average on the other side uses one common interval and one common phase. The proof does not claim the corresponding estimate for independently chosen intervals at each level or for noncoprime $(\mathfrak{a}, z)$. Appendix E specifies the theorem versions, cusps, normalizations and coefficient conditions used at every such interface.

2.2 From the local estimate to prime products

For a fixed finite smooth weight W_N on k labelled original primes, all at least cN^{a_*} for fixed $a_* > 0$, put

$$X_W^{(q)} = \sum_{(p_1\cdots p_k,q)=1} W_N(p_1,\dots,p_k), \qquad X_W = \sum W_N(p_1,\dots,p_k).$$

The product is $O(N)$. The allowed geometric families have fixed polylogarithmic mixed Euler-derivative bounds and no additional arithmetic filter. The conclusion used throughout the paper is

$$\sum_q \rho_N(q)\left\{\sum_{p_1\cdots p_k\equiv N \pmod q} W_N(p_1,\dots,p_k) - \frac{X_W^{(q)}}{\varphi(q)}\right\} = O_{A,W}\big(N(\log N)^{-A}\big). \tag{15}$$

The same signed coefficient ρ_N occurs in both terms. Its admitted classes are the actual upper tree, the actual lower trees, the marked queries and ordinary descendants of the finite H/h construction, and the positive switched families on which the upper coefficient is used.

The coefficient proof in Appendix B first freezes one finite causal policy. The physical level N^λ determines its density and its residual Q/B. A slightly larger auxiliary support cover $N^{\ell_{\mathrm{aux}}}$ may be used to prove its factorization; it does not change λ or the integer coefficient. Every full square

$$S_{j-1} + 2t_j \le \ell_{\mathrm{aux}}$$

and every selected-complement square

$$\sum_{i\in I} t_i + \sum_{\substack{i<j\\ i\notin I}} t_i + 2t_j \le \ell_{\mathrm{aux}} \quad (j\notin I)$$

is retained. After all selected slots have appeared, the second test becomes the full-square test. Descendant closure and the physical prime ceiling therefore persist through the finite ordinary queries.

There is an exact tensor identity before estimating discrepancies. For a common prime-label alphabet with multiplicities m_I, split the slots into three groups and divide each group convolution by $M_j = \prod_I m_{I,j}!$. The resulting γ_j is one-bounded. With $M_{\mathrm{all}} = \prod_I m_I!$, the full normalized convolution is

$$\frac{M_1M_2M_3}{M_{\mathrm{all}}}(\gamma_1 * \gamma_2 * \gamma_3).$$

Marked multiplicities remain in the external scalar. The small-prime beta factor attaches by the unique small/rough factorization. The partition is fixed on a whole atom box before any arithmetic discrepancy is inspected. Global squarefreeness remains in the physical coefficient; its removal only for analytic comparison has the fixed-power error $O((N+N^{\theta_*})N^{-\omega+o(1)})$.

Appendix C makes a smooth complete-product trim below $2N^{1-\kappa_0}$ and then an exact smooth dyadic partition. The local scale is the complete product, with

$$cN^{1-\kappa_0} \le x \le CN, \qquad c_0x \le N \le x^{1+\kappa}.$$

It is never a residual quotient. On an original group at scale R, the exact order-three Heath–Brown identity is obtained from $\Lambda * (\delta - 1 * \mu_Z)^{*3} = 0$ below $2R$, with $Z = (2R)^{1/3}$:

$$\Lambda = \sum_{j=1}^{3} (-1)^{j-1}\binom{3}{j}\mu_Z^{*j} * 1^{*(j-1)} * \log. \tag{16}$$

The smooth original chart is retained; no sharp coupled group boundary is inserted. The Fourier separation of the geometric profile is charged by its integrable Fourier norm. The actual rough

slots have the auxiliary SW estimate. Medium branches use Proposition 2.1; low ranks use below-half Bombieri–Vinogradov. The one-long and two-long alternatives are the direct Brun arguments of Appendix D, keeping the exact coprime mean and both Poisson zero-axis cancellations. In particular, negative lower Brun polynomials are not multiplied as though they were positive.

Choose the fixed moment order K_{mom} before narrowing

$$0 < \kappa \le \min\left(\kappa_{\mathrm{old}}, \frac{\sigma}{200}, \frac{1}{2K_{\mathrm{mom}}}\right), \qquad 0 < \kappa_0 < \frac{\kappa}{4(1+\kappa)}.$$

The complete local support has exponent $\Theta_{\mathrm{loc}} < 3/5$; the long-branch parameter ζ is selected with $20\zeta + \Theta_{\mathrm{loc}} < 1$ and $40\zeta + 3\Theta_{\mathrm{loc}}/2 < 1$. All finite atom scalar masses and the fixed logarithmic number of charts are paid by stronger local logarithmic accuracy. This proves (15) for the licensed families and all sufficiently large even targets. The assertion is not extended to arbitrary high-level coefficients or arbitrary signed sharp coupled domains.

3 A positive upper sieve with its own density

The upper construction uses

$$\rho_0 = \frac{19}{96}, \quad \mu_0 = \frac{25}{128}, \quad \Theta = \frac{299}{500}, \quad \Lambda = \frac{153}{256}, \quad v(t) = \min\left(\frac{1+t}{2}, \frac{57}{64} - \frac{3t}{2}\right).$$

Its causal cap changes are made on the actual prefix; every retained ancestor decision is preserved. Fourth, fifth and sixth changes are successive replacements, whose raw density differences telescope. Later eighth-rank changes occupy $30 + 15 + 13 + 29 = 87$ specified domains. Same-stage domains have disjoint interiors; successive stages use new-minus-current-parent differences. Improving an enclosure of one integral does not create another cap change or another copy of its gain. Appendix F supplies the cap formulas and separation proofs.

3.1 Exit identity and literal residual densities

For a finite allowed prime universe and $0 \le h(p) \le 1$, the upper tree, whose first exits have odd rank, satisfies the exact identity

$$\sum_d \mu(d)\mathbf{1}_{d \text{ retained}} h(d) = \prod_p (1 - h(p)) + \sum_{e \text{ odd first exit}} h(e) \prod_{p < p_{\mathrm{last}}(e)} (1 - h(p)). \tag{17}$$

Partition the excluded divisors by their first excluded prefix; the remaining smaller-prime subsets give the displayed Euler product. Taking $h(p) = \mathbf{1}_{p|n}$ proves a rough-indicator majorant for every integer n, including zero and negative integers.

For the raw density, take $g_N(p) = \mathbf{1}_{p>2,,p\nmid N}/(p-1)$ and $V_N(y) = \prod_{2<p<y,,p\nmid N}(1 - 1/(p-1))$. At a fixed rank the normalized exit for g_N lies between zero and the corresponding exit for $g_0(p) = \mathbf{1}_{p>2}/(p-1)$: a surviving prefix acquires the multiplier

$$\prod_{\substack{p_{\mathrm{last}} \le p < y \\ p \mid N}} (1 - g_0(p)) \le 1.$$

The preceding square tests bound the last prime of a rank-j exit below by $N^{\beta_{\min}/(3\cdot 2^{j-1})}$ on positive compact parameter sets. Deleting target primes thus changes each fixed-rank integral by $o(1)$ uniformly in N. Positive exit-series convergence, the one-sided bounds for the same raw Rosser weights, and then a finite-rank sandwich yield

$$\frac{V_N^+(R,y)}{V_N(y)} = F(s) + o(1) \quad (s > 1), \qquad \frac{V_N^-(R,y)}{V_N(y)} = f(s) + o(1) \quad (s \ge 2), \quad s = \frac{\log R}{\log y}. \tag{18}$$

The upper identity for $1 < s \le 3$ also follows directly by moving its first cube to $R^{1/3}$. This proves the density of the literal coefficient; it is not obtained by subtracting two unrelated sieve bounds. The ordinary exit-series inputs and their normalization are specified in Appendix F.

Mertens gives, uniformly in the target,

$$V_N(N^u) = \frac{2e^{-\gamma}C_N}{u\log N}(1+o(1)). \tag{19}$$

For example the exact ratio to $2C_N\prod_{p<z}(1-1/p)$ is

$$\prod_{p\ge z}\left(1-\frac{1}{(p-1)^2}\right)^{-1}\prod_{\substack{p\mid N\\ p\ge z}}(1-1/p),$$

whose logarithm tends uniformly to zero at $z = N^u$, $u > 0$ fixed. With $\overline{F} = F/(2e^{\gamma})$ and $\overline{f} = f/(2e^{\gamma})$, this is the normalization producing $4\overline{F}(s)/u$ and $4\overline{f}(s)/u$ in the density functionals.

For an adopted eighth cell, if $S_8 = t_1 + \cdots + t_8$ and the current parent is bounded above by R_6, its new cap M satisfies

$$1 < \frac{\lambda_{6,\mathrm{current}} - S_8}{t_8} \le \frac{R_6 - S_8}{t_8} < \frac{M - S_8}{t_8} < 3.$$

The same residual identity therefore gives the lower saving

$$\frac{4(M-R_6)}{(R_6-S_8)(M-S_8)\prod_{i=1}^{8}t_i}. \tag{20}$$

Using the larger parent R_6 is conservative. This density has no Chen quarter or ordering factorial. The adopted domains have $t_8 > 53/13824$, and all full and selected-complement squares remain in force.

3.2 Finite compilation and the two upper reserves

The ideal policy is approximated on a finite causal partition before N grows. Small-last-prime tails tend to zero; proper affine walls have vanishing mass, whereas open inherit/equality regions are retained. At even rough rank take outer endpoint tests and at odd rank inner tests, preserving ancestry. This produces a coefficient $a_N(d)$ above the raw upper coefficient. Attach the small-prime beta upper polynomial to positive $a_N(d)$ and the beta lower polynomial to negative $a_N(d)$. If $B^- \le I_w \le B^+$, then $a_N(d)B_{\mathrm{sign}}(n) \ge a_N(d)I_w(n)$ term by term. The resulting physical ρ_N has unique small/rough decomposition, bounded squarefree support, $(q, N) = 1$, and

$$R_N(n) := \sum_{q\mid n}\rho_N(q) \ge \mathbf{1}_{(n,P_N(N^{\rho_0}))=1} \ge 0, \qquad P_N(y) = \prod_{\substack{p<y\\ p\nmid N}} p. \tag{21}$$

The finite sum also defines $R_N(0)$.

Restricting (17) to primes above the presieve gives the upper projection inequality

$$V_N(w)\sum_{d\ \mathrm{rough}}\lambda_N(d)g_N(d) \le \sum_{d\ \mathrm{full}}\lambda_N(d)g_N(d).$$

Together with the absolute beta and box-error estimates this controls the density of the same ρ_N from above. Support inflation can be kept below $1/1000$, giving $\theta_* = 599/1000$ before the small complete-scale adjustment. The auxiliary support cover used in Section 2 does not change this density.

Let E be the selected raw upper endpoint and u each of its two distinct reserves, for finite density approximation and compilation:

$$E = \frac{9979702576710911234850619807816847864907359959}{14759135509622012279467806190993531521513984000}, \tag{22}$$

$$u = \frac{14500796122704831221059342753012800078468 1}{11807308407697609823574244952794825721118720000}. \tag{23}$$

The numerical lemma in Appendix I obtains E by four base terms minus twenty-two selected positive savings, with the full noncompact third tail replaced once. Exact subtraction gives

$$K - E - 2u = \frac{712292867275421312487585745470509085724 31}{737956775481100613973390309549676607569920000000} > 0. \tag{24}$$

Both finite operations fit strictly within their reserves. The remaining surplus exceeds $9.6522 \cdot 10^{-8}$. Thus the physical coefficient in (21) satisfies

$$S_N := \sum_q \frac{\rho_N(q)}{\varphi(q)} \le (K + o(1)) \frac{C_N}{\log N}, \tag{25}$$

and it is precisely the coefficient to which (15) applies.

Proposition 3.1 (Positive product consequence)**.** *Fix a finite nonnegative labelled prime-product family of bounded rank, with every original prime at least N^{a_*} for some fixed $a_* > 0$, product $O(N)$, and only the admitted fixed geometric restrictions. Suppose its unsifted weighted mass X_W satisfies*

$$\limsup_{\substack{N\to\infty \\ 2\mid N}} \frac{\log N}{N} X_W \le J < \infty.$$

Then the weighted count with product $m < N$ and $N - m$ prime is at most $(KJ + o(1))H_N$ in the limsup sense. Errors are absolute also when $J = 0$.

Proof. For a prime ℓ, squarefree support and boundedness give

$$\sum_{q:\ell\mid q} \frac{|\rho_N(q)|}{\varphi(q)} \ll \frac{\log N}{\ell - 1}.$$

Hence, by a union bound over the fixed number of original prime slots,

$$\left| \sum_q \frac{\rho_N(q)}{\varphi(q)} (X_W^{(q)} - X_W) \right| \ll X_W N^{-a_*} \log N = o(H_N). \tag{26}$$

No factor N^{θ_*} is introduced. Small prime remainders cost $N^{\rho_0 + o(1)}$. The smooth complete-product trim costs $N^{1-\kappa_0+o(1)}$ after the full modulus sum; its product is strictly below N, so no $\tau(0)$ occurs. Positive majorization before distribution gives

$$\sum_{m<N,\, N-m \text{ prime}} W_N(m) \le \sum_m W_N(m) R_N(N-m) + o(H_N) \le X_W S_N + o(H_N).$$

Outer approximations with $m \ge N$ are harmless because $R_N(n) \ge 0$ for every integer. Now apply (25). Fixed geometric boundaries are handled by positive smooth outer approximation, with N tending to infinity before the approximation error tends to zero. At each precision the same ρ_N is used in the pointwise inequality, density and distribution formula. It need not be the same coefficient at different precisions. No division by J is made. □

4 The full Chen weight and paired deletion

Put

$$a=\frac{417}{5000},\qquad h=\frac{19}{125},\qquad d=\frac{59}{250},\qquad \mu=\frac{25}{128}.$$

For $n\le N$ define $Q_b(y;n)=\mathbf{1}_{b|n}\mathbf{1}_{P^-(n/b)\ge y}$. All prime prefixes below have distinct entries. Suppressing the argument n, the eleven terms are

$$\begin{aligned}
Y_1&=Q_1(N^a), & Y_2&=Q_1(N^h),\\
Y_3&=\sum_{N^a\le p<N^{1/3}}Q_p(N^a), & Y_4&=\sum_{N^a\le p<N^d}Q_p(N^a),\\
Y_5&=\sum_{N^a\le p_1<p_2<N^h}Q_{p_1p_2}(N^a), & Y_6&=\sum_{N^a\le p_1<N^h\le p_2<N^d}Q_{p_1p_2}(N^a),\\
Y_7&=\sum_{N^d\le p_1<p_2<(N/p_1)^{1/2}}Q_{p_1p_2}(p_2), & Y_8&=\sum_{N^a\le p_1<N^{1/3}\le p_2<(N/p_1)^{1/2}}Q_{p_1p_2}(p_2),\\
Y_9&=\sum_{N^h\le p_1<N^d\le p_2<(N/p_1)^{1/2}}Q_{p_1p_2}((N/(p_1p_2))^{1/2}),\\
Y_{10}&=\sum_{N^a\le p_1<p_2<p_3<p_4<N^h}Q_{p_1p_2p_3p_4}(p_2),\\
Y_{11}&=\sum_{N^a\le p_1<p_2<p_3<N^h\le p_4<N^d}Q_{p_1p_2p_3p_4}(p_2).
\end{aligned}$$

In particular Y_{11} has its full last range. Set

$$W=3Y_1+Y_2-Y_3-Y_4+Y_5+Y_6-2Y_7-Y_8-Y_9-Y_{10}-Y_{11}. \tag{27}$$

Lemma 4.1 (Pointwise full-weight inequality)**.** *For squarefree $n\le N$,*

$$W(n)\le 4\mathbf{1}_{\Omega(n)\le 2}.$$

Proof. If n is not N^a-rough, all eleven terms vanish. For Y_9, the only less immediate case, its quotient cutoff exceeds $N^{(1-d)/4}>N^a$. Suppose therefore that n is squarefree and N^a-rough. Let r,s,t count its prime factors below $N^h,N^d,N^{1/3}$, respectively. Its first six counts are

$$(Y_1,Y_2,Y_3,Y_4,Y_5,Y_6)=\left(1,\mathbf{1}_{r=0},t,s,\binom{r}{2},r(s-r)\right).$$

If $r\ge 2$, the quotient sifted at the second selected prime forces the first two primes in Y_{10} and Y_{11} to be the two least prime factors of n. Thus

$$Y_{10}=\binom{r-2}{2},\qquad Y_{11}=(r-2)(s-r),$$

with the binomial zero at $r=2,3$. Substitution in the first six weighted terms minus $Y_{10}+Y_{11}$ gives $s-t\le 0$; the other subtractions help.

For $k=\Omega(n)\ge 3$, let $q_1<q_2<\cdots$ be the factors of n. Then $q_1q_2^2<n\le N$, so $t\ge 1$, $Y_7=\mathbf{1}_{s=0}$ and $Y_8=\mathbf{1}_{t=1}$. When $r\le 1$, the remaining upper bounds for W are

case	upper bound
$r=1$	$2-t-\mathbf{1}_{t=1}$
$r=0,\ s\ge 2$	$4-s-t$
$r=s=0$	$2-t-\mathbf{1}_{t=1}$
$r=0,\ s=1,\ t\ge 3$	$3-t$
$r=0,\ s=1,\ t\le 2$	$3-t-1-\mathbf{1}_{t=1}$.

In the last case, $k \geq 4$ would imply $n \geq N^{h+d+2/3} > N$, so $k = 3$; moreover $q_1 q_2 q_3^2 \geq N^{h+d+2/3} > N$. The pair (q_1, q_2) therefore contributes to Y_9, giving the extra subtraction displayed there. All the bounds are nonpositive. For $k \leq 2$ direct substitution gives $W \leq 4$; the unit has $W(1) = 4$. □

Let E_0 be the selected 6091 zero-floor rectangles and E_1 the disjoint selected 11481 rectangles in the pair-512 partition, in coordinates

$$(t, u) = (\log_N p_2, \log_N p_1), \qquad a \leq u < h \leq t < d.$$

Thus $E = E_0 \cup E_1$ contains 17572 rectangles. Write $Y_6(E)$ for its pair terms and $Y_{11}(E)$ for the four-prefix terms whose last two selected primes (p_3, p_4) have $(\log_N p_4, \log_N p_3) \in E$. The current weight is

$$W_E = W - Y_6(E) + Y_{11}(E). \tag{28}$$

Lemma 4.2 (Paired injection)**.** *For squarefree $n \leq N$, $Y_{11}(E) \leq Y_6(E)$, and hence $W_E(n) \leq 4\mathbf{1}_{\Omega(n)\leq 2}$.*

Proof. The quotient roughness in a contributing four-prefix tuple forces its first two selected primes to be the two least factors of n. Send this tuple to (p_3, p_4). It lies in the prescribed pair rectangle, and the residual after removing $p_3 p_4$ is still N^a-rough. The first two factors are uniquely determined by n, so the map is injective into the pairs counted by $Y_6(E)$. □

The fixed prime-factor bound $\lfloor 1/a \rfloor = 11$ bounds $|W_E|$ by 1016 on rough n. The number of $n \leq N$ with a repeated prime at least N^a is at most $2N^{1-a}$ for large N. After restricting to primes $p < N$ with $(p, N) = 1$ and removing the unit, the squareful and unit costs are at most $2032N^{1-a} + 4 = o(H_N)$. It follows that

$$4D_{1,2}(N) \geq \sum_{\substack{p<N \\ p \text{ prime} \\ (p,N)=1}} W_E(N-p) - o(H_N). \tag{29}$$

This establishes the count inequality independently of a contracted last-range Chen weight.

5 Lower compilation, finite feedback and its current insertion

5.1 A genuine lower estimate for the original prime sequence

The ordinary lower trees start at $w = a$ and $w = h$, with cap $\nu = 249/448$ and fixed small buffers. Their first exits have even rank. The analogue of (17) is

$$\sum_d \mu(d)\mathbf{1}_{d \text{ retained}} q(d) = \prod_p (1 - q(p)) - \sum_{e \text{ even first exit}} q(e) \prod_{p<p_{\text{last}}(e)} (1 - q(p)). \tag{30}$$

This identity proves both the pointwise lower inequality and the literal raw-density formula. The selected fourth own-cube restorations remain legal even exits: their cap is chosen before the fourth cube, $S_3 + 3t_4 < \lambda_4$, and their ordinary residual level exceeds p_4^2. A restored slab receives no old fifth gain again.

For a presieve $z_0 = N^\omega$, even exits give the exact projection direction

$$V_N(z_0) R_{\text{rough}} - V_{\text{full}} = \sum_{\substack{e \text{ full even exit} \\ p_{\text{last}}(e)<z_0}} g_N(e) V_N(p_{\text{last}}(e)) \geq 0. \tag{31}$$

Round positive even coefficients inward and negative odd coefficients outward, retaining the causal history. Attach beta-minus to positive coefficients and beta-plus to negative coefficients. If the

rounded coefficient is $a(d)$, then $a(d)B_{\mathrm{sign}}(n) \le a(d)I_0(n)$ term by term. Therefore the physical lower coefficient satisfies

$$\sum_{q\mid n} \rho_N(q) \le \mathbf{1}_{(n,P_N(N^w))=1}, \qquad \sum_q \frac{\rho_N(q)}{\varphi(q)} \ge V_{\mathrm{full}} - \text{compiler error}.$$

Beta-minus need not be nonnegative. The second inequality uses (31) and absolute error control, not coefficient order alone. The coefficient theorem supplies (15) for this very lower coefficient.

The lower construction needs a sharp count of the original prime, which requires an endpoint argument beyond sharp individual Heath–Brown slots. We include the correction from Appendix H.

Lemma 5.1 (Sharp original-prime endpoint)**.** *Suppose the actual coefficient has bounded squarefree support $q \le N^\theta$, $\theta < 1$, $(q, N) = 1$, and satisfies the permitted smooth-singleton instance of (15). Then*

$$\sum_q \rho_N(q)\left(\pi(N;q,N) - \frac{\pi(N)}{\varphi(q)}\right) = O_A(N(\log N)^{-A}). \tag{32}$$

Proof. For a prescribed A, set $\delta = (\log N)^{-B}$ with fixed $B > A + 3$, and smooth the original endpoint on $[N-\delta N, N+\delta N]$. Its Euler derivatives have the fixed polylogarithmic size allowed in (15). Apply the already permitted smooth lower trim below $2N^{1-\kappa_0}$. The endpoint contribution after summing moduli is bounded absolutely by

$$\sum_{\substack{p \text{ prime} \\ 0<|p-N|\le \delta N}} \sum_{q\mid N-p} |\rho_N(q)| \ll 2 \sum_{1\le n\le \lceil \delta N\rceil} \tau(n) \ll \delta N \log N + \log N.$$

The zero difference $p = N$ is impossible because $N > 2$ is even. The exact mean boundary has the same bound, using $\sum_{q\le N^\theta} 1/\varphi(q) \ll \log N$. The lower trim has a fixed power saving. Finally, the exact-coprime mean differs from $\pi(N)/\varphi(q)$ by the prime divisors of q; squarefreeness gives total error

$$\ll \sum_{\ell\le N^\theta} \frac{1}{\ell - 1} \sum_{e\le N^\theta/\ell} \frac{1}{\varphi(e)} \ll \log^2 N.$$

Taking the local logarithmic accuracy after these fixed derivative orders proves (32). Omitting the $O(\log N)$ primes dividing N also has a negligible compiled divisor cost. □

Combining the pointwise lower coefficient with Lemma 5.1, the raw density (18), and PNT gives the required ordinary lower prime counts. The positive-product upper coefficient K is not being used to infer an ordinary prime lower bound.

5.2 Finite defects and actual descendants

Put $A(s) = sF(s)/(2e^\gamma)$ and $\overline{a}(s) = sf(s)/(2e^\gamma)$. For a nonnegative finite marked family $\sigma(B)$ and one common $Q = N^\lambda$, normalize by

$$\Theta_Q = 4\,\mathrm{li}(N) \sum_B \frac{\sigma(B)C(BN)}{\varphi(B)\log(Q/B)}.$$

Here $C(BN)$ is the same singular-factor function as in (1). A defect $H(s)$ means an upper bound $(A(s) - H(s) + o(1))\Theta_Q$; a defect $h(s)$ means a lower bound $(\overline{a}(s) + h(s) + o(1))\Theta_Q$. These are estimates averaged over the positive marked family, not pointwise in B and not assertions for a signed outer measure.

For residual exponent $L = \lambda - \log_N B$, write $\alpha = (1 - \log_N B)/L \ge 2$. A singleton child with exponent Lu has

$$L' = L(1-u), \qquad \alpha' = 1 + \frac{\alpha - 1}{1-u};$$

a two-prime child replaces u by $u+v$. All children retain the same total Q, every full and selected square, and their physical prime ceiling. The parameter α increases and is never reset to 2 in a switched mass.

The numerical seed is finite: four applications of $J \mapsto 0.020249 + 0.13J$, starting at $J = 0.02$, give $0.023273777353 > 2327/100000$. Together with its analytic seed inequality this supplies $H \geq 2327/100000$ on $[1, 11/5]$. The action bank consists of 70 one-step and 36 repaired two-step actions, together with the positive Buchstab transitions

$$H_{\mathrm{new}}(s) = \int_{\max(2,s-1)}^{5} \frac{h(w)}{w}\, dw, \qquad h_{\mathrm{new}}(s) = \int_{s-1}^{5} \frac{H(w)}{w}\, dw \quad (s \geq 2).$$

The one-step switched cost is $KI/8$. A two-step action has the positive defect contribution

$$\frac{1}{5}\left[4H(s_0) + H(\phi_1) + \sum_{\mathcal{D}_1} \int \frac{h(\rho)}{u(1-u)}\, du + \sum_{\mathcal{D}_2} \iint \frac{H(\rho)}{uv(1-u-v)}\, du\, dv\right] \tag{33}$$

in addition to its proved ordinary and positive-switching base. There are three singleton and four good pair domains. With $a_i = 1/\phi_i$ and $a_0 = 1/s_0$, the pair domains are

	domain	ρ
D_5^+	$a_0 < u < v < a_2,\ v + 2u < 1$	$s_0(1-u-v)$
D_6	$(a_0, a_1) \times (a_2, a_3)$	$s_0(1-u-v)$
D_7	$a_0 < u < v < a_1$	$(1-u-v)/u$
D_8	$(a_0, a_1) \times (a_1, a_2)$	$(1-u-v)/u.$

All bank gates, including $2v < 1$ and $v + 2u < 1$, are enforced. The excluded portion of D_5 receives only a positive product cost, not ordinary feedback. Unit surfaces in the switching terms stay in their masses. The envelope used for every actual $\alpha \geq 2$ is $\omega(s) \leq \max(1/s, 70893/125000)$ for $s \geq 1$, with the specified conservative extension below 1.

Every right-hand side uses only a preceding finite generation. Alternatives are maximized, not added, and crossing cells use closed support minima. There is therefore an induction on actual positive counts. The selected arrays have 200 closed cells with endpoints $1+i/50$; a new seed-only replay of 430 simultaneous rounds gives all 400 final H, h entries exactly. The 248 and 312 arrays are read only for comparison, not as generating inputs. Only the two upper costs U_3, U_4 use the 430 table. The designated 248 occurrences elsewhere are unchanged.

The finite induction also needs a common approximation, not just finite depth. Positive transition mass is at most 12 and the root coefficient norm at most 7/2. A local oriented error δ propagates by at most $\delta(12^{441} - 1)/11$ before root normalization. Choose δ inside the prescribed final reserve and use one nested finite refinement for all queries. Residual exponents after the finite transitions remain above a fixed positive multiple of 101^{-440}, so every positive leaf satisfies the fixed-cutoff hypothesis of Proposition 3.1. Zero-mass cells are controlled by absolute errors.

5.3 The selected eleven occurrences and paired accounting

The root functional is

$$\mathcal{C} = \frac{1}{4}(3L_1 + L_2 + L_5 + L_6 - U_3 - U_4 - 2U_7 - U_8 - U_9 - U_{10} - U_{11}). \tag{34}$$

The lower starts are the causal starts at a, h. In replacing the old second start, its entire former H_{248} least-prime correction is removed; it is not added to the new start. In U_3, U_4 the H_{430} argument is $\max(2, L/a)$ with $L = v(t) - t$ in the enhanced range. The clamp at 2 is a real state of positive measure. The pair lower terms use h_{248} where $L/a \geq 2$ and zero below 2. The positive costs U_7, U_8, U_9 retain their labelled multiplicities and are estimated by Proposition 3.1.

On one common positive four-prefix partition, U_{10}, U_{11} use the minimum of the switched kernel $K\omega((1-S_4)/z)/z$ and each eligible ordinary kernel $4(1-H_{248}(\max(2,L/w)))/L$, with decreasing exponents $x>y>z>w$. Every ordinary option passes all full and selected squares. Priority refinements replace their parents; alternatives on overlapping domains are not summed. Appendix G specifies all the actual states and the simultaneous causal approximation.

We first record the baseline accounting, before the refinement in Section 5.5. The crop in (28) requires a lower functional argument as well as the pointwise injection. On a pair rectangle $B=[t_l,t_r]\times[u_l,u_r]$, put

$$R(t,u)=\frac{57}{64}-\frac{5}{2}t-u, \qquad R_{\max}=R(t_l,u_l), \quad s_{\max}=R_{\max}/a, \quad M_B=\log(t_r/t_l)\log(u_r/u_l).$$

The saved lower floors are zero on E_0. In E_1 there are 227 guarded crossing cells with zero saved floors and 11254 positive cells with $2\le s_{\min}\le s_{\max}\le 9/4$. Since $\log(s-1)/s$ is increasing there, and H_B denotes the saved h_{248} minimum over the full closed argument support, the discarded saved lower estimator, in Chen-quarter units, is bounded above by

$$\sum_{B\in E_1 \text{ positive}} \frac{M_B}{R_{\max}}\bigl(\log(s_{\max}-1)+H_B\bigr)\le 0.002372094. \tag{35}$$

This is not an upper bound for the actual removed Y_6 count.

For the negative term, the same selected parents are switched in the current $n48$ cover and avoid the priority-40000 replacement. If A_B is the old saved density mass and s_B its fixed K feedback slope, the certificate proves $0\le s_B\le A_B/K_0$, $K_0=6.7619$. Therefore

$$A'_B=A_B-(K_0-K)s_B\ge (K/K_0)A_B.$$

Dividing by its positive reference mass gives an explicit pointwise density majorant of the current switched kernel. Restricting this majorant to the complementary positive family removes at least $KI_E/8$ in Chen units, where

$$I_E=\int_{(t,u)\in E}\int_{a<w<z<u}\frac{dt\,du\,dz\,dw}{tuz^2w}. \tag{36}$$

The use of $\omega\ge 1/2$ is valid since $(1-S_4)/z\ge 77/38>2$ throughout this family. For one rectangle the integral is exactly

$$\log(t_r/t_l)\left[\frac{\log(u_r/u_l)}{a}+\frac{\log(u_r/a)+2}{u_r}-\frac{\log(u_l/a)+2}{u_l}\right].$$

The finite lower bounds for the gross improvements are 0.002095161 on E_0 and 0.003799514 on E_1. Subtracting (35), the additional raw improvement from E_1 is

$$0.003799514-0.002372094=0.001427420=\frac{71371}{50000000}. \tag{37}$$

This removes mass from a known positive majorant, while paying the discarded saved lower floors. It does not subtract an upper bound for a removed count in the wrong direction. The two masks are disjoint and no previous crop gain is counted twice.

5.4 The baseline endpoint

The complete baseline finite inequalities give the conservative raw endpoint

$$R_{\text{base}}=\frac{1345575129530003739571681432301188857312754737}{685738127494661151198444592354557952000000000}. \tag{38}$$

Each baseline lower reserve is

$$B_{\mathrm{base}} = \frac{250474831809194280822969421937979930755098 7}{548590501995728920958755673883646361600000000 0}. \tag{39}$$

One pays the common finite density approximation; the other pays compilation and smoothing. These already include all allowances adopted before the present same-mask refinement. They gave the former safe coefficient $c_{\mathrm{base}} = 1.96131568$ and the positive slack

$$R_{\mathrm{base}} - 2B_{\mathrm{base}} - c_{\mathrm{base}} = \frac{248545088694905535260056510703831852 1}{274295250997864460479377836941823180800000000 0} > 0. \tag{40}$$

We retain this entire accounting and make only the following two local changes on the already deleted pair cells.

5.5 Refinement on the same paired mask

The outer arithmetic weight in (28), the 6091 cells of E_0, the 11481 cells of E_1, their parent assignments and the 248/430 table policy remain exactly the same. The total is 17572 cells. The value 17772 in some inherited summary fields was an addition or transcription error; it did not change the mask or the sums. We strengthen the lower bound for the removed negative majorant on a fixed inner fibre, and reduce the upper payment for the discarded saved positive estimator. In particular this step does not postulate a stronger distribution theorem.

Lemma 5.2 (A global Buchstab lower bound)**.** *For every $s \geq 7/3$ one has $\omega(s) \geq 11/20$.*

Proof. The definition $\omega(s) = 1/s$ on $[1, 2]$ and the delay equation $(s\omega(s))' = \omega(s-1)$ first give $\omega(s) \geq 1/2$ for all $s \geq 1$. Indeed the assertion holds on $[1, 2]$, and the derivative of $s(\omega(s) - 1/2)$ is nonnegative on each successive unit interval on which the preceding assertion is available.

For $7/3 \leq s \leq 3$ the exact formula is $\omega(s) = (1 + \log(s-1))/s$. The inequality

$$\log x \geq \frac{2(x-1)}{x+1} \qquad (x \geq 1)$$

follows by differentiating their difference: its derivative is $(x-1)^2/[x(x+1)^2] \geq 0$ and the difference is zero at $x = 1$. Consequently

$$\omega(s) \geq \frac{3s-4}{s^2}.$$

After multiplication by $20s^2$, the difference of this bound and $11/20$ has numerator $60s - 80 - 11s^2$. It is a concave quadratic and is positive at both endpoints. Equivalently the endpoint differences are

$$\frac{27}{49} - \frac{11}{20} = \frac{1}{980}, \qquad \frac{5}{9} - \frac{11}{20} = \frac{1}{180}.$$

This proves the assertion on $[7/3, 3]$ without a monotonicity assumption on ω.

For $3 \leq s \leq 10/3$, the global $1/2$ bound and $\log 2 \geq 2/3$ give

$$s\omega(s) \geq 1 + \log 2 + \frac{s-3}{2} \geq \frac{s}{2} + \frac{1}{6}, \qquad \omega(s) \geq \frac{1}{2} + \frac{1}{6s} \geq \frac{11}{20}.$$

Thus the desired bound holds on the full unit interval $[7/3, 10/3]$. For later s the derivative of $s(\omega(s) - 11/20)$ is $\omega(s-1) - 11/20$. Induction on successive unit intervals, starting with the nonnegative endpoint value at $10/3$, proves the assertion for all $s \geq 10/3$. □

The extra positive fibre. The already selected parent majorant dominates $K\omega((1-t-u-z-w)/z)/z$ on $a<w<z<u<h<t<d$. For each rectangle $B=[t_l,t_r]\times[u_l,u_r]$ of the same mask, put

$$Z_B=\min\left(u_l,\frac{3(1-t_r-u_r)}{13}\right). \tag{41}$$

The finite exact geometry check gives $Z_B>a$ for every cell. Moreover the definition gives $1-t_r-u_r-2Z_B\ge(7/3)Z_B$. Hence on the inner fibre $a<w<z<Z_B$ one has $z<u$ and

$$1-t-u-z-w\ge 1-t_r-u_r-2z\ge\frac{7}{3}z.$$

Lemma 5.2 supplies an additional $1/20$ in ω, beyond the $1/2$ already paid in (36). The remainder of every fibre keeps its original bound. These two nonnegative contributions hold simultaneously; they are not two alternative estimates of one saved component.

Integrating the added $1/20$ and including the Chen factor $1/4$ gives at least the following additional coefficient from B:

$$\begin{aligned}G_B&=\frac{K}{80}\log\frac{t_r}{t_l}\log\frac{u_r}{u_l}\int_a^{Z_B}\frac{\log(z/a)}{z^2}\,dz\\&=\frac{K}{80}\log\frac{t_r}{t_l}\log\frac{u_r}{u_l}\left[\frac{1}{a}-\frac{1+\log(Z_B/a)}{Z_B}\right].\end{aligned} \tag{42}$$

The inner w integral is $\log(z/a)$, and the remaining primitive is $-(1+\log(z/a))/z$. Thus there is no permutation factor and no extra factor 2. The fixed rational walls in (41) remain within the same licensed finite positive family. Restriction is applied to the proved positive parent majorant, with its original ownership, rather than to an unproved decomposition of a global maximum.

The two separately implemented directed evaluations described in Appendices K and L give the rigorous safe lower bound

$$\sum_{B\in E_0\cup E_1}G_B>\frac{558583}{10^9}=0.000558583=:g. \tag{43}$$

The enclosed totals are approximately $1.9318091250\cdot10^{-4}$ on E_0 and $3.6540308174\cdot10^{-4}$ on E_1; only the rational lower bound in (43) is adopted. This is the version using threshold $7/3$. It replaces the prospective threshold-$12/5$ estimate of the same increment; those two versions must not be added together.

Literal discarded floors and an independent upper proxy. Let $Q_{80}=2^{80}=1208925819614629174706176$. The selected historical L_6 integer is $365077417315690091410168 5$. Its equality with the recomputed author-512 sum fixes the required decomposition into cells. That calculation uses the ordinary mesh 16384 and the h_{248} table; its enhanced half is unchanged here. Replaying the saved nonenhanced estimator gives zero on all 6091 cells of E_0 and on the 227 crossing cells of E_1. The remaining 11254 cells have the same support restrictions stated above. The discarded integer sum is 11290806845175652020270, so the literal discarded estimator in Chen-quarter units is exactly

$$\begin{aligned}L_{\mathrm{lit}}&=\frac{11290806845175652020270}{4Q_{80}}=\frac{5645403422587826010135}{2417851639229258349412352}\\&<\frac{466977}{200000000}=0.002334885=:\lambda.\end{aligned} \tag{44}$$

These are saved lower estimators; their zero values do not assert that the corresponding arithmetic counts vanish.

There is also a separately implemented upper proxy for the same saved floors. To specify it, on a positive cell use the residual function $R(t,u) = 57/64 - (5/2)t - u$ from the baseline accounting, and put

$$r_B = R(t_r, u_r), \quad R_B = R(t_l, u_l), \quad s_l = r_B/a, \quad s_h = R_B/a,$$
$$f_l = \frac{\log(s_l - 1)}{s_l}, \qquad f_h = \frac{\log(s_h - 1)}{s_h}.$$

Here $2 \le s_l \le s_h \le 9/4$. Define

$$\begin{aligned} M_B &= \log(t_r/t_l)\log(u_r/u_l), \\ M_{R,B} &= \int_B R(t,u)\frac{dt\,du}{tu} \\ &= \frac{57}{64}M_B - \frac{5}{2}(t_r - t_l)\log(u_r/u_l) - (u_r - u_l)\log(t_r/t_l). \end{aligned}$$

The two old ordinary alternatives, after division by the Chen factor 4, are bounded above by

$$\frac{f_l M_B}{a}, \qquad \frac{f_l(R_B M_B - M_{R,B}) + f_h(M_{R,B} - r_B M_B)}{a(R_B - r_B)}. \tag{45}$$

Indeed the exact endpoint values f_l, f_h dominate the historical lower endpoints. Both coefficient integrals are nonnegative because $r_B \le R(t,u) \le R_B$, and the historical fixed-point lower roundings can only decrease the saved value. The historical procedure takes a maximum of its constant and chord alternatives; taking the maximum of the two upper bounds in (45) preserves the direction.

For the separate defect term, take H_B to be the minimum of the h_{248} closed-cell values meeting the exact interval $[s_l, s_h]$, including both adjoining cells at a knot. This minimum is at least the one over the old outward-rounded lookup interval. Therefore $H_B M_B/R_B$ is an upper bound for the saved defect floor, and can be added to the preceding maximum. All relevant h_{248} entries are nonnegative. If the old outward lookup crosses below 2 and returns zero, the same upper bound remains valid. On the 227 crossing cells the constant, eligible chord alternatives and lookup all give zero. Every E_0 cell has $s_h \le 2$ and likewise has zero saved floor.

A decimal integer interval implementation, using 320-term alternating Mercator sums after range reduction to $[1, 3/2]$ and the next-term remainder, encloses the sum of these proxies strictly below λ. Its displayed value is $0.00233488413635981310147442734879 04\ldots$. It imports no project interval or sieve evaluator. Separately, the final review compares the upper proxy to the literal saved integer on every one of the 11481 cells of E_1. Thus both the replay and the proxy support replacing the baseline payment (35) by λ. This is not an upper bound for the true removed Y_6 count, and the proxy is not claimed to reconstruct the historical delay mesh.

The recovered payment is exactly

$$0.002372094 - \lambda = \frac{37209}{10^9} = 0.000037209. \tag{46}$$

The retained L_6 lower endpoint is the old whole L_6 lower endpoint minus 4λ, namely

$$\frac{14216731883232723828945058469141}{47223664828696452136960000000}.$$

The total removed negative gross in Chen units is now the safe value $0.002095161 + 0.003799514 + g = 0.006453258$. Together with the retained old allowances, these substitutions account for the entire change; no new crop or table improvement is included.

Feedback and reserve allocation. Combining (43) and (46), the adopted raw increment is

$$\Delta = g + (0.002372094 - \lambda) = \frac{37237}{62500000} = 0.000595792. \tag{47}$$

For possible later changes in K, only the extra gross is linear in K. The future raw feedback slope is consequently

$$\begin{aligned} A_{\mathrm{new}} &= A_{\mathrm{base}} - \frac{g}{K} = A_{\mathrm{base}} - \frac{558583}{6761736800} \\ &= \frac{25892776044271626542645156082021325504082078 7}{66239724740624887279841241469783045025300480 0}. \end{aligned} \tag{48}$$

The loss recovery is independent of K and is not divided by K in this calculation. This slope is recorded for consistency; the present theorem keeps K fixed as in (5).

We conservatively add $\Delta/8 = 37237/500000000 = 0.000074474$ to each of the two baseline reserves. All earlier reserves are retained once, and no previous allowance is paid twice. The net increment after both extra reserves is

$$\Delta - 2(\Delta/8) = \frac{3\Delta}{4} = \frac{111711}{250000000} = 0.000446844. \tag{49}$$

The new walls and queries are finite, so their local accuracy can be chosen inside these positive additional allowances by the same common approximation and compilation argument used for the baseline.

5.6 The refined lower endpoint and common quantifiers

With the substitutions just proved, the conservative raw endpoint is

$$R = R_{\mathrm{base}} + \Delta = \frac{13459836868204600387282762560017569641040927 21}{68573812749466115119844459235455795200000000 0}. \tag{50}$$

Each of the two current lower reserves is

$$B_0 = B_{\mathrm{base}} + \frac{\Delta}{8} = \frac{29133056085482419648245179199479060988889 71}{54859050199572892095875567388364636160000000 00}. \tag{51}$$

From (29), the actual lower compilation, finite induction, signed root insertion and the same-mask refinement,

$$\liminf_{\substack{N\to\infty \\ 2\mid N}} \frac{D_{1,2}(N)}{H_N} \geq R - 2B_0. \tag{52}$$

Here

$$R - 2B_0 = \frac{53810214416732919129482805060870799503174819 13}{27429525099786446047937783694182318080000000 00}, \tag{53}$$

and exact subtraction gives the strict slack

$$R - 2B_0 - c = \frac{13457260926863633771775678584711245753}{27429525099786446047937783694182318080000000 00} > 4.9061 \cdot 10^{-9} > 0. \tag{54}$$

In particular $D_{1,2}(N) \geq cH_N$ eventually, with the new $c = 1.96176252$.

The order of choices is part of the argument. Fix the query graph, tail and compact accuracies, causal caps and support reserves. Then choose the common beta/rough parameters, fine labels, finite divisor orders, permitted κ, and local logarithmic accuracy. Only then let N grow through all even integers. There are finitely many constants and thresholds to maximize. Since $C_N \geq C_2 > 0$, fixed power savings and $O_A(N/\log^A N)$ with $A > 2$ are uniformly $o(H_N)$. Neither an average over targets nor an infinite fixed point is used.

6 Finite numerical lemma and reproducibility scope

The baseline numerical assertion concerns one selected set of analytic functions, domains and action constants. Its explicit specification is `global-selected-numeric-completion-20261010-obligations.json`; the exact reconstruction and complete selected outputs are indexed by `global-selected-numeric-completion-20261010.json`. Appendix I states all 29 obligations, with their enclosure proofs and complete result families. The adopted supplement is specified by `corpus-renewal-improvement-assembly-20261010.json`, which binds the version-2 paired certificate, the separate interval calculation and the final reverse review. These additional inequalities are derived in Section 5.5; their complete texts appear in Appendices K–L.

The upper calculation is the direct expression

$$E = \frac{4}{\Theta} + I_1 + I_{3,\mathrm{compact}} + I_{3,\mathrm{tail}} - \sum_{j=1}^{22} G_j,$$

where each G_j is a lower bound for a selected positive saving. The full tail is replaced by its new bound, not subtracted both as an old tail correction and as a new gain. Same-domain log-moment, curvature and mixed improvements are only increments over the already adopted bound. The final value is (22), with reserves (23).

The lower reconstruction starts from the complete eleven-term functional (34), using lower endpoints for positive terms and upper endpoints for negative terms. It adds the selected complete priority replacements, the changed ordinary starts with the full obsolete correction removed, the six current I_3, I_4, I_5 values, and the selected restoration. The slanted restoration contains 98668 boxes and 789344 layers and replaces the older whole-box restoration. Fixed K feedback, only the U_3/U_4 table differences, and the two paired masks give the baseline raw value. A direct reconstruction is slightly above the conservative R_{base} in (38); the larger value is not adopted as another improvement. The current endpoint then adds exactly Δ from (47), and enlarges each reserve by exactly $\Delta/8$ as in (51).

Complete coverage has different meanings for the two signs. A positive saving may use a legal subset and discard the rest. A negative-cost upper integral must cover its whole retained domain. For example the $n48$ minimum cover includes all 1190700 ordered rows, including tied-bin simplices; the 40000 priority parents have all 516779 children. The parent refinement is a replacement by a maximum saving. The paired crop uses the explicit restrictable parent majorants after feedback, not an unproved cell decomposition of a global maximum.

The directed arithmetic uses rational endpoints, explicit series remainders, and outward floor/ceiling operations. The finite polytope method solves all selected active bases exactly, verifies every original facet inequality, proves boundedness, reconstructs the face lattice and uses a pulling triangulation with exact moments. It does not accept a floating-point hull as a certificate. Ordinary delayed-function queries have numerator-chord and curvature/interpolation proofs; the shared three lower profiles have 23199750 checked node intervals, together with their query interface. That number is not a count of all grids in the project.

The short reconstruction reruns direct rational ledgers, the final finite action map from the seed, budgets and logarithm margins. It checks all 400 final entries after 430 rounds and source identities. It does not rerun every historical large quadrature or every polytope calculation. Those selected finite inequalities remain supported by their complete records and the identified enclosure and partition proofs. The archive distinguishes fresh replays, earlier complete runs, shared arithmetic backends and separately implemented checks. In the adopted supplement, the literal ordinary-floor replay is explicitly distinguished from the independent upper proxy. The final reverse review checks every one of the 17572 inner cutoffs and its exact geometry, every E_1 proxy against the saved floor, the intersection of the two E_1 extra-mass intervals, and the 53 source and dependency bindings of those two ledgers. This does not claim another independent execution of the old delayed function mesh. Neither numerical agreement nor separate program authorship substitutes for an analytic density identity or an admissible finite action.

Released reproduction materials. Source and the compact verification package are available at https://github.com/UyNewNas/chen-imbalanced-semiprimes. The full numerical evidence and checksums are attached to release v2026.10.10. The compact package contains the 49 JSON inputs used by the baseline selected ledger and the adopted refinement's additional records and replay programs. From the repository root, run

```
python verification/run_verification.py --work-dir replay-quick
```

The output directory must be new. This command validates the compact manifest, reconstructs the 22 upper savings and the lower ledger, checks both reserves and the logarithm margins, and applies the 106 finite actions for 430 simultaneous rounds from the seed, comparing all 400 final entries. It then regenerates the adopted refinement on all 17572 cells using both interval implementations, compares their saved-floor and mass estimates, and checks the new reserves and logarithm margins. The documented compact run was repeated successfully during preparation of this release.

The separate full evidence archive adds the inherited records and their binding audit. Follow its README for the full replay. It verifies the public manifest and inherited source bindings and repeats the selected ledger and finite iteration; it does not regenerate all quadratures. This full audit was also rerun successfully for the publication copies. Nonmathematical private provenance has been removed from a small set of publication copies. The original and public hashes and exact permitted transformations are documented. The adapted checker accepts only these listed hash substitutions; it does not claim the changed files are byte-identical to their private originals. The source library separately records its text-only provenance edits and the mask-count errata.

The baseline replays alone would not certify the current increment. The integrated command therefore also runs the version-2 paired certificate and the separate interval and final-review checks; their generators, outputs and source bindings are supplied. The final assembly keeps the baseline upper endpoint and budgets, adds exactly the lower increment and reserves proved above, and checks the new rational logarithm margins. The superseded threshold-12/5 candidate is historical material, not an additional selected input.

7 Ordered bad semiprimes and the final subtraction

Appendix J contains the full transfer derivation, including its separate scalar certificate. We give the count and normalization argument here to fix the ordering and all-even quantifiers.

Fix $1/2 < b < 1$ and set $\beta = b/(1+b)$. Products below $2N^{1-\kappa_0}$ have a negligible count. For a remaining bad product $m = rq$, the inequality $r > q^b$ implies $r > m^\beta$; both primes therefore have a fixed positive exponent. Use the coordinates

$$t = \frac{\log r}{\log N}, \qquad y = \frac{rq}{N}.$$

On a fixed radial strip $\eta \le y \le C$, PNT in $q = Ny/r$ followed by reciprocal-prime partial summation in r gives the unsifted mass element

$$\frac{N}{\log N}\,\frac{dt\,dy}{t(1-t)}. \tag{55}$$

The PNT errors are uniform on positive exponent compacta. A prime upper bound controls $y < \eta$ by $O_a(\eta N/\log N)$; N tends to infinity first, then η tends to zero.

The exact ordered bad inequalities are

$$0 < y < 1, \qquad t \le \frac{1}{2} + \frac{\log y}{2\log N}, \qquad t > \beta + \frac{b\log y}{(1+b)\log N}.$$

Their limiting unsifted mass is

$$J_b = \int_0^1 dy \int_\beta^{1/2} \frac{dt}{t(1-t)} = \left[\log \frac{t}{1-t}\right]_\beta^{1/2} = \log(1/b). \tag{56}$$

The ordering $r \le q$ counts each composite once. There is no factor 2 and no further singular factor in J_b. The equal-prime diagonal has size $O(\sqrt{N})$ and can be retained or restored after a squarefree argument.

Smooth the complete-product wall in y and the remaining walls in t. For any fixed $\epsilon > 0$ a positive outer approximation has limiting mass at most $J_b + \epsilon$ and stays in the licensed finite family. A partition of unity repeats no mass. Proposition 3.1, followed by $N \to \infty$ and then $\epsilon \downarrow 0$, gives

$$\limsup_{\substack{N\to\infty \\ 2|N}} \frac{B_b(N)}{H_N} \le K \log(1/b). \tag{57}$$

Unique factorization gives the exact identity

$$D_b(N) = D_{1,2}(N) - B_b(N). \tag{58}$$

The unit was excluded in (2); a convention including it would change the count by at most one. Zero is excluded by $p < N$. A square is a bad remainder when $b < 1$, and a prime remainder is a good one. Combining (52), (54) and (57) proves Theorem 1.1.

For completeness, all final scalar signs can be checked without any numerical quadrature. For rational b put $z = (1-b)/(1+b)$. The positive atanh series and its geometric tail give

$$2\sum_{j=0}^{95} \frac{z^{2j+1}}{2j+1} < \log(1/b) < L_{96}(b) := 2\sum_{j=0}^{95} \frac{z^{2j+1}}{2j+1} + \frac{2z^{193}}{193(1-z^2)}. \tag{59}$$

Every endpoint is rational. Substituting (5) gives

$$\begin{aligned} c - KL_{96}(74816964/10^8) &> \frac{1001149712517311}{10^{20}} > 10^{-5}, \\ c - KL_{96}(74816865/10^8) &> \frac{106416285358807}{10^{20}} > 10^{-6}, \\ c - KL_{96}(9/10) &> \frac{124934244400949969098}{10^{20}} > 1.24889. \end{aligned} \tag{60}$$

The respective surpluses exceed $1.1497 \cdot 10^{-8}$, $6.4162 \cdot 10^{-8}$ and $4.5244 \cdot 10^{-4}$. First choose the fixed realization and positive-smoothing errors inside these strict margins; then let the remaining uniform asymptotic errors tend to zero. Taking the maximum of the three resulting thresholds proves Corollary 1.2.

8 Comparison, scope and authorship

The explicit comparison recorded in the proof sources is with Li–Liu [9], arXiv:2606.05224v2, Theorem 1.1 and its following remark. Their factor parameter is $1 + b$; the stated unconditional values correspond to $b = .9$ and the indicated refinement .894, with the all-sufficiently-large-even quantifier. The exponents in Corollary 1.2 impose a smaller factor ratio under the same quantifier. Counts at different exponents should not be compared by their leading coefficients alone. At the common exponent .9, (9) provides the separate count comparison.

This comparison does not extend to their conditional $b = .4$ statement under WEH(.999), to fixed-gap twin-prime statements, or to assertions for almost all even targets. It is a comparison with explicitly identified statements, not a proof of exhaustive literature priority or of superiority to every possible sieve optimization. The complete texts of Cai (2008) and Fouvry–Grupp (1989)

were not part of the recorded full-text comparison, and no claim about having checked every theorem in those papers is made.

The current proof is modular: the local distribution derivation, its coefficient and product insertion, the same-weight upper density, the full lower-weight argument, and every selected finite inequality have distinct stated responsibilities. The complete analytic modules are included for mathematical review; the new distribution derivation is part of the submitted argument, not an externally accepted black-box theorem. A remaining error in one of those modules would require revising the conclusion. The numerical checks alone do not establish that analytic chain.

Generative-AI assistance through OpenAI ChatGPT/Codex was substantive in the research and preparation of this manuscript. It included candidate proof constructions, parameter exploration, organization and checking of arguments, development of generators and checkers, and writing, translation and integration. Separate implementations and AI-assisted internal reviews are not described as independent human peer review. A complete machine formalization is not supplied, and the verification record does not claim a fresh recomputation of every historical calculation or a reproof of all classical analytic inputs. Exact model identifiers for all earlier sessions were not preserved and are not inferred from interface names. The human author remains responsible for the mathematical content and its publication. No AI system is listed as an author.

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[8] J. D. Lichtman, *Primes in arithmetic progressions to large moduli, and Goldbach beyond the square-root barrier*, with an appendix with S. Drappeau, arXiv:2309.08522v1 (2023). The unrestricted moving-residue interface is not an input to the present proof.

[9] Jiamin Li and Jianya Liu, *Theorem* $(1 + 1.9)$ *on Goldbach Conjecture*, arXiv:2606.05224v2 (2026). The comparison in Section 8 is limited to the explicitly identified statements in that version.

[10] Yuta Suzuki, *The Rosser–Iwaniec sieve*, unpublished notes, 14 March 2023, 102 pp. Author-hosted PDF. Proposition 9.4(i), pp. 22–23; Proposition 11.8, pp. 64–65; and Lemma 14.7, pp. 97–98.

Reading the complete proof modules

The following twelve appendices reproduce the frozen consolidation modules in their original ASCII mathematical notation. Their internal equation numbers are local to each module. Fixed-width notation such as `sum`, `prod`, `<=` and `1_condition` denotes sums, products, inequalities and indicator functions, respectively. These are complete source texts, rather than abbreviated extracts. Further cited derivations and finite records are indexed in the accompanying `proof-library/` and verification packages. Publication copies of those libraries explicitly record any redaction of private provenance; mathematical text is not silently rewritten.

Statements in the frozen modules about the main manuscript, internal review, or publication status describe the state when that module was written. They do not supersede the current scope and reproduction statement in the main text. Appendices A–J provide the baseline analytic construction and numerical assembly. The adopted same-mask refinement in Appendices K–L replaces the baseline lower coefficient and its derived exponents. The old values printed in the baseline source texts are preserved as provenance, not as the current selected result. The main text states and proves the combined conclusion. Likewise, an internal "completed" or "PASS" label is not external validation.

The references to Suzuki in Appendices F–G mean [10, Propositions 9.4(i) and 11.8, and Lemma 14.7]; the bibliography identifies the dated notes and page locations.

Editorial correction. The count "17772" in the coefficient and upper modules is an arithmetic typographical error: the two disjoint families have $6091 + 11481 = 17572$ cells. The selected masks and numerical certificate have not changed. The same typographical error occurs in the supporting positive-pair improvement note; the source library records all identified locations.

A The restricted medium-variable distribution estimate

Source: `distribution-medium-completion-20261010.txt`.

```
Completion of the local medium convolution theorem, with a Poisson-tail repair
10 October 2026. New proof; the old records, manuscript and state are unchanged.

1. The theorem actually established

Fix theta=7/32, 0<sigma<1/100, c0>0, a fixed divisor order B, and
0<=kappa<=sigma/100. Put x=NM and assume
 c0*x<=|a|<=x^(1+kappa),
 Q1<N*x^(-sigma),
 N^2*Q2*Q3^2<x^(1-8sigma),
 |a|^theta*N*Q1*Q2^5*Q3^2<x^(2-15sigma),
 N^(1+2theta)*Q1*Q2^(5-2theta)*Q3^(2+2theta)<x^(2-15sigma).  (1)
All lengths are at least one, with fixed endpoint multiples allowed using
the strict margins. A bounded nominal group may carry delta_1.

Let alpha be one-bounded, supported on one interval in [N,2N] and on
Pminus(n)>z=x^(1/(loglog x)^3). Assume full auxiliary Siegel--Walfisz:
 for each A>0 and every k,q>=1, (b,q)=1,
 |sum_(n=b mod q,(n,k)=1)alpha_n
   -phi(q)^(-1)sum_((n,kq)=1)alpha_n|
      <=C_A*N*tau(k)^C0*log(x)^(-A),                        (2)
with fixed C0 and uniform family constants. Alpha is fixed while the
moduli are summed. Let |beta_m|<=tau(m)^B, m in [M,2M]. Let gamma_i be
three separate one-bounded sequences on qi~Qi with (qi,a)=1. Define
 Delta_l=sum_(n,m)alpha_n beta_m
       [1_(nm=a mod l)-1_((nm,l)=1)/phi(l)].
Then, relative to the finite list of established classical inputs in
section 10, for every A>0,
 |sum_(q1,q2,q3)gamma_1(q1)gamma_2(q2)gamma_3(q3)Delta_(q1q2q3)|
                          <<_A x/log(x)^A.                  (3)

The proof below derives (3); it does not assume preliminary moving-a
dispersion, a bare L51 bound, or an unrestricted large-residue theorem as
additional analytic inputs. It retains the exact coprime mean. Its
constant is uniform in a and all coefficient families with the stated
uniform bounds. The assertion is a signed factorized-modulus estimate;
it is not an absolute discrepancy sum over arbitrary moduli.

For the current application alpha is an individual chart-rough 1 or mu
slot with an individual interval and a fixed-polylogarithmic Mellin twist.
The explicit rough-SW derivation supplies (2) with C0=1 from classical
Mobius SW. Beta's order is fixed by the finite number of remaining HB
slots. The currently adopted elementary q=1 alternative also requires
kappa<=1/(2*K_mom), with the already fixed maximum moment order; this
only narrows the application of (3), whose more general proof may retain
classical Shiu at q=1. No new constraint on K,c,b follows here.

2. A small omitted specification and its constructive repair

The preliminary notes call the majorant merely a fixed smooth function,
then truncate Poisson at H=(modulus/M)*log(x)^5 with a power-small tail.
An arbitrary C-infinity majorant supplies Schwartz decay, which alone
does not prove that stated power-small tail at this logarithmic cutoff.
Maynard Lemma 13.4 expressly distinguishes its general-smooth cutoff
from its derivative-controlled cutoff. The free dispersion majorant
must be chosen in the latter class. This is a specification repair,
not a counterexample to the intended local theorem.

Here is a sufficient choice with no new external theorem. Choose positive
lengths a_j=c/j^2 with sum_j a_j=1/2. Let u_j be the uniform probability
```

```
density on [-a_j/2,a_j/2] and let phi be their infinite convolution.
This probability measure has support [-1/4,1/4]. Differentiating the
first j box factors as finite signed measures gives total variation
at most prod_(i<=j)(2/a_i)=C^j*(j!)^2. The remaining convolution is a
probability measure. Additional undifferentiated box factors give
continuous densities, so this also proves the required differentiability
of the final convolution to every fixed order. Set
                psi=1_[3/4,9/4]*phi.
Then 0<=psi<=1, psi=1 on [1,2], support(psi) subset [1/2,5/2], and
                ||psi^(j)||_infinity<=C^j*(j!)^2.         (4)
The construction and its constants are fixed, independently of x and a.

Integrating by parts j times and choosing j proportional to sqrt(|t|)
in (4), with j!<=j^j, gives
                |psihat(t)|<=C1*exp(-c1*sqrt(|t|)).        (5)
For any polynomial-sized modulus l and scale M, exact Poisson followed
by (5), with H>=C2*l*log(x)^5/M, gives
 (M/l)*sum_(|h|>H)|psihat(hM/l)|=O_D(x^(-D))               (6)
for every fixed D. A decreasing-integral comparison bounds the left
side by a polynomial prefactor times exp(-c2*log(x)^(5/2)). If H<1,
the first omitted integer already has |h|M/l>c*log(x)^5, so (6) still
holds. This argument also allows polynomial l>x; no truncated source
lemma with l<=x is being overextended.

Use this same psi for the positive q and m majorants in dispersion.
E1 needs only its fixed first few seminorms after completion; E2's five
coordinate derivatives also use only fixed seminorms. Thus (4)--(6)
validate their unchanged logarithmic H cutoffs and introduce no new
power loss. The smooth majorants used much later in the spectral return
may remain ordinary C-infinity: there the cutoff is x^epsilon and its
tail is paid by a fixed derivative order. These are distinct operations.

3. Preliminary lengths, smooth-factor deletion, and beta normalization

Write n=log_x N, q=log_x Q1, u=log_x Q2, v=log_x Q3. From (1),
 n>sigma, n<=1/2-4sigma,
 q+u+v<7/10-13sigma/2.                                    (7)
The last follows from q<n-sigma, 2n+u+2v<1-8sigma,
and u<(2-15sigma)/5. In particular the complete modulus is eventually
at most x^(4/5), and both N and M are fixed positive powers of x.

Put y=x^(1/loglog x) and split q2=u0*s, q3=v0*t into their z-rough
and z-smooth parts. Terms with s>y or t>y are negligible. Indeed, the
collapsed product coefficient is divisor bounded. Primitive Shiu for
integers of size x, with modulus l<=x^(4/5), bounds count plus coprime
mean by x*tau(l)^C*log(x)^C/l, uniformly in the residue represented by a.
The modulus multiplicity is included in this fixed C. Rankin with
xi=1/log z gives
 sum_(l<=x^(4/5),sm(l;z)>=y)tau(l)^C/l
   <<exp(-(loglog x)^2+O_C(loglog x)).                    (8)
This is stronger than every requested logarithmic saving. A pointwise
x^epsilon divisor bound would not suffice in (8), and is not used.

Dyadically fix s,t,u0,v0 and their products d=st~D, w=u0*v0~Rcal.
There are a fixed power of log x such blocks,
 D<=y^2=x^o(1), D*Rcal comparable to Q2*Q3.
Define literally
 Lambda_d(w)=sum_(st=d,u0*v0=w)gamma_2(u0*s)gamma_3(v0*t)
                  *1_(all the fixed intervals/supports).
It is independent of q1 and |Lambda_d(w)|<=tau(d*w).
All d in this dyadic block remain inside ONE preliminary dispersion
estimate. We never sum D separate arbitrary-logarithmic remainders.
With Q=Q1,Y=QD and delta=sigma/3, (7) gives eventually
 N,M>=x^delta, Y<=N*x^(-delta), Y*Rcal<=x^(1-delta).      (9)
```

```
It suffices initially to prove (3) for one-bounded beta. For general
beta, truncate at tau(m)<=log(x)^C and normalize by log(x)^(BC).
Alpha and (2) remain unchanged. The discarded coefficient at v=nm is
at most log(x)^(-C)*tau(v)^(B+2). For v!=a, summing the three-factor
modulus coefficient over l|v-a costs at most tau(|v-a|)^3. Fixed shifted
moments therefore make the discarded count O(x*log(x)^(E-C)), E fixed
independently of C. The coprime mean has the same form by ordinary
divisor means and sum_l |rho(l)|/phi(l)<<log(x)^O(1).
The exact v=a instead costs x^(4/5+o(1)); never write tau(0).
Choose C first, then the one-bounded theorem's log exponent to pay BC.
The current elementary mixed-moment lemma supplies this q=1 cleanup
under its additional fixed kappa cap, and is independent of log precision.

4. Shifted progressions and the exact-product order

The nonprimitive shifted progression bound used here is, for fixed
C,K,delta and I of length at most L in [-x^K,x^K],
 sum_(v in I,v=r mod f,v!=0)tau(|v|)^C
       <<(L/f)*tau(f)^C*log(x)^O(1), f<=L*x^(-delta).       (10)
To derive it put g=(r,f), v=g*w and f'=f/g. The w progression is
primitive and has L/g divided by f' equal to L/f>=x^delta.
Values |w|<=f'*x^(delta/2) cost x^(3delta/4) by a fixed divisor bound.
For the rest, split signs and dyadic sizes; cover by intervals of length
H comparable to the smaller of their scale and L/g. They satisfy
H>=c*f'*x^(delta/2), H>=c*x^(delta/2), and location X<=C*x^K.
Primitive Shiu applies with fixed alpha,beta<=delta/(8K), giving
(L/f)*tau(g)^C*log(x)^O(1). This proves (10), including a translated
interval at a polynomially large shift. Cauchy gives the mixed version
for tau(v)^B*tau(|v-a|)^C on a progression v=r mod f, v!=a.

Delete nm=a from every original congruence indicator BEFORE Cauchy;
leave its coprime mean unchanged. The first-moment correction is
O(Y*Rcal*x^o(1)), power-small by (9). The sequences are unchanged;
no m-dependent SW hypothesis is introduced. The square has unchanged
pure-mean S1, singly deleted S2', and doubly deleted S3'.

In S2' removal of (m,s)>1 uses a rough common factor f>z. The r-sum
contributes Rcal/f times a divisor power; collapsing v=mn with f|m
and applying (10) supplies x/f times a fixed logarithmic/divisor cost.
Thus the error is V*log(x)^C/z, V=MN^2/Y, by the divisor tail
sum_(f>z)tau(f)^C/f^2<<log(z)^O(1)/z. Only afterwards restore nm=a
for the one-congruence Poisson step. Its cost is Rcal*N*x^o(1).

In S3' first discard large coefficients or large tau(mn_i-a), using
(10) at f=1 and the n2 progression length O(N/Y+1). The error is
(x*N/Y+x)*log(x)^(C0-Ccut); C0 is fixed before Ccut. Under the remaining
logarithmic cutoffs, the exclusions (r,s)>1, (n1,n2)>1 and (s,ell)>1,
ell=qd, cost V*log(x)^C/z+x*log(x)^C. The counting reasons are concrete:
 * a common rough f in r,s gives n1=n2 mod ell*f, since m is a unit;
 * a common rough f in n1,n2 is coprime to ell; one n1 multiple count
   and one progression count give the two factors 1/f;
 * f|(s,ell) forces mn2=a mod ell*f, again one primitive class.
Their +1 terms contribute x*log(x)^C, absorbed by Y/N<=x^(-delta).
The close-pair restriction |n1-n2|>=N/log(x)^Csep costs
V*log(x)^(C-Csep)+x*log(x)^C. Choose Csep after Ccut. Restore large
coefficient cutoffs while the shifts are still nonzero.

Now n1!=n2. A single equality mn1=a leaves x^o(1) choices of m,n1,
O(N/Y+1) choices of n2 for each q,d, and only x^o(1) possibilities
for the other rough modulus dividing the NONZERO mn2-a. Its total
restoration cost is Rcal*(N+Y)*x^o(1)=Rcal*N*x^o(1). Both equalities
cannot hold. Restore these single zeros now, immediately before CRT
and full smooth Poisson. A premature double-zero restoration, of
```

```
size Y*Rcal^2*x^o(1), is neither made nor covered by (9).

5. The exact principal terms and their positive variance

Fix q,d; let ell=qd, eta_r=Lambda_d(r),
 B_r=sum_((n,ell*r)=1)alpha_n,
 A_r(b)=sum_(n=b mod ell,(n,r)=1)alpha_n, (b,ell)=1,
 d_r(b)=A_r(b)-B_r/phi(ell),
 c_ell(r,s)=phi(ell*r*s)/(ell*r*s*phi(ell*r)*phi(ell*s)).
With h0=integral psi and w_q=psi(q/Q), the pure mean is
 MT=M*h0*sum_(q,d)w_q*sum_(r,s)eta_r*conj(eta_s)
                                          *c_ell(r,s)*B_r*conj(B_s).
The S1 counting error is O(N^2*log(x)^C/Y), by Mobius inversion and
ordinary divisor means. It is V times a fixed inverse power by (9).

For any common multiple L of ell and its finitely many supported r,
 c_ell(r,s)=(1/L)*sum_(v mod L)
    [1_((v,ell*r)=1)/phi(ell*r)]*[1_((v,ell*s)=1)/phi(ell*s)].  (11)
This is an exact positive Gram kernel. L is never used as an error scale.
In the restored S2 zero mode the coefficient is
1/(ell*r*phi(ell*s)); its ratio to c_ell(r,s) is 1+O(log(x)/z).
Hence S2'=MT+M*E1+O_A(V/log(x)^A+Rcal*N*x^o(1)).

On the S3 clean CRT set (r,s)=(s,ell)=1,
                  phi(ell)*c_ell(r,s)=1/(ell*r*s)          (12)
exactly, even if r shares prime factors with ell. The h=0 CRT term
may have its rough common-factor and close-pair exclusions restored;
ordinary progression counts give the same V/log(x)^A errors, with
own conditions (n1,ell*r)=(n2,ell*s)=1 retained. It becomes MT+P, where
 P=M*h0*sum_(q,d)w_q*phi(ell)*sum_(r,s)eta_r*conj(eta_s)*c_ell(r,s)
                                      *sum_b d_r(b)*conj(d_s(b))>=0.   (13)
The equality follows from sum_b A_r(b)=B_r; positivity follows from
(11). Thus the three main terms leave P, not zero.

For any fixed w and A, auxiliary SW (2) and the primitive character
large sieve imply
 sum_(ell<=8Y)tau(ell)^w*sum_b |d_r(b)|^2
                 <<tau(r)^(2C0)*N^2/log(x)^A.              (14)
Here is the essential derivation, including the imprimitive characters.
Normalize alpha*1_((n,r)=1) by tau(r)^C0; auxiliary SW constants stay
uniform because tau(kr)<=tau(k)tau(r). Character orthogonality gives
variance phi(l)^(-1)*sum_(chi!=principal)|sum alpha_n chi(n)|^2.
Write l=f*k with primitive conductor f>1 and
chi(n)=chi_star(n)*1_((n,k)=1). Since phi(fk)>=phi(f)phi(k), the
large sieve on dyadic f~R0 for a FIXED k gives
 O(N/phi(k)*(N/R0+R0)). Summing f>log(x)^b and k gives
O(N^2*log(x)^2/log(x)^b+N*Y). Small conductors are bounded by (2),
with the explicit cost tau(k)^C0; their total is
O(N^2*log(x)^(3b+C1-2Bsw)), C1 fixed before Bsw. Choose b, Bsw last.
For the weight tau(l)^w use variance<=O(N^2/l), ordinary divisor
means, and Cauchy in l with the unweighted arbitrarily strong saving.
This proves (14) since Y<=N*x^(-delta), without an x^o(1) on a log error.

Use Cauchy on the O(Rcal) terms in (11), phi(ell*r)^(-1)<<log(x)/(Y*Rcal),
and the at most tau(ell) representations ell=qd. With a fixed common
divisor order B0 dominating C0 and the actual Lambda bound,
 P<<M*log(x)^2/(Y*Rcal)*sum_(r~Rcal)tau(r)^(2B0)
         *sum_(ell<=8Y)tau(ell)^(2B0+1)*sum_b|d_r(b)|^2
   <<MN^2/(Y*log(x)^A).                                   (15)
The last step keeps tau(r)^(2C0) from (14) and sums it by an ordinary
divisor MEAN. This is uniform in r, not an unproved filter-stability claim.

Outer Cauchy has multiplier M*Y*log(x)^C. Equations (13)--(15), the two
Poisson remainders, and section 4 prove
```

```
 |S_block|<<_A x/log(x)^A
    +M*sqrt(Y)*log(x)^C*(sqrt(|E1|)+sqrt(|E2|)).            (16)
The single-zero cost becomes sqrt(MY*Rcal*N)*x^o(1)
=sqrt(x*Y*Rcal)*x^o(1)<=x^(1-delta/2+o(1)). This verifies its uniform
absorption in the logarithmic term. Equation (6) pays all Fourier tails.

6. E1: complete inversion sums and explicit scale margins

For a fixed smooth w, complete inverse phases satisfy
 |sum_((n,l)=1)w(n/N)e(A*inverse(n)/l)|
   <<tau(l)^2*sqrt(l)+(N/l)*gcd(A,l).                       (17)
Poisson gives N/l times complete Kloosterman sums. The zero frequency
is the displayed Ramanujan term. For the other frequencies use Weil,
sqrt(gcd(k,A,l))<=sum_(d|gcd(A,l),d|k)sqrt(d), and
sum_(k!=0,d|k)(1+|k|/V)^(-2)<=2V/d. This proves (17) for every A.

Cauchy on the actual E1 sum, with its original unconstrained r,s support,
reduces the phase to A=a*(h*r'-h'*r), l=q*r*r'. On primitive moduli
gcd(A,l)=gcd(h*r'-h'*r,l). Write j=h*r'-h'*r. The j=0 tuples number
O(R*H*x^o(1)); no tau(0) is used. For j!=0,
sum_(q~Q)gcd(j,q)<=3Q*tau(|j|),
sum_(|h|<=H,h!=0)gcd(h,r)<=2H*tau(r),
sum_(r,r'~R)gcd(r,r')^2<<R^3.
Factoring gcd(r,r') in j shows that these bound the Ramanujan term.
The complete result, for H=O(QR/M*log(x)^5), is
 |Z_E1|^2<<x^epsilon*N^4/Q^2
                *[NQ/x+NQ^(5/2)R^3/x^2+N^2*Q*R/x^2].    (18)
Every epsilon here is freely chosen before x; no log remainder is involved.

For fixed d use Q'=d*Q1, R=Rcal and restrict q' to multiples of d in
the permitted bounded q'-dependent coefficient. This substitution is
allowed in E1 only. Absorb Lambda's fixed divisor bound at this power
stage. All three brackets have a strict power gap from (1). Explicitly,
 n+q<1-9sigma, 2n+q+u+v<17/10-29sigma/2,
 n+(5/2)q+3u+3v
  =(35/16)(q-n)+(23/16)(2n+u+2v)
     +(5/16)(n+q+5u+2v)-v/2
  <1021/512-(147/8)sigma+O(1/log x).                       (19)
The final line uses |a|>=c0*x and theta=7/32. All exponents of D cost
only genuine vanishing subpowers. Formula (18) therefore gives, with
spare power, |E1(d)|<<N^2/(d*Q1)*x^(-sigma+o(1)). Summing all d in
the dyadic block uses sum_d 1/d=O(1); then pay one D=x^o(1) to get
                  |E1|<<N^2/(D*Q1)*x^(-sigma/2).          (20)

7. E2: the exact phase, coefficient class and spectral return

For fixed d the actual normalization is E2=sum_(d~D)E3(d)/d, and
E3 has denominator q*w1*w2, with w_i~Rcal. Its two phases satisfy
 e(ah*inverse(n1*w2)/(q*d*w1)+ah*inverse(n2*q*d*w1)/w2)
  =e(ah*(n1-n2)*inverse(n2*q*d*w1)/(n1*w2))
                *e(ah/(q*d*w1*w2*n1)).                  (21)
All inverses are licensed by the clean CRT coprimalities. For
H=QD*Rcal^2*log(x)^5/M, the last phase has normalized amplitude
(a/x)*(Q/q0)*(D/d)*log(x)^5 on each q0~Q block. Five-coordinate
partial summation, in q,w1,w2,n1,h, consequently costs at most
                        x^(5kappa+o(1)).                (22)
This charge is paid once. The resulting endpoints are frozen scalars,
and the h weight is independent of all remaining summation variables.
The smooth q support is split into a bounded number of common intervals;
there is no arbitrary q weight in the resulting bare sum.

Partition the rough coprime pairs (n1*w2,n2) by the named coprime-coloring
lemma into exp(O((loglog x)^4)) cross-coprime classes. Expand both copies
of Lambda_d independently, including both st=d choices. Put S=rough Q2
```

length, R=rough Q3 length. Product windows on w1,w2 are differences of
two product upper cutoffs. Non-Hermitian polarization makes the two
R, the two S and the two n coefficients diagonal at cost at most 4^3.
No SW is reapplied after polarization. The fixed signed h interval
still supplies only one scalar one-bounded coefficient omega_h.

The bare sum is reduced by f=(n1-n2)/(dq), then positive Cauchy, to a
smooth reciprocal kernel with outside factor
P=N/(Q*R*S^(3/2)),
up to subpowers, and variables b=n2*r1, c=n1*r2*s2. Only after Cauchy
is f enlarged to O(N/Q); a favorable d^(-1/2) is dropped. Fixed smooth
majorants remove the b,c arithmetic coefficients at a divisor cost.
After squaring, the kernel coefficient is dominated by the positive
count of z=s1*s2, y=f*(h1*s1-h2*s2)!=0; (a,z)=1 is retained.
A unit phase for each y,z aligns the kernel, leaving no dependence on
the future completed m variable. The separate zero difference costs
x^epsilon*H*S*Bspec*F*Cspec.

For nu>=1 its actual counting norm obeys
B_nu^2<=x^epsilon*(1+H/S)*(HS)^2*F/nu. (23)
Indeed fix z,y and use Cauchy over the divisor-bounded decompositions
(s1,s2,f,k); then count colliding h pairs with nonzero k. Equal pairs
cost (HS)^2*F/nu. For unequal pairs put g=gcd(s1,s2), s1=g*u,s2=g*v;
the differences are m1=j*v,m2=j*u with nonzero |j|<=2Hg/S. Nonzero
multiples k/g of nu/(c*gcd(g,nu/c)), c|nu, number O(HSc*gcd(g,nu/c)/(g*nu)).
Factoring the remaining nonzero h2*v costs only a divisor factor.
The total unequal contribution is
x^epsilon*H^3*S*F/nu*sum_(g<=2S)gcd(g,nu)/g,
and that sum is <=tau(nu)*(1+log(2S)). This proves (23). Counting
nonzero multiples from zero has no +1. Beyond the nonzero y support
B_nu=0. The actual H/S<=x^o(1)*(Q/N)*D0<1, D0=N^2*R^2*S/x.

For completeness the exact spectral return used at this point is
written here, to avoid assuming the desired bare L51 estimate. At generic
scales C,D,U,Z with (a,z)=1, Poisson in b gives
(D/c)*sum_m psihat(mD/c)*S(m*inverse(z),a*y;c).
The m=0 term contributes D^2*U*Z*B_all^2 to the squared bound. Truncate
nonzero m at the GLOBAL cutoff (C/D)*x^epsilon, split common dyadic m
intervals, and separate the smooth amplitude on the fixed normalized
coordinates. The change c/C=sqrt((m/M)*(y/U)/(z/Z))/t cancels a exactly;
its Fourier norm costs a selectable small power only. It leaves a common
m interval and a common additive phase at every level z.

At level z and cusps infinity,1 the phase conversion is e(-a*y/z).
The coprime normalized ABL packet has transformed coefficient
beta_nu(n,z)=1_((n,a)=1)*beta(nu*n,z)*e(-n/z), cost (|a|nu)^(theta/2).
The transformation is common to every oldform dilation, since (a,z)=1.
The continuous part uses its formal Plancherel quadratic form, not an
L2 vector for one Eisenstein series. The regular cumulative large sieve,
fixed-level exceptional estimate F and common-interval exceptional J
listed in section 10 therefore apply with these precise coefficients.

Their algebra yields, with actual common m scale V,
|L_V|^2<<x^epsilon*P0*{
sum_nu (|a|nu)^theta*(U/nu+Z+T0)*(V+Z+T0)*B_nu^2
+(C^2*Z)^theta*(U+Z)^(1-theta)*(V+Z)^(1-2theta)*B_all^2},
P0=V*C^4*Z/(|a|*V*U+C^2*Z), T0=|a|*V*U/C^2. (24)
Here the regular transform is summed over heights max(1,2Y0), then
dyadic shells: its high-height bounds t^(-5/2)+Y0*t^(-3) multiplied
by the two cumulative norm factors grow as 2^(-j/2)+2^(-j), a convergent
sum. The level Cauchy costs sqrt(Z), not Z. For exceptional forms use
1+X^tau<=2*(1+X/sqrt(X0))^tau*(1+X0)^(tau/2), X0=1+Z/(U/nu).
The packet power cancels |a|nu only in the improved exceptional term.
These operations give exactly (24) from the stated classical norms.

```
Multiplying by (D/C)^2 and using
 Z*(L+Z+T0)*(V+Z+T0)/(Z+T0)<=(L+Z)*(V+Z)+Z*T0
for L=U/nu, V<=C/D, gives the returned squared polynomial
 sum_nu (|a|nu)^theta*[C*(U/nu+Z)*(C+D*Z)+|a|*U*Z]*B_nu^2
 +[C*(C*D*sqrt(Z))^(2theta)*(U+Z)^(1-theta)*(C+D*Z)^(1-2theta)
                                  +D^2*U*Z]*B_all^2.     (25)
The actual cutoff V<=x^epsilon*C/D costs at most x^(2epsilon) here.
It is valid also when C/D<1/2: that real value is only a comparison
point, and spectral estimates are invoked on nonempty actual intervals.

Now substitute Bspec=N*R, Cspec=N*R*S, Z=S^2,
U=N*R^2*S^3/M, F=N/Q, H=x^o(1)*Q*R^2*S^2/M, D=Bspec.
Apply (25) first on actual dyadic blocks and then enlarge positive
scales. Since U/Z=D0<1, Bspec*Z>=Cspec, Bspec^2<x, and
Cspec*Bspec*Z=x*U, equations (23)--(25) and the two square roots give
 |W_bare|<<N^2/Q*x^epsilon*
     {D0^(1/2)+[(1+|a|/x)*D1+D2]^(1/4)},                 (26)
 D1=|a|^theta*N*Q*R^2*S^5/x^2,
 D2=N^(1+2theta)*Q*R^(2+2theta)*S^(5-2theta)/x^2.
The weighted auxiliary sum is
sum_(nu|a^infinity)nu^(theta-1)=x^o(1), uniformly for polynomial a.
Its logarithm is O((log x)^theta) by splitting primes at log x.
Thus neither nu nor the zero Poisson mode was silently dropped.

8. Finish E2 and the local theorem

All powers of R,S in (26) are positive. Hence the original four
inequalities imply D0<<x^(-8sigma), D1,D2<<x^(-15sigma), uniformly
over every small-factor choice and fixed q0~Q block. The latter needs
only the relaxed q0<N*x^(-3sigma/4), eventually. Since |a|/x<=x^kappa,
(22) and (26), including every endpoint/coloring/normalization loss,
give a bound of the form
 |E3(d)|<<N^2/Q*
 [x^(-4sigma+5kappa+epsilon)
     +x^(-15sigma/4+21kappa/4+epsilon)].                 (27)
Choose all selectable classical epsilons after sigma and all finite
derivative orders. Their total, every genuine vanishing subpower, and
ONE extra factor D needed below fit, for example, x^(sigma/10).
For kappa<=sigma/100 both savings in (27) then exceed sigma/2.
Sum E2=sum_d E3(d)/d using sum_(d~D)1/d=O(1), and express it in the
normalization of (16). This proves
                     |E2|<<N^2/(D*Q)*x^(-sigma/2).       (28)
No second D factor and no second 5kappa phase charge occur.

Equations (20),(28) in (16) give an exponential-sum contribution
x^(1-sigma/4) times fixed logarithms. The preliminary exact-product
restoration is weaker, x^(1-sigma/6+o(1)), but still power-small.
Choose the preliminary SW/logarithmic precisions to pay the fixed
number of dyadic length choices, and the beta normalization of section 3.
The smooth-part error (8) has arbitrary logarithmic saving as well.
Together these prove (3). No overall x^(1-sigma/5) assertion is made.

9. What this completion discharges

This note supplies a single local derivation from the exact Delta to
the restricted medium estimate (3), after the explicit majorant choice
(4). Its intersections have been checked: the q-independent Lambda,
paid r-filtered BDH, primitive phase inverses, common spectral m interval,
nonzero shifted moments, and harmonic d sum all use compatible classes.
The extra majorant choice is implementable within the proof and changes
no coefficient class, support test, numerical margin or selected constant.
There is no remaining unspecified local-medium analytic proposition in
this derivation beyond the precise established inputs of section 10.
```

```
It does not establish that EVERY compiled atom satisfies (1), perform
HB/Mellin recombination, prove low-rank or long-variable alternatives,
or derive the pointwise/density/numerical Chen constructions. Those are
separate finite-family obligations. It also does not prove unrestricted
L23, arbitrary q-dependent Lambda, noncoprime (a,z) spectral reduction,
or arbitrary level-dependent m intervals. Formalization, external review
and a reproof of all classical spectral theory are not new hypotheses.

10. Finite established-input boundary and source map

The established analytic inputs are exactly:
 (i) classical Mobius Siegel--Walfisz only for the actual rough-slot
     derivation of (2); alternatively (2) itself is the theorem hypothesis;
 (ii) classical primitive multiplicative large sieve, ordinary divisor
      means, primitive Shiu in fixed polynomial interval/modulus ranges;
 (iii) Fourier/Poisson summation and complete Kloosterman Weil;
 (iv) the rough cross-coprime partition, Maynard I Lemma 13.2;
 (v) normalized Kuznetsov at level z, cusps infinity/1, trivial character,
     spectral arithmetic singleton s=1; Bessel transforms and the regular
     cumulative large sieve in the matching coefficient convention;
 (vi) ABL Lemmas 3.1/3.2 only when (a,z)=1, with oldform packet and
      continuous formal Plancherel normalization; Kim--Sarnak theta=7/32;
 (vii) exceptional norms as printed in Pascadi Theorems A/J, using
       fixed-level arbitrary n coefficients, and one common m interval
       and additive phase across levels. Exact finite Fourier completion
       extends J to that common partial interval at logarithmic norm cost.

In (v)--(vii) use R(n)=sqrt(|n|)*rho_ABL(n). The continuous measure is
dt/cosh(pi*t), up to Kuznetsov's fixed 1/(4pi). The required exceptional
input forms, with tau_j<=theta, are
 sum_exc X^tau_j*|sum_(n~L)b_n R_j,1(n)|^2
       <<(zL)^epsilon*(1+L/z)*||b||_2^2,
            X<<max(1,z/L),
 sum_(z~Z)sum_exc X^tau_j*|sum_(m in I)e(m*omega)R_j,infty(m)|^2
       <<(ZM)^epsilon*(Z+M)*M,
            X<<max(M,Z^2/M), I common, omega common.
The continuous n-removal has no extra height power: ABL's continuous
bound uses theta=0 and the same t-independent transformed coefficient.
No unrestricted moving-target distribution theorem is on this list.

Principal sources read in this audit:
 * Maynard I v2, actual saved lines 3940--4089 (Poisson distinction),
   4542--4903 (all third-variance cutoff/CRT operations), and the prior
   local equations bound below. Live primary page opened:
   https://arxiv.org/html/2006.06572v2 .
 * Pascadi, primary live page opened; the exact A/J, cusp and regular
   conventions were checked against the fresh source-bound spectral
   audit and regular-polynomial derivation:
   https://arxiv.org/html/2404.04239v3 .
 * The current local Delta statement, preliminary Gram and filtered BDH
   proofs, shifted-zero proof, full E1 completion, fixed-d E2 interface,
   multiplicity proof, spectral return, regular polynomial, and fresh
   normalized spectral boundary were read as equations, not PASS labels.
 * The current assembly identifies the actual scope and parameter
   overrides. It is not evidence for any analytical inequality above.

The companion JSON binds every principal local source and this report.
No numerical certificate was rerun or modified. No old source, state,
main TeX, formalization project, GitHub object or external message changed.
```

B The actual finite coefficient families

Source: distribution-coefficient-completion-20261010.txt.

```
Completion of the actual finite coefficient interface
10 October 2026. A coefficient theorem, with an explicit analytic premise.

1. The result being proved

Let T tend to infinity through even integers. Assume the restricted local
prime-product proposition of restricted-moving-target-distribution-20261010.txt:
it is uniform for its three separated one-bounded modulus sequences, exact
coprime mean, individual rough HB slot, four size inequalities and complete
moving-residue range. Its analytic proof is NOT a premise inferred from this
note. Classical below-half BV and its stated fixed-family convolution version
are also analytic premises here.

Then every actual signed coefficient required by the following CURRENT finite
construction belongs to that proposition's class, after an arbitrarily small
fixed finite-policy/compiler approximation and a fixed power-small analytic
collision removal:

 (U) the extended third upper tree, its adopted fourth/fifth/sixth changes,
     the above-mu second/third fallbacks, and the 87 eighth domains;
 (L) the ordinary lower starts at a=417/5000 and h=19/125, their causal
     third/fourth/fifth changes and the adopted fourth own-cube restorations;
 (H) the ordinary marked queries in the finite 248/430 H/h constructions,
     including all their admitted ordinary descendants;
 (P) the positive switched leaves in those constructions and the current
     paired-cropped U11 family, using (U)'s coefficient on each smooth family.

The resulting error is O_A(T/log(T)^A) for the SAME signed coefficient in
the count and exact mean. There is no additional factorization assumption on
an individual integer modulus, no changing of the density cap, and no claim
for arbitrary arithmetic filters. The conclusion is conditional only on the
stated analytic proposition and the explicit finite-policy constructions,
whose coefficient/gate closure is established below. It does not assert the
numeric integral inequalities, the pointwise Chen weight, or the global theorem.

The current 17772-cell deletion only restricts the fixed geometric L6/U11
queries. It creates no new arithmetic coefficient class: lower queries are
omitted on selected pair boxes and upper positive queries use their complement.
Nor do later Jensen/log/transverse integral refinements create new sieve
policies. Only actual cap changes enter (U).

2. Finite realization precedes the distribution argument

Use rho=19/96, mu=25/128, Lambda=153/256, Theta=299/500 and
v(t)=min((1+t)/2,57/64-3t/2). The complete carrier is Theta, not Lambda.
The latter bounds licensed high-rank caps. For any requested positive
approximation error, upper-finite-compact-margin-audit supplies a finite
causal upper realization with actual frozen caps lambda and a positive
external licence reserve. Its uniform third-envelope estimate is
  0 <= L_endpoint-lambda <= 4e+6chi.
The noncompact third tail is O(h0*(1+log^2(rho/h0))) as h0 tends to zero;
the higher adopted actions have positive last-prime cutoffs. Proper affine
walls can be trimmed, while identical inherit labels must be retained.
The 87 eighth domains have t8>53/13824 and bounded kernels. Ancestor caps
are frozen on their own prefixes before refining children.

For lower starts the corresponding third/fourth/fifth tails are
O(epsilon*(1+log^(r-1)(h/epsilon))), r=3,4,5. Lower own-cube restoration
has separate strict history and full/selected gates; it restores an old
fourth first exit with a legal residual lower support. It is not obtained
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```
by deleting a boundary node from an ordinary lower tree. These constructions
are set out in lower-joint-typed-density-bridge and the lower compiler.

This finite approximation produces actual coefficients; it is not a claim
that all ideal closed domains have a positive margin. Weak residual squares
are allowed and are handled by endpoint inflation below. The same frozen
coarse history is used in the full tree, rough tree and fine compiler.

3. The physical cap and the analytic covering cap are different data

The density, ancestor history and residual Q/B always use Q=T^lambda.
To prove a factorization one may use a larger auxiliary covering level
T^ell_aux. Only support and full/selected squares are enlarged to that
cover; the integer coefficient is unchanged. This is correction (C0) of
upper-finite-compact-margin-audit and is essential at a W gate.

Here is the concrete base W resolution. Write A(t)=(1+t)/2 and
W(z)=Lambda-3z/8. A coarse winning rescue entry has
inf_box max(A,W)=Gamma and lambda=Gamma-2e, with chi<e/10.
If inf A>=lambda+e/2, use the largest-singleton licence and
ell_aux=lambda+e/4. Otherwise sup A<lambda+11e/20, hence W>=Gamma
throughout the box. Set ell_aux=lambda+e. This has at least 9e/20
above the W gate and e below W. The other initialization square has
reserve 1-Lambda-2mu=3/256, and the other third-slot bound at W has
reserve 525/8704+49z/272. Both are positive. Thus a W gate above
lambda does not require changing lambda or assuming an illegal raw tree.

For a winning ordinary tent use ell_aux=lambda+e/4. A needed nonleading
pair has uniform remaining-square gate surplus 3/256; a dominated pair
can use the baseline label. The actual adopted higher actions are kept
on compact positive-gate pieces, so their own buffered covers are legal.
On every fine box permit O((J+2)tau) endpoint inflation within this
external reserve, where J is the later fixed rough-rank bound.

4. Full and selected squares close under every actual descendant

Use decreasing exponents t_i and S_j=sum_(i<=j)t_i. A full square is
S_(j-1)+2t_j<=ell_aux. An admitted cube at j gives its own square and
the next square, since S_j+2t_(j+1)<=S_(j-1)+3t_j. Upper trees have
odd first exits; ordinary lower trees have even first exits and a legal
initial square. These parities supply all ordinary tail squares.

For a fixed selected subset I among the marked slots, retain ALL tests
 b_I+sum_(i<j,i notin I)t_i+2t_j<=ell_aux, j notin I,
where b_I=sum_(i in I)t_i. Once j is later than every selected slot,
the left side is exactly S_(j-1)+2t_j. Hence every subsequent residual
full-square test is also the needed selected test. The selected set itself
does not change as smaller primes are appended. For a W licence the first
three slots and its ceiling remain unchanged. Merely checking b_I would
not prove this closure.

The initial current upper gates are explicit. Below mu the robust third
envelope gives the selected or W alternative from section 3. Above mu,
v(x)>3x for x<rho. The second fallback selecting y<x has 2x+y<3x;
the third fallback selects y+z<x and has 2x+y+z<3x. Their initial full
squares and preceding cube are likewise below 3x. At higher adopted nodes
all initial full/selected tests are retained. For example mask138 selects
{2,4,8} and has complement indices {1,3,5,6,7}; mask129 selects {1,8}
and has {2,3,4,5,6,7}. No complement test disappears at descendants.

Untouched lower starts have largest t1<=h<mu and endpoint licence
max((1+t1)/2,Lambda-3t1/8)>=249/448. A root buffer below 249/448 gives
a strict covering interval. Their restored fourth nodes check the new
own cube and full/selected gates before attaching any residual. Thus the
```

```
same closure works despite their different first-exit parity.

In the typed graph write S=log_T B, L=lambda-S, alpha_state=(1-S)/L>=2.
New primes are below the old ceiling. A singleton Lu has new residual
L(1-u); a pair Lu,Lv has L(1-u-v). In either case alpha_state increases.
The singleton square is 2u<1; the pair tests are 2v<1 and v+2u<1.
Those are exactly the new complete full squares after multiplying by B.
The actual Buchstab, W1 and W2 domains satisfy these inequalities, or
approach their allowed structural boundary by smaller-cutoff comparison.
Their selected tests then follow from the displayed identity, not from
the state ratio alone. The W ceiling is always retained.

There are at most 440 elementary transitions after unrolling the actual
finite seed and 430 rounds. If L_root>=ell>0, the action-bank inequalities
give L>=ell*101^(-440) and every appended prime exponent>=L_min/6.
Thus all marked cutoffs and switched-product cutoffs have a common fixed
positive lower bound before T grows. Finite compact refinement of each
query is still necessary; depth alone does not discretize its parameters.

5. Endpoint factorization is a uniform coefficient representation

Use local scale x, qscale=log_x T, theta=7/32 and source residue parameter
alpha_src>=qscale. This alpha_src is not alpha_state above. Put
vartheta=qscale*ell_aux+2zeta and D=x^(vartheta-2zeta).
For a medium length U let A=U*x^(-zeta). Set
 v0=2vartheta-1,
 u0=min((2-(2+2theta)vartheta)/(3-4theta),
        (2-2vartheta-alpha_src*theta)/3).
The local three-line tent licence says v0<=b_I<=u0; its middle line
must be retained after local rescaling even though it is dominated at
alpha_src=1. Full and selected tests are applied to ORDERED UPPER endpoints.

If C=D/(A*x^b_I)>=1, fix all selected slots in group 2. Greedily assign
each remaining slot to capacities A,C. Its selected square says its
square is at most the product of the remaining capacities; therefore
at least one accepts it. The resulting fixed group endpoints B1,B2,B3
satisfy B1<=A, B2=x^b_I, B3<=C and
 A^2 B2 B3^2 <= x^(1-4zeta),
 T^theta A B1 B2^5 B3^2 <= x^(2-4zeta),
 A^(1+2theta) B1 B2^(5-2theta) B3^(2+2theta)
                        <= x^(2-(4+4theta)zeta).
If C<1, reassign ALL slots into A,D/A using the full squares. Here
1<D/A<x^b_I. With B3=1 the last two inequalities follow by replacing
x^b_I by D/A; the first follows from A^2 B2<=AD<=x^(23/24-5zeta/2).
No putative integer factor of length below one is introduced.

For W initialize capacities A,x^v0,D/(A*x^v0) using only the first two
slots. One fits x^v0; the other fits one of the remaining capacities
because its square is at most their product x^(1-vartheta-2zeta).
All capacities are at least one. Continue inserting whole slots. At the
first slot p that fits none, append it to group 2. Its new product B is
in (x^v0,x^u0], since p<=p3<=x^(u0-v0). Failure of the first capacity
and the full square imply A*B*c<D. Thus the two residual capacities are
valid; the full-square greedy argument finishes. If no failure occurs,
the capacity product bounds directly give the same three monomials.

These algorithms run on endpoint data once per atom and HB length U,
not on individual primes or observed progression errors. Coordinatewise
reduction preserves the bounds. Their strict licences survive qscale,
alpha_src close to one and the fine inflation in section 3.

6. The exact tensor coefficient, including marked colours and beta

Refine every finite interval label to a common disjoint half-open alphabet.
```

For a slot list with m_I copies of label I let M=prod_I m_I!. Its raw
prime convolution T_J(n) is exactly M/prod_p nu_p(n)! on its support.
Hence the sorted rough coefficient is 1_squarefree*T_J/M.
For any fixed three-group partition let M_j=prod_I m_(I,j)! and
gamma_j=(convolution of group j)/M_j. Then 0<=gamma_j<=1 on ALL
integers, and the exact unrestricted identity is
 T_J/M = (prod_j M_j/M)*(gamma_1*gamma_2*gamma_3).
The restoring scalar is fixed and <=1. Empty groups are delta_1.

A raw MARKED convolution sigma is different: its multiplicity was already
present in the state. Only ordered RESIDUAL slots receive the denominator
M_res. The combined atom is epsilon*1_squarefree*T_all/M_res, with its
original sign epsilon. Regrouping gives restoring scalar
epsilon*prod_j M_j/M_res, bounded by |epsilon|*M_mark for disjoint
marked/residual supports. Normalizing marked sigma additionally is valid
only if both Phi(sigma) and Theta_Q(sigma) are scaled identically. No
one-sided extra factorial is admitted.

Choose w=T^omega below every marked exponent. For p<w, p divides BT iff
p divides T. The small-prime local factors and excluded primes are
therefore identical for every marked prefix B. Choose ONE beta_T+/- pair
with support below the smallest residual reserve. The same pair works
throughout the fixed finite query graph; it is not a coupled beta(B,d).
The sign/upper-lower rule determines which fixed beta is attached on an
atom. Since small and rough prime universes are disjoint, the factorization
n=n_small*n_rough is unique and |(gamma_3*beta)(n)|<=1.

The original global squarefree restriction remains in the actual count
and density. Remove it only in the analytic tensor expansion. If support
D_*<=T^theta_* with theta_*<1, the discrepancy difference is bounded by
 T^o(1) sum_(1<=D_*, exists p>=w:p^2|l)(T/l+1+T/phi(l))
                   << (T+D_*)T^(-omega+o(1)).
The fixed outer tuple multiplicity is absorbed in T^o(1). The +1 includes
the complete product T; tau(0) is never used. This does not replace the
original coefficient by an unrestricted pointwise sieve weight.

7. Low ranks, small losses and parameter order

Before beta/endpoints/local conversion, rough ranks at most two have
the following complete support exponents and positive half-level gaps:
 upper: 19/48,   gap 5/48;
 lower start: 38/125,   gap 49/250;
 marked: 1527/3584,   gap 265/3584.
For marked rank two the largest x<57/224 and full square x+2y<=Lambda
give x+y<=(Lambda+57/224)/2=1527/3584. Descendants append smaller
primes, so this remains valid. Classical states have their own buffered
complete level below one half. The coefficients are bounded by a FIXED
scalar mass, not necessarily one in marked states. The absolute BV
premise handles them directly. High rank is not licensed by BV.

Choose in this order:
 (i) finite geometric query graph, proper-wall/tail tolerances, coarse
     causal cap policy, analytic covering licences and positive reserves;
 (ii) fixed zeta below those reserves and a_star/100, and sigma=zeta/2000;
 (iii) common beta support ell_beta=nu0*eta, omega=nu0*eta^2, with eta
       small enough for support, beta-density and third-tail allowances;
 (iv) fixed J>theta_*/omega+2, fine tau with (J+2)tau below every reserve,
      and the actual finite divisor/moment orders;
 (v) kappa<=min(sigma/200,1/(2*K_mom)), where
     K_mom=max 2^(2*max(R,S,1)) over the actual shifted-moment orders,
     including the twice-squared tau^(4J) boundary requirement;
     then kappa0<kappa/[4(1+kappa)], also below the support reserves.
Selectable epsilons and logarithmic accuracies come afterwards, then T.
Making kappa smaller leaves the already fixed coefficient and mesh intact.

```
Take total beta/fine support inflation below 1/1000, so theta_*=599/1000
is a safe complete carrier ceiling. Choose kappa0 so
theta_*/(1-kappa0)<3/5 and all three low-rank supports remain below half.
At the medium slot, replacing A by U leaves savings zeta,2zeta,9zeta/16.
After two reserves zeta/16, the minimum is 7zeta/16>15sigma. Consequently
 Q1<U*x^(-sigma), U^2 Q2 Q3^2<x^(1-8sigma),
 T^theta U Q1 Q2^5 Q3^2<x^(2-15sigma),
 U^(1+2theta)Q1 Q2^(5-2theta)Q3^(2+2theta)<x^(2-15sigma).
Beta enlarges only group 3, with costs 2ell_beta,2ell_beta,
(2+2theta)ell_beta, already reserved. The complete support also satisfies
the stated one/two-long alternatives. At retained complete scales
c*T^(1-kappa0)<=x<=C*T, the required residue range is c0*x<=T<=x^(1+kappa).

There are finitely many atoms and their total absolute restoring scalar
is fixed before T. The partition may vary with atom, T and HB length,
because every such representation is the exact same coefficient and the
local proposition is uniform over all its one-bounded sequence class.
It may not vary with the arithmetic tuple or discrepancy sign. Fixed
polylog chart counts and Fourier norms are paid by a stronger requested
log exponent. The collision and complete-product trimming terms have
fixed power savings. This proves the claimed insertion with absolute
O_A(T/log(T)^A) error for each actual finite family.

8. Exact scope of this completion

The theorem removes the conditional phrase 'if the actual coefficients
have a fixed separated representation' for these finite policies: the
representation, coloured scalar, uniform beta, endpoint partition,
descendant closure and admissible hierarchy have been constructed above.
Its inputs still include the restricted analytic proposition itself and
the independent pointwise/density/integral results when one passes from
this distribution statement to K, c and b. A certificate hash is provenance,
not a proof of any of those inputs. No numeric K/c/b is changed here.

The three important exclusions are substantive: an arbitrary high-rank
weight below T^.598, an arbitrary prefix-dependent small-prime beta, and
a per-integer factorization do not meet this theorem. The actual common
label construction needs none of them. Historical assertions equating
ell_aux and lambda, dividing every marked factorial, or fixing kappa
before the actual moment orders must be read with the corrections above.
No new hole in the coefficient interface was identified. The main TeX,
earlier artifacts, research-state, formalization and GitHub are unchanged.
```

C The prime-product distribution theorem

Source: distribution-prime-product-completion-20261010.txt.

```
From the restricted medium estimate to the actual smooth prime-product estimate
10 October 2026. New proof; all earlier files and the manuscript remain unchanged.

Conclusion

This note proves the prime-product INSERTION, with the restricted medium
convolution estimate and the recorded finite whole-box licences as explicit
premises. It does not prove that medium estimate by naming it. The new
substantive step is a direct elementary Brun reduction for BOTH one-long
and two-long branches. It keeps the exact coprime mean, permits the actual
complex smooth profiles, and handles sharp individual intervals without a
moving-shift divisor moment, a product=T exception, or freezing a small
sieve divisor. Its smooth cores are uniform in every primitive residue.
The resulting long-branch argument needs no fundamental lemma of the sieve.

The physical upper/lower compiler still has its own fundamental-lemma and
density obligations. The elementary q=1 shifted moments are still used for
the medium beta truncation and the geometric Fourier tail. No global
K,c,b, density allowance, numerical table, or selected domain changes here.
The older common kappa cap, including the now unnecessary long-boundary
orders, may be retained unchanged; this note does not enlarge that range.

1. Exact assertion and premises

For even T, a fixed smooth geometric weight W_T on k<=k_star labelled
original primes, and (q,T)=1, define
 E_q(T;W)=sum_(p_1,...,p_k) W_T(p_1,...,p_k)
       [1_(p_1...p_k=T mod q)-1_((p_1...p_k,q)=1)/phi(q)].
All original primes are >=c*T^a_star, with fixed a_star>0; their product is
O(T). The fixed smooth family has polylogarithmic mixed Euler derivative
norms after the changes of variables below. Its actual geometric labels
are retained. Arbitrary additional arithmetic filters are not allowed.

The fixed finite compiler provides the SAME rho_T in this count and mean:
 (i) its low-rank/classical part is bounded by a fixed constant and has
     complete support <=x^(1/2-epsilon_L) on retained local scales;
 (ii) each high-rank part is a fixed finite scalar combination of labelled
     rough-prime convolutions with a one-bounded small-prime beta factor;
     the original global rough squarefreeness is retained until its
     separately estimated collision remainder is removed;
 (iii) for every medium length, a partition fixed on each WHOLE atom box
     gives three one-bounded factors at Q_1,Q_2,Q_3 satisfying (M) below;
 (iv) the complete local support is <=x^Theta_loc, Theta_loc<3/5, with
     20*zeta+Theta_loc<1 and 40*zeta+3*Theta_loc/2<1.
The number of atoms and their total restoring scalar norm are fixed before
T grows. All coarse/fine labels and finite ancestor choices are fixed.
The auxiliary support envelope ell_aux may exceed the physical density
cap lambda: this is correction (C0) in the bound source. It licenses the
same coefficient; it does not replace lambda in rho or its density.

Premise M is the signed local medium convolution estimate for one individual
rough HB slot alpha, a fixed-divisor-bounded complement beta, and three
separate one-bounded gamma_i, with exact coprime mean. Write x=NM, a=T,
theta=7/32 and require
 Q_1<N*x^(-sigma),                N^2*Q_2*Q_3^2<x^(1-8*sigma),
 |a|^theta*N*Q_1*Q_2^5*Q_3^2<x^(2-15*sigma),
 N^(1+2*theta)*Q_1*Q_2^(5-2*theta)*Q_3^(2+2*theta)<x^(2-15*sigma), (M)
 c0*x<=|a|<=x^(1+kappa),          0<kappa<=sigma/100.
Its conclusion is O_A(x/log^A x), uniform in this coefficient class and in
the permitted intervals/twists. The individual alpha must have auxiliary
```

SW uniformly in all auxiliary coprimality integers, with fixed tau(k)^1
loss. Section 5 below supplies that property, rather than assuming a
general arbitrary rough coefficient is SW. Root's separate local-medium
audit owns the proof of M itself.

Under these premises the conclusion proved here is
sum_q rho_T(q) E_q(T;W_T)=O_A(T/log^A T). (PP)
It holds for every sufficiently large even T, with an absolute error.
No almost-all-target conclusion, relative error in a cell's mass, sharp
signed coupled-domain theorem, or arbitrary high-level rho theorem is used.

2. Complete scales, an explicitly SMOOTH trim, and the exact HB identity

Choose eta in C-infinity([0,infinity)), 0<=eta<=1, equal to zero on [0,1]
and one on [2,infinity). Before charting multiply W_T by
eta(product p_i / T^(1-kappa0)).
This makes the complete-product trim smooth. All its Euler derivatives
are bounded uniformly, since they vanish off 1<=product/T^(1-kappa0)<=2.
The discarded complete products are <2*T^(1-kappa0), hence are not T for
large T. Their fixed divisor multiplicity and the modulus-divisor sum
give O(T^(1-kappa0+o(1))); their exact means have only fixed logarithmic
losses. This supplies the trim without inserting a new sharp coupled wall.

Insert an exact smooth nonnegative dyadic partition
sum_j psi(p_i/2^j)=1, psi=chi(t)-chi(2t),
where chi is one on t<=1 and zero on t>=2. Then psi is supported in
[1/2,2]. Set R_i=2^j, X=product R_i, Xplus=2^k X. On each original chart
fix Zchart=f(2^(6*k_star+2)*Xplus), f(v)=v^(1/(loglog v)^3), BEFORE HB.
Every original prime exceeds Zchart eventually. Replace primes by the
rough sequence Lambda(n)/log n and remove proper prime powers absolutely.
The sum of reciprocals of proper powers n>=c*T^a_star is
T^(-a_star/2+o(1)), so their total tuple mass is T^(1-a_star/2+o(1)).
For nonzero complete product minus T the modulus sum costs T^o(1);
the exact product T is bounded separately by T^(Theta+o(1)), Theta<1.
The mean has only a reciprocal-totient logarithmic loss. All are power-small.

For an original group at scale R put Z_HB=(2R)^(1/3), mu_Z=mu*1_(n<=Z_HB).
With Dirichlet convolution identity delta, E=delta-1*mu_Z is supported
above Z_HB. Hence Lambda*E^3 vanishes at n<=2R, and expansion gives
Lambda=sum_(j=1)^3 (-1)^(j-1) binom(3,j)
mu_Z^{*j}*1^{*(j-1)}*log. (HB)
Multiplication by psi(n/R) makes this identity valid everywhere needed.
The same chart-rough projector becomes the product of its projectors on
ALL HB variables. Units use Pminus(1)=infinity; a unit log slot vanishes.
There is no residual sharp indicator R<=product(HB group)<=2R.

Retain only individual HB rectangles meeting the smooth support. Each
group has at most six variables and product of lower endpoints >=R_i/128.
Thus 2^(-7k)X<=x_HB<=Xplus. After dyadically grouping a complementary
product the local complete scale satisfies
2^(-7k-1)X<=x<=2^(6k-1)Xplus,
f(x)<=Zchart<=x^(2/(loglog x)^3). (S)
In particular c*T^(1-kappa0)<=x<=C*T. Retain the current choices
kappa<=min(kappa_old,1/(2*K_mom)),
0<kappa0<min(kappa0_old,kappa/[4*(1+kappa)]).
Then c0*x<=T<=x^(1+kappa) eventually. All scale constants depend only on
the fixed maximum rank; strict power margins absorb them.

3. Same coefficient, finite tensor identity, and global collisions

Refine interval labels to a common disjoint half-open alphabet. For a
slot list J with m_I occurrences of label I, its unrestricted convolution
at n equals product_I m_I! / product_p nu_p(n)! on the correct support.
For a fixed partition into three groups set M_j=product_I m_(I,j)! and

gamma_j=(group convolution)/M_j. Each gamma_j is one-bounded on ALL integers. With M_all=product_I m_I!,

$$(\text{full convolution})/M_{\text{all}} = (\text{product_j } M_j/M_{\text{all}})*(\gamma_1*\gamma_2*\gamma_3). \qquad (T)$$

For a raw marked convolution divide only by the ordered residual-slot factorial. Its existing marked multiplicity stays in the external scalar. This changes neither side of the count/mean pair. The small/rough prime universes are disjoint, so attaching the bounded small-prime beta to one group still gives a one-bounded coefficient by UNIQUE factorization into its small and rough parts. A single beta pair works for the finite marked family because every marked prime is above its fixed small-prime cutoff.

Within-group squarefreeness does not imply global squarefreeness. Keep the latter in rho itself. For discrepancy estimation only, dropping it adds moduli divisible by p^2 with p>=T^omega. Summing complete integers first, including the +1 for any exact product T, bounds this difference by

$$(T+T^{\Theta})*T^{(-\omega+o(1))}. \qquad (C)$$

Indeed sum_(p>=T^omega)1/p^2<<T^(-omega), and each modulus contributes T/q+1+T/phi(q), up to fixed divisor powers. This is a fixed power error.

On each length block the whole-box partition in (T) is fixed before any integer or error sign is inspected. It may change with LENGTH or atom, since (T) remains the same coefficient identity. Dyadic restrictions of the gamma_j cost only a fixed logarithmic number of pieces. Smaller Q_i preserve every inequality (M), whose exponents on Q_i are positive.

4. Geometric separation and the exhaustive HB alternatives

For individual coordinates z_j=log(n_j/U_j), extend the smooth geometric factor with a compact cutoff equal to one on the individual rectangle. If D=sum_(S subset [m])||partial_(2S)w||_1, then Fourier integration gives

$$|\hat{w}(\xi)| \le D\prod_j (1+\xi_j^2)^{-1},\quad \int |\hat{w}| \le (2\pi)^m D/2^m,\quad \sup|w-w_H| \le mD/H, \qquad (F)$$

where w_H keeps |xi|_infinity<=H. Here D is a fixed logarithmic power. The new factors psi(c_i*exp(sum_group z_j)) have uniformly bounded mixed Euler derivatives, as does the smooth trim in section 2. No hard group cutoff is being differentiated. All finite arithmetic sums commute with the absolutely integrable Fourier representation.

The Fourier tail is bounded AFTER the complete modulus sum. For b!=T it is a pointwise tail error times sum_b tau(b)^R tau(|b-T|)^S. The elementary near-linear mixed moment in the bound source is O(x*log^E x), for fixed R,S and the retained kappa cap. At b=T use T^(Theta+o(1)), without tau(0). For the mean use sum_q |rho(q)|/phi(q)<<log^E T. Choose a fixed logarithmic H after these fixed orders. The Fourier norm, rather than H^m, is charged when applying uniform block estimates to the retained modes.

Select one medium slot, if present, in the ORIGINAL window

$$X^{(3\zeta)} \le U \le X^{(1/3+\zeta/4)}.$$

By (S) its local length lies in [x^(2*zeta),x^(1/3+zeta/2)]. If none is present, every group must have a long slot: six short slots have product <X^(18*zeta), contrary to R_i/128 and zeta<a_star/100. Three long slots would exceed the complete scale O(X). A truncated mu slot is <=O(X^(1/3)) and cannot be long. Thus the no-medium alternatives are exactly one or two unweighted integer long slots, apart from their logarithms, separated smooth twists, intervals and the common rough projector. The conservative short-product bounds are x^(20*zeta) and x^(40*zeta), respectively.

5. The actual medium coefficient has full auxiliary SW

The selected individual slot is

$$e(n)*1_I(n)*1_{(P^-(n)>Z)}*h(n)*n^{it},\ e=1 \text{ or } \mu,$$

with normalized |h|<=1 and fixed polylogarithmic variation/frequency.

It is one-bounded. For U>=x^xi, fixed xi>0, the rough-SW derivation gives
 |sum_(n=b mod q,(n,k)=1)alpha_n
       -phi(q)^(-1)sum_((n,kq)=1)alpha_n|
              <<_A U*tau(k)*log^(-A)x   (SW)
uniformly in EVERY k,q, primitive b, interval and admitted frequency.
Here are the essential actual identities and exponents, not a SW label.
Use even Brun order h=2 floor((loglog x)^2). Its next divisor support
Zchart^(h+1) is <U^(1/4) eventually, and its gap density is smaller than
every logarithmic power. For squarefree ell, mu(ell v)=mu(ell)mu(v)1_(v,ell)=1.
The exact identity
 mu(v)1_((v,E)=1)=sum_(w|v, p|w implies p|E) mu(v/w)
reduces to classical unrough mu SW at w<=V^(1/4). The smallest inner
scale is U^(9/16); the large-w remainder is <=2*V^(7/8)*tau(E)^2.
After summing ell<=U^(1/4) this is O(U^(29/32)*log^4 U).
Small q therefore gives arbitrary log saving; large q follows from the
one-bounded interval count O(U/sqrt(q)+1). Auxiliary k removes only primes
p|k above Zchart and costs <=2*(U/Zchart+1)*omega(k), bounded as in (SW).
Partial summation pays the fixed variation/frequency by a stronger initial
SW exponent. No modulus-dependent filter is introduced into alpha.

The complement of at most 6*k_star-1 slots is bounded by a FIXED divisor
power, independent of accuracy and frequency. If M is provided first for
one-bounded beta, the bound source's beta-only truncation proves the stated
extension: deleted nonzero products cost log^(-Ccut)x times the elementary
mixed moment; exact b=T costs Dtot*x^o(1); the mean costs fixed log powers.
Choose Ccut after the fixed divisor orders, then ask for the stronger local
log exponent needed to restore beta's normalization. Alpha is never
truncated. With (T), (M), and (F), all medium blocks give O_A(T/log^A T).

6. Explicit elementary Brun weights, uniform in deleted modulus primes

The next construction applies to either long alternative. Write L2=loglog x,
h=2 floor(L2^2), z=Zchart and P(z)=product_(p<=z)p, using the same endpoint
convention as the rough projector. Put
 lambda^+(d)=mu(d)*1_(d|P(z),omega(d)<=h),
 lambda^-(d)=mu(d)*1_(d|P(z),omega(d)<=h+1),
 y=z^(h+1)=x^o(1),
 U_z(n)=sum_(d|n)lambda^+(d), L_z(n)=sum_(d|n)lambda^-(d),
 G_z(n)=U_z(n)-L_z(n), r_z(n)=1_((n,P(z))=1).
The coefficients are one-bounded and d<=y. For s=omega(gcd(n,P(z))),
the finite alternating binomial identity gives
 L_z(n)<=r_z(n)<=U_z(n), U_z(n)>=0,
 G_z(n)=binom(s,h+1)>=0.   (B1)
This includes s=0, when both polynomials equal one. L_z may be negative;
its positivity is NEVER assumed.

For every q define
 A_q=sum_((d,q)=1)lambda^+(d)/d,
 g_q=sum_((d,q)=1)(lambda^+(d)-lambda^-(d))/d,
 V_q=product_(p<=z,p not|q)(1-1/p).
Finite inclusion-exclusion (equivalently Bonferroni for independent
Bernoulli variables with probabilities 1/p) gives
       0<=A_q-V_q<=g_q<=H(z)^(h+1)/(h+1)!=:epsilon_x,   (B2)
where H(z)=sum_(p<=z)1/p. This is UNIFORM in q, regardless of its factors.
One elementary bound is H(z)<=e*log(1+log z), by comparison with
log zeta(1+1/log z). Thus epsilon_x<=exp(-c*L2^2*log L2) eventually,
while log y/log x<=2*(h+1)/L2^3=O(1/L2). In particular A_q<=2 eventually.
No fundamental lemma, distribution theorem or growing divisor-moment
order is required for (B1)--(B2). The growing h is paid only through
y=x^o(1) in POWER-saving remainders; not through a log-saving term.

7. Two smooth cores with the EXACT coprime mean

Let alpha_m be a fixed complex short sequence with

```
 sum|alpha_m|<<M*log^B x; write the complete scale as x~M*U (one long)
or x~M*U*V (two long). Let f,g be compactly supported C^2 profiles on
[1/2,5/2] whose relevant norms are fixed logarithmic powers. For any
(a,q)=1 and fixed m with (m,q)>1 both discrepancy terms vanish.

For one long, ordinary progression counting and Mobius inversion give
 sum_(n=b mod q) f(n/U)=(U/q)*integral f+O(N1(f)),
 sum_((n,q)=1) f(n/U)=U*phi(q)/q*integral f+O(tau(q)*N1(f)),
where N1(f)=||f||_1+||f'||_1. The exact discrepancy for each fixed m is
therefore O(N1(f)*(1+tau(q)/phi(q))), uniformly in U and primitive b.
This applies at U/d, with m replaced by m*d; nonunits again vanish.

For two long, write fhat(t)=integral f(s)e(-ts)ds. Double periodization
gives the EXACT fixed-m discrepancy
 U*V/q^2 sum_(j,k in Z) fhat(jU/q)ghat(kV/q)
 [S(a*m^(-1)*j,k;q)-c_q(j)c_q(k)/phi(q)].                    (P1)
Both coordinate axes vanish identically, including their nonzero Fourier
masses: S(0,k)=c_q(k), S(bj,0)=c_q(j) for primitive b, c_q(0)=phi(q).
For nonzero frequencies use the composite Weil bound
 |S(bj,k;q)|<=tau(q)*sqrt(q)*gcd(j,k,q)^(1/2)
and |fhat(t)|<<N2(f)*(1+|t|)^(-2), N2(f)=||f||_1+||f''||_1.
The elementary sum
 sum_(j!=0,d|j)(1+|j|/H)^(-2)<=2H/d
holds for all H>0. Expanding the gcd through divisors gives the uniform
fixed-m bound
 N2(f)N2(g)*[sqrt(q)*tau(q)+tau(q)^2/phi(q)].               (P2)
The first term uses sum_(d|q)d^(-3/2)<=zeta(3/2); the second uses
|c_q(j)|<=gcd(j,q) and sum_(d|q)phi(d)/d<=tau(q).
For q=1 the discrepancy is exactly zero. There is NO size bound on a.

If |rho(q)|<=tau(q)^J*log^B1 x and q<=D, summing fixed divisor means
turns these smooth core bounds into
 one long: O(M*D*log^C x),
 two long: O(M*D^(3/2)*log^C x).                            (P3)
For fixed d_1,d_2<=y the profiles are exactly f(n_1/(U/d_1)),
g(n_2/(V/d_2)); their norms in (P2) are unchanged. The short product
becomes m*d_1*d_2 only inside the primitive residue. The core was proved
pointwise in that product, so its L1 mass is still sum|alpha_m|. Sum the
d pairs by their number, at most y^2. No dyadic freezing, new common
interval, or regrouped high-order divisor sequence is required. Thus
the expanded two-variable Brun errors cost M*y^2*D^(3/2)*log^C x.
The product-divisor support is d_1*d_2<=y^2=x^o(1), explicitly retained.

8. Passing from these smooth cores to rough smooth variables

First, for any nonnegative smooth F and any q, exact Mobius inversion gives
 sum_((n,q)=1) F(n/U)*U_z(n)
 =U*phi(q)/q*(integral F)*A_q+O(y*tau(q)*N1(F)),
 sum_((n,q)=1) F(n/U)*G_z(n)
 =U*phi(q)/q*(integral F)*g_q+O(y*tau(q)*N1(F)).             (E1)
The left sides are nonnegative by (B1); the second upper bound uses
g_q<=epsilon_x. This is a comparison of EXACT coprime sums, not a
replacement by an unrestricted Euler mean. The factor phi(q)/q cancels
against the exact mean's 1/phi(q) where appropriate.

For two variables abbreviate U_i=U_z(n_i), L_i=L_z(n_i), r_i=r_z(n_i),
G_i=U_i-L_i. The valid product sandwich is
 L_1*U_2+U_1*L_2-U_1*U_2 <= r_1*r_2 <= U_1*U_2,
 0<=U_1*U_2-r_1*r_2<=B:=G_1*U_2+U_1*G_2.                 (E2)
Indeed U_i>=r_i>=0 and U_i-r_i<=G_i. This proves (E2) even if either
L_i is negative; using L_1*L_2 as a lower bound would be invalid.

For complex f,g choose fixed nonnegative smooth majorants F>=|f|,
G>=|g| with fixed polylog norms, and replace alpha by |alpha| only in
```

```
the following absolute error bound. Write C_q and M_q for the counting
and exact coprime-mean functionals, respectively, and Delta_q=C_q-M_q.
Then
 |Delta_q(alpha*f*g*r_1*r_2)-Delta_q(alpha*f*g*U_1*U_2)|
 <= C_q(|alpha|*F*G*B)+M_q(|alpha|*F*G*B)
 <= |Delta_q(|alpha|*F*G*B)|+2*M_q(|alpha|*F*G*B).         (E3)
Every count on the right is nonnegative; no order on a complex sequence
has been asserted. Expanding U_1*U_2 and B into their d pairs uses (P3)
with at most y^2 terms and coefficients of bounded absolute value.

Use (E1) for each factor of the mean in (E3) and then
sum_(q<=D)|rho(q)|*tau(q)^j/phi(q)<<log^C x for each FIXED j. The complete
rough smooth two-variable discrepancy is bounded by
 O(log^C x)*[M*U*V*epsilon_x
             +M*(U+V)*y+M*y^2+M*y^2*D^(3/2)].             (E4)
For one variable the same argument with 0<=U_z-r_z<=G_z gives
 O(log^C x)*[M*U*epsilon_x+M*y+M*y*D].                     (E5)
Here C depends on fixed profile/divisor norms, not on h. The epsilon_x
term is multiplied only by fixed logarithmic factors. All y factors
occur in errors with a fixed power reserve. In the actual alternatives
M<=x^(20*zeta) or x^(40*zeta), min(U,V)>=x^nu for fixed nu>0, and (iv)
therefore makes (E4)--(E5) O_A(x/log^A x) for every fixed A.

9. Sharp INDIVIDUAL intervals, with no shifted-moment step

Fix delta=log^(-B)x. A sharp interval I subset [U,2U] has a smooth
approximation h_I(n/U), 0<=h_I<=1, whose pointwise error is bounded by
a nonnegative smooth k_I(n/U), supported in O(delta)-width strips at
the two endpoints. This also works when |I|<delta*U: take h_I=0 and a
single strip covering I. The profiles satisfy
 integral k_I<<delta, N1(k_I)<<1, N2(k_I)<<delta^(-2).
Constants are uniform in interval endpoints. Multiplying by the smooth
logarithm/twist costs only its fixed polylog norms. The product interval
error is bounded by a sum of two such strips times fixed positive smooth
envelopes in the other variable.

Apply the ALREADY PROVED rough smooth bound (E4), or (E5), to each
nonnegative strip weight with |alpha| and |rho|. Its exact mean is bounded
using r_z<=U_z and (E1): in the two-variable case the total is
 O(log^C x)*[x*delta+M*(U+V)*y+M*y^2],                     (I)
and for one variable O(log^C x)*[x*delta+M*y]. Its counting sum equals
this mean plus the rough smooth discrepancy. The latter may have a
larger fixed log power from N2(k_I), but all non-epsilon remainders still
have a fixed power reserve. Choose B after the fixed coefficient/profile
orders and the desired accuracy. Since epsilon_x beats every log power,
the interval error is O_A(x/log^A x).

This controls both count and exact mean and is uniform in EVERY primitive
a. No tau(|product-a|) estimate occurs, including at product=a. In this
new long-branch proof, the old boundary Cauchy, its doubled tau^(4J)
requirement, intermediate product=a isolation, and small-divisor freezing
are ALL unnecessary. The global Fourier tail and medium beta truncation
still need their separate elementary shifted moments and zero handling.

10. Low rank and final assembly

Before HB, low-rank moduli have a complete fixed gap below x^(1/2).
For at least two original primes choose one prime interval as alpha,
the others as beta; both lengths are positive powers of x. Prime SW
with auxiliary coprimality has fixed tau(k)^1 loss, because inserting
(p,k)=1 deletes only omega(k) primes. The bound source's explicit
primitive-character proof supplies
 sum_(q<=x^(1/2-epsilon)) max_((a,q)=1)|Delta_q(alpha,beta)|
                          <<_A x/log^A x.
```

In that proof induced characters retain coprimality on BOTH factors;
the large-conductor k-sum uses sum 1/(k*phi(k))<infinity, and the small
conductors pay their number and sum tau(k)^1/phi(k) by fixed log powers.
Its fixed coefficient norm (not x^o(1)) permits the actual bounded rho.
For one original prime use ordinary maximal prime BV instead. A singleton
unit product, if present, has elementary T^o(1) discrepancy, and is not
confused with a unit quotient in a nonempty prime-product family.

The order is now: same-coefficient tensor/collision reduction and smooth
trim; exact original charts and HB; Fourier separation; medium M with
the actual rough-SW slot OR direct Brun one/two-long estimates; low-rank
BV separately; restore fixed scalar factors, prime powers and all tails.
There are only fixed/logarithmic numbers of atoms/charts/modes and their
integrated coefficient norm is polylogarithmic. Request sufficiently
strong local log exponents before summing. The finite common parameter
choice fixes a positive reserve in every POWER error, and kappa/kappa0
are reduced only after their actual finite divisor orders are known.
One eventual T threshold therefore works for all members of the family.
This proves (PP) under the explicit premises of section 1.

New closure supplied by this note is the direct Brun long-branch theorem,
with exact mean and sharp individual endpoints, and its complete insertion.
The local medium theorem, actual finite licence construction, positive
same-rho majorant, raw Rosser density, Chen weight/table inequalities and
numerical certificates remain separate proof components. No conclusion
about their truth is inferred from this insertion alone. No old files,
selected constants, paused formalization, GitHub objects or external
communications have been changed.

D The direct Brun treatment of long variables

Source: direct-brun-long-branches-root-20261010.txt.

```
Direct Brun treatment of one/two long variables: independent root derivation
10 October 2026. New local proof; numerical and physical sieve policies unchanged.

Purpose and scope

The prime-product insertion agent proposed replacing the two-long roughness
removal by explicit Brun polynomials. I independently derive the replacement
below, including negative lower weights, exact coprime means, interval edges,
and the one-long extension. The Poisson identity is also given, so the remaining
non-elementary input at this step is the classical complete Kloosterman Weil
bound (plus the elementary prime reciprocal bound). No shifted divisor moment,
fundamental-lemma error estimate, or small-divisor freezing is used here.
This is only the no-medium branch of PP, not its medium spectral theorem or
the separate physical small-prime compiler.

1. Explicit polynomials and uniform density gaps

Put L=log log x and fix the common chart threshold z with
 log z <= 2 log x/L^3, z->infinity.
Let h=2 floor(L^2), r=h+1, y=z^r=x^o(1), and P=prod_(p<=z)p.
For squarefree divisors d of P define
 beta+(d)=mu(d)1_(omega(d)<=h), beta-(d)=mu(d)1_(omega(d)<=r),
and zero otherwise. Both supports lie in [1,y], with |beta|<=1.
For a positive integer n define U(n)=sum_(d|n)beta+(d),
Lw(n)=sum_(d|n)beta-(d), I(n)=1_((n,P)=1), G(n)=U(n)-Lw(n).

If k=omega(gcd(n,P)), the alternating binomial partial sums give
 Lw(n)<=I(n)<=U(n), U(n)>=0, G(n)=binom(k,r)>=0.            (1)
At k=0 both polynomials equal1; for k>0 the partial sum through j is
(-1)^j binom(k-1,j), interpreted as zero when j>=k. This proves (1),
including the fact that Lw is allowed to be negative.

For any q, put V_q=prod_(p<=z,p does not divide q)(1-1/p) and
B_q^+=sum_((d,q)=1)beta+(d)/d, B_q^-=sum_((d,q)=1)beta-(d)/d.
Apply (1) to independent Bernoulli variables of means 1/p. It yields
 0<=B_q^+-V_q<=B_q^+-B_q^- = e_r({1/p:p<=z,p not dividing q})
                    <=H(z)^r/r! =:epsilon_x,              (2)
where H(z)=sum_(p<=z)1/p. In particular B_q^+<=1+epsilon_x and the gap
is nonnegative, uniformly in every deletion set of q-primes. No theorem
with constants depending on the number or sizes of those primes is used.
The prime harmonic estimate gives H(z)<=2L eventually. Since r~2L^2,
r!>=(r/e)^r shows epsilon_x <=exp(-c L^2 log L) for a fixed c>0,
eventually. It is smaller than log(x)^(-A) for every fixed A.

2. Smooth progression and exact coprime means

Let f be a compactly supported smooth profile on a fixed positive interval,
with N1(f)=||f||_1+||f'||_1 and N2(f)=||f||_1+||f''||_1.
One-dimensional summation in a residue class and Mobius inversion give
 sum_(n=b mod q)f(n/U)=(U/q) int f+O(N1(f)),
 sum_((n,q)=1)f(n/U)=U*(phi(q)/q) int f+O(tau(q)N1(f)).      (3)
The constants are independent of the primitive class b and all positive U.
The first estimate needs no coprimality; that condition is used in solving
the actual congruence after a sieve divisor is inserted.

For B=U or G, expand its at most y divisor coefficients. Terms with
(d,q)>1 vanish from the coprime sum. Equation (3) then gives
 S_B(f):=sum_((n,q)=1)f(n/U)B(n)
 =U*(phi(q)/q)*b_q int f+O(y*tau(q)*N1(f)),                (4)
where b_q=B_q^+ for U, and b_q=B_q^+-B_q^- for G. For nonnegative f,
```

these sums are nonnegative. Thus (2) bounds their main terms by respectively
(1+epsilon_x)U phi(q)/q int f and epsilon_x U phi(q)/q int f.
Equation (4) is about the exact coprime sum, not an assumed density for a
different coefficient. Every finite expansion is legal although h grows.

3. One long variable

Let a be any integer, (q,a)=1. A short coefficient alpha_m has
sum_m |alpha_m|<=Z log^C x, on a fixed-multiple interval at scale Z;
put x=ZU up to fixed constants. For smooth f consider the discrepancy
 E_q(I)=sum_(m,n)alpha_m f(n/U)I(n)
             [1_(mn=a mod q)-1_((mn,q)=1)/phi(q)].
If (m,q)>1 both indicators vanish. Otherwise every inserted (d,q)=1
leaves one primitive class for n/d. Equation (3) cancels the two main
terms exactly, with error O((1+tau(q)/phi(q))*N1(f)) per d and m.
Consequently expansion of U or G costs
 O(Z*y*log^C x*(1+tau(q)/phi(q))*N1(f)).   (5)

For a complex f choose a nonnegative smooth majorant F>=|f| of fixed support
and polylogarithmic N1 norm. The difference between I and U has magnitude
at most G. Its congruence sum is at most its exact coprime mean plus the
absolute discrepancy of G with |alpha|,F. Formula (4) bounds that mean;
use (5) for the discrepancy. Thus, for |rho(q)|<=tau(q)^J log^C x,
 sum_(q<=D,(q,a)=1)|rho(q)E_q(I)|
 << epsilon_x*x*log^C1 x + Z*y*D*log^C1 x.   (6)
Ordinary divisor/totient MEANS absorb all fixed powers; epsilon_x is not
multiplied by a pointwise x^o(1). Here and below C1 can increase by a fixed
amount for derivative norms or short coefficient bounds.
For Z<=x^(20*zeta), D<=x^Theta, Theta<3/5, (6) has arbitrary logarithmic
saving, since 20*zeta+Theta<1 with the actual zeta<=10^-6.

4. Two long variables: the smooth core

Put x=ZUV up to fixed constants. For any fixed m coprime to q let b=a/m mod q.
Double periodization gives the exact smooth discrepancy
 UV/q^2 sum_(j,k in Z) fhat(jU/q)ghat(kV/q)
       [S(bj,k;q)-c_q(j)c_q(k)/phi(q)].   (7)
This includes the Fourier expansion of the exact coprime mean. All terms
with j=0 OR k=0 vanish because b is a unit. For q=1 the discrepancy is zero.
For q>=2 use complete Weil,
 |S(bj,k;q)|<=tau(q)*sqrt(q)*gcd(j,k,q)^(1/2),
and |c_q(j)|<=gcd(j,q). Integration by parts twice gives
 |fhat(t)|<<N2(f)(1+|t|)^(-2).
For every H>0 and d>=1,
 sum_(j!=0,d|j)(1+|j|/H)^(-2)<=2H/d.
The common-divisor sums are bounded by sum_(d|q)d^(-3/2)<=zeta(3/2)
and sum_(d|q)phi(d)/d<=tau(q), respectively. Therefore (7) is
 <<N2(f)N2(g)[sqrt(q)tau(q)+tau(q)^2/phi(q)].   (8)
This is uniform in ALL positive scales U,V and primitive residues a.
There is no frequency cutoff, discarded real phase or exact-product exception.

On inserting d1|u,d2|v, use scales U/d1,V/d2 in (8) and replace m by md1d2.
If that replacement is not coprime to q both indicators vanish; otherwise
(8) applies unchanged. No freezing of d1,d2 is required. Sum the at most y^2
pairs absolutely and then use divisor means in q. Each expanded pair of
Brun polynomials has total discrepancy
 << Z*y^2*D^(3/2)*log^C2 x.   (9)
The x^o(1) factor y^2 multiplies only this fixed-power-saving term.

5. Two rough indicators, including a possibly negative lower polynomial

By (1), for I_i=I(n_i), U_i=U(n_i), G_i=G(n_i),
 P^+=U_1 U_2 >= I_1 I_2 >= P^-=U_1 U_2-G_1 U_2-U_1 G_2. (10)
For example U_1U_2-I_1I_2=(U_1-I_1)U_2+I_1(U_2-I_2), which lies between

zero and G_1U_2+U_1G_2. This proves (10) without assuming Lw_i>=0.
For complex f,g and alpha, use nonnegative smooth majorants F>=|f|,Gf>=|g|
and |alpha| to control the gap. Its exact coprime mean is bounded by
 [S_G(F)S_U(Gf)+S_U(F)S_G(Gf)]/phi(q),
times the short absolute mass. From (2),(4), summing |rho(q)| gives
 O(epsilon_x*x*log^C3 x
   +Z*(U+V)*y*log^C3 x+Z*y^2*log^C3 x). (11)
In deriving this, the main product has phi(q)/q^2<=1/q; cross terms use
tau(q)/q and the final error product uses tau(q)^2/phi(q). Their sums cost
only fixed logarithms. The congruence gap is its exact mean plus the
expanded discrepancy, bounded by (9). Triangle inequalities now yield
 sum_(q<=D,(q,a)=1)|rho(q)E_q(I_1 I_2)|
 << epsilon_x*x*log^C4 x
    +[Z*y^2*D^(3/2)+Z*(U+V)*y+Z*y^2]*log^C4 x. (12)
There is no unlicensed estimate for a signed rough function. Positivity is
used only for the explicit gap majorants, before their finite expansion.

For actual no-medium HB blocks, Z<=x^(40*zeta), min(U,V)>=x^nu for a fixed
nu>0, and D<=x^Theta, Theta<3/5. The leading power exponent is at most
40*zeta+3Theta/2+o(1)<0.90004+o(1); the two cross terms are
x/min(U,V)*x^o(1) and Z*x^o(1). All are fixed power savings. Thus (12)
has arbitrary logarithmic saving. This holds even if a equals an intermediate
product: no divisor of zero was introduced. No assumption on the size of a
is needed in this branch beyond primitive coprimality for the summed modulus.

6. Sharp individual intervals and what is actually replaced

For an interval within a fixed positive scale, replace its indicator by a
smooth cutoff differing only on two normalized strips of width
delta=log(x)^(-B). Choose positive smooth outer strip envelopes with
integral O(delta), N1=O(1), N2=O(delta^(-1)), and fixed higher derivatives
bounded by powers of delta^(-1). Twists/log factors multiply these by fixed
logarithmic norms. Positive U or U_1U_2 bounds the rough strip count.
Equations (3)--(4) and (8)--(9) give its exact mean/main mass
O(delta*x*log^C x), with the same fixed-power errors times logarithms.
The exponent C in this boundary MAIN term is independent of B: it uses
envelope amplitudes/integrals, short mass and divisor means. B-dependent
derivative norms occur only in the fixed-power errors. Choose B after the
requested error exponent and those fixed main-term orders.
This proves the interval extension by positive boundary estimates. It does
not approximate an arbitrary sharp COUPLED domain by separated smooth weights.

Consequently the actual one/two-long branch needs neither the old fundamental-
lemma roughness removal nor small-divisor freezing, shifted-moment boundary
Cauchy, or separate removal of intermediate product=a. Its internal rank h
grows slowly only in auxiliary Brun weights; this is not a growing physical
modulus tree or a new numerical density policy. All its coefficients are
handled explicitly by support size y=x^o(1) and uniform gap (2).
The q=1 elementary moments remain needed for other steps, notably medium
beta truncation and global Mellin tails. Keep the previously certified common
kappa/kappa0 cap, including its former boundary moment order; a redundant
stricter bound is harmless. No relaxation of K, c, b or a physical cap follows.

The full HB classification, chart projector window, local medium theorem,
actual coefficient licensing, physical density, lower induction and numerical
inequalities are separate. Sources consulted: lichtman-two-long-exact-axes-
20261010.txt; lichtman-hb-long-branches-20261009.txt; restricted-moving-target-
distribution-20261010.txt; q1-elementary-shifted-moments-20261010.txt.
No old artifact, main document, state, formalization or external service was
changed by this proof. The accompanying assembly records adoption and hashes.

E Correspondence with the classical spectral inputs

Source: final-classical-input-correspondence-20261010.txt.

```
Final classical-input correspondence: the coprime medium spectral route
10 October 2026. Independent narrow source audit; new report only.

1. Finding and scope

For the actual coefficient class in distribution-medium-completion-20261010.txt
lines 349--380 and 463--483, the named classical spectral statements exist
with the required quantifiers. I found no missing spectral inequality in
that interface. This finding is more precise than a blanket completion:
it applies to trivial character, level z, cusps infinity and 1, (a,z)=1,
arbitrary complex n coefficients independent of spectral height, and ONE
common m interval and additive phase across levels. It does not verify
every preliminary dispersion estimate, every compiled atom, or the final
prime-count conclusion. It does not restore an unrestricted L23 theorem.

The Lemma 4.6 replacement in the current medium note is an actual retained-
phase derivation for |a|<=x^(1+kappa), kappa<=sigma/100. It does not simply
insert the restricted moving-residue theorem as a new black-box assumption.
Its phase and final saving calculation are checked explicitly in section 5.

2. Exact normalization and the moving integer a

The relevant ABL source is Assing--Blomer--Li, Uniform Titchmarsh divisor
problems, arXiv:2005.13915v1, (3.1)--(3.8), Lemmas 3.1 and 3.2, PDF pp.7--11.
https://arxiv.org/pdf/2005.13915v1
Saved primary extraction:
external-source-audit-20261009/assing-blomer-li-2005.13915v1-large-a-20261009.txt
especially lines 575--1010. The paper's variables correspond to
 q_ABL=1, r_ABL=z, s_ABL=1, Q_ABL=z, Q_bad=z, Q_good=1, conductor=1.
The dispersion factor d in E2=sum_d E3(d)/d is NOT q_ABL or a level.

Use R(n)=sqrt(|n|)*rho_ABL(n), the DI normalization. At level z the scaling
matrices are sigma_infinity=Id and sigma_1=[[sqrt(z),0],[sqrt(z),1/sqrt(z)]].
Drappeau Lemma 4.3 (PDF p.11) gives the Kloosterman cusp conversion; in this
case its phase is e(ay/z). Reversing that conversion gives e(-ay/z) on the
n coefficient of the ordinary Kloosterman sum in the medium argument.
https://drappeau.perso.math.cnrs.fr/documents/divtm-siegel.pdf

Write y=nu*n, nu|a^infinity, (n,a)=1. In ABL Lemma 3.1 the exceptional
integer K is 1 for EVERY oldform dilation because (a,z)=1. Thus
 R_v(a*nu*n)=lambda_v(a*nu)*e(n*(a*nu-1)/z)*R_v(n).
Multiplication by the retained cusp phase e(-a*nu*n/z) leaves EXACTLY
e(-n/z). Therefore the transformed functional is
 beta_nu(n,z)=1_((n,a)=1)*beta(nu*n,z)*e(-n/z),
common to all dilations, and independent of height and future m,c variables.
The negative sign version is its corresponding signed identity. This is
an algebraic identity, with no small-a expansion and no frozen-a constant.

One must not pretend lambda_v is common to an oldform packet. ABL (3.3)
instead writes the orthonormal packet using coefficients xi_g(v). Its
Frobenius norm is z^epsilon after divisor losses. Each normalized dilation
has norm one, so Cauchy bounds evaluation on it by the norm of the same
linear functional. Bounding lambda_v then costs (|a|*nu)^(7/64+epsilon).
In squared norms this is (|a|*nu)^theta, theta=7/32. The unchanged m
functional can now use the level average. This is the packet reasoning in
fixed-d-abl-packet-input-audit-20261010.txt lines 117--157, and agrees with
the printed ABL Lemma 3.2 rather than assuming a general n-removal theorem.

ABL Lemma 3.2 explicitly includes the holomorphic and continuous quadratic
forms, without the a and nu powers in those two cases. Its continuous
```

```
proof invokes the character-pair oldclasses with theta=0. Thus the actual
cumulative cusp-summed continuous form has the SAME t-independent beta_nu
and NO additional power of the height. This is not a claim that individual
Eisenstein series are square-integrable cusp forms. Young's formal
Plancherel basis identity (Prop.8.2, arXiv:1710.03624v2) matches this use.
https://arxiv.org/pdf/1710.03624v2

Although ABL initially writes a length >=1, its algebra is valid also for
the endpoint block 1/2<=Y<1. Every nonempty transformed interval has actual
length Y/nu>=1/2; the regular theorem below expressly allows that length.
The possible integer n=1 must stay in its actual interval, not be moved
to (1,2]. No source claim for an empty sub-unit integer interval is needed.

3. Exceptional norms, complex weights and intervals

Pascadi, arXiv:2404.04239v3, gives the needed classical formulations:
https://arxiv.org/html/2404.04239v3
Saved primary TeX-preserving extraction:
external-source-audit-20261009/pascadi-2404.04239v3-classical-spectral-inputs-20261009.txt
Theorem A: lines 512--556; Theorem J: lines 2886--2943.

Let tau_j=2*sqrt(1/4-lambda_j), so tau_j<=theta=7/32. Theorem A at level z
and cusp 1, whose mu is 1/z, allows ANY complex n sequence and any scaling
matrix. With L=Y/nu and X0=1+Z/L, z~Z, its allowed X range includes X0:
 sum_exc X0^tau_j*|sum_(n~L)b_n R_j,1(n)|^2
       <<(zL)^epsilon*(1+L/z)*sum_n|b_n|^2.
Hence b may depend on z. No level-averaged arbitrary-coefficient version
is being inferred here. The stronger z^2/L^3 range in the source is unused.

Theorem J, with sigma_infinity=Id, states the level average for e(m*omega),
uniformly in ONE common omega, for X<<max(M,Z^2/M). Its proof explicitly
records the extension from DI's phase-zero printed statement. Consequently
the correct citation is this explicit uniform-phase theorem, not the claim
that DI's original printed special case itself contained all quantifiers.

Here is the additional partial-interval derivation, rather than an assumed
arbitrary-weight extension. For I contained in (M,2M], choose integer
P>4M. On residues modulo P write the exact finite expansion
 1_I(m)=sum_(0<=k<P)c_k*e(k*m/P),
where sum_k|c_k|=O(log(2+M)). Minkowski in the Hilbert norm indexed by
(z,j), followed by Theorem J at each omega+k/P, gives
 sum_(z~Z,exc) X^tau_j*|sum_(m in I)e(m*omega)R_j,infty(m)|^2
       <<(ZM)^epsilon*(Z+M)*M.
The extra logarithmic square is absorbed by a selectable epsilon. The same
I must be used for all z; this proof does not interchange a level-dependent
maximum with a spectral sum. A common I of bounded size is immediate.

For larger X, tau_j<=theta gives the continuation factor
 1+(M*X/(M+Z)^2)^theta.
Using X=(1+X_kernel/sqrt(X0))^2 and the n-side X0 split gives the exceptional
term in medium equation (24). The positive imaginary cosh weights are
bounded above and below in this exceptional range, so their omission only
changes constants. No arbitrary level-dependent m coefficients are used.

4. Regular spectrum and the summation over height

Drappeau Proposition 4.7, PDF pp.15--16, explicitly states the three
holomorphic/Maass/continuous cumulative inequalities for T>=1, N>=1/2
and arbitrary complex coefficients in the sqrt(n)*rho normalization.
The continuous measure is dt/cosh(pi*t), with a cusp sum; the bound is
 (T^2+q0^(1/2)*mu(cusp)*N^(1+epsilon))*||b||_2^2.
Here q0=1 and mu(cusp)=1/z. This matches ABL (3.2),(3.8), including its
continuous convention. Pascadi Lemma H is the DI-normalized formulation.
```

```
Pascadi Lemma D (saved lines 2400--2426) supplies both transform bounds
 |H(t)| << (1+|log Y|)/(1+Y),
 |H(t)| << |t|^(-5/2)+Y*|t|^(-3) for |t|>=1.
The local regular proof first uses cumulative norms up to max(1,2Y), then
dyadic shells. The two norm factors together grow by at most 2^(2j),
while these two transform terms leave 2^(-j/2) and 2^(-j). Thus the shell
sum converges, with no extra hidden power of a or the spectral height.
At t=0 use the limiting transform; for holomorphic forms use height k-1.

At each level both cusp densities are 1/z. Cauchy in z costs sqrt(Z),
not Z. Carrying these factors through gives the first term of medium
(24); combining section 3 gives its second term. The needed arithmetic
singleton is s=1, not a claim for general r,s,q in a larger theorem.
The explicit polynomial calculation is in
spectral-regular-polynomial-derivation-20261010.txt lines 55--155.

5. Lemma 4.6: why this replacement actually pays for the large phase

The original problematic L23 source is retained at
external-source-audit/l23-v1-joint-current.txt lines 1697--1795.
Its proof replaces a reciprocity factor by 1 plus an error of size
(|a|/x)*log(x)^O(1). Such an error cannot be discarded uniformly when
a is of order x. The current argument uses the exact factor instead.

In medium lines 299--310 the actual normalization is
 E2=sum_(d~D)E3(d)/d,
and exact reciprocity leaves e(a*h/(q*d*w1*w2*n1)). With
 H=Q*D*Rcal^2*log(x)^5/M, x=N*M,
its dimensionless amplitude on q~q0~Q is
 (a/x)*(Q/q0)*(D/d)*log(x)^5.
After normalizing q,w1,w2,n1,h to bounded intervals, any mixed derivative
using each of those five coordinates at most once is bounded by
O((1+|a|/x*log(x)^5)^5). Iterated partial summation therefore costs at
most x^(5*kappa+o(1)). This is an explicit fixed power, not an arbitrarily
small epsilon or a Taylor remainder. Endpoint rectangles are fixed before
the later spectral sums; the resulting h coefficient is independent of
their other variables. There is no arbitrary q coefficient left over.

The current note also keeps the |a|/x from the returned spectral polynomial:
 |W_bare| << N^2/Q*x^epsilon
      *{D0^(1/2)+[(1+|a|/x)*D1+D2]^(1/4)},
where D0<=x^(-8*sigma), D1,D2<=x^(-15*sigma). Thus the full exponents are
 -4*sigma+5*kappa+epsilon,
 -15*sigma/4+21*kappa/4+epsilon.
At kappa<=sigma/100, allowing sigma/10 for every other selectable loss
including ONE normalizing D factor, the respective savings are at least
 (77/20)*sigma=3.85*sigma and (1439/400)*sigma=3.5975*sigma.
Both exceed sigma/2. The harmonic sum sum_(d~D)1/d=O(1) introduces no
second D factor; the phase is not paid a second time. This checks the
claimed narrow rescue, including its actual a dependence, directly.

6. Necessary boundaries and meaning of this finding

The source correspondence would fail as written if (a,z)=1 were removed:
the ABL integer K can then depend on the dilation, destroying the simple
common beta_nu used above. It also does not justify coefficients depending
arbitrarily on spectral height, a level-dependent choice of m interval,
or an arbitrary q-dependent Lambda before the bare-sum reduction.

The current medium note explicitly retains these restrictions and charges
the phase as above. Within that class I found a precise correspondence
to established primary spectral results, rather than a circular invocation
of the desired moving-residue estimate. This audit has not re-opened the
separately replaced long-variable cleanup and has not independently rerun
```

```
the numerical certificates or verified the entire global distribution
assembly. It changes no theorem constants or main-manuscript claims.

The companion JSON records the exact source hashes and this report hash.
No old file, state, main TeX, formalization, GitHub object or external
communication was changed.
```

F The positive upper sieve and its density

Source: global-positive-upper-completion-20261010.txt.

```
The current positive prime-product upper theorem: completion of (U)
10 October 2026. New proof; no earlier file or research state is changed.

1. The derived theorem and its exact inputs

Put K=8452171/1250000=6.7617368 and H_T=C_T*T/log(T)^2, where
C_T=C_2*prod_(p|T,p>2)(p-1)/(p-2) and
C_2=prod_(p>2)(1-1/(p-1)^2). Let T tend to infinity through ALL even
integers. Fix a>0, a finite maximum k of original prime slots, bounded
nonnegative fixed label weights, and finitely many geometric families
defined by ordering and monomial conditions, with product O(T) and
every original prime >=T^a. There is no extra arithmetic filter.
Let w_T(m) count the labelled tuples with their weights and suppose

 limsup_(even T) (log T)/T * sum_m w_T(m) <= J < infinity.

Then, on domains m<T,

 limsup_(even T) H_T^(-1)*sum_(m<T,T-m prime)w_T(m) <= K*J.       (U*)

The result also holds directly for each fixed nonnegative smooth tuple
weight of product O(T), bounded size and normalized derivatives, and
for finite positive sums. Repeated prime factors can be kept or omitted.
The error is absolute o(H_T), including J=0. Constants and the eventual
threshold depend on the fixed family. No target averaging is used.

For each fixed finite set of smooth queries in this theorem, the actual
construction below gives ONE coefficient rho_T satisfying all three
parts (i)--(iii) of (U) in current-downstream-proof-spine-20261010.txt.
In particular (U) is discharged by the present theorem; it is not an
additional unproved distribution hypothesis at the end of this argument.

Here are the proof inputs, distinguished by what they assert.
 A. The derived restricted PP theorem: distribution-medium-completion,
    distribution-prime-product-completion and distribution-coefficient-
    completion, all dated 20261010, with the Gevrey dispersion majorant
    and direct Brun long-branch replacements in the current assembly.
    Its exact statement is (8) below. This is a local proved lemma with
    the named classical input list in section 9, not unrestricted L23.
 B. The classical ordinary cubic Rosser exit-series convergence and
    raw one-sided bounds, PNT/Mertens and the dimension-one fundamental
    lemma. Section 3 proves the target-uniform literal-density bridge.
 C. The retained outward numerical lemmas: the reference upper functional,
    the actual same-tree rank4/5/6 and fallback differences, the ten later
    upper ledger differences on the 87-domain policy, and the replacement
    of its single original third-tail allowance. Their precise endpoint
    is E in (6). The certificate-coverage audit owns these enclosures;
    they are mathematical inequalities, not inferred from PASS strings.

Subject to these explicitly supplied proved lemmas and established
classical results, no unmatched analytic or same-coefficient obligation
remains in (U). This proof does not use any lower Chen count, H/h table,
Li's 1.9728 claim, or the disputed unrestricted large-residue theorem.

2. One causal upper tree, including every adopted change

Use rho0=19/96, mu0=25/128, Theta=299/500 and Lambda=153/256.
Write A(t)=(1+t)/2, D(t)=57/64-3t/2, v=min(A,D), S_j=sum_(i<=j)t_i.
The reference first stop is rho0. Below t1=mu0 its third endpoint is
```

```
 L3=max(v(t1),v(t1+t2),v(t1+t3),v(t2+t3),v(S3),Lambda-3t3/8);

above mu0 it is v(t1). The ordered gated-envelope identity is used:
the displayed nonleading pair terms are not ungated permissions.

The rank4 action retains the fourth node and replaces its later odd
tests. It completely replaces the old partial-fourth and fourth-tail
bounds. The rank5 action replaces its own fifth cube and then freezes
the increased cap, legitimately restoring formerly excluded fifth nodes.
Its old-parent subdivisions are whole-parent maxima, and its new covers
are disjoint from those parents; they are not added to an old bound on
the same parent. Rank6 retains its node and changes later odd tests.
Its old third/fifth histories are retained against L3 and its comparison
R5 bounds the ACTUAL fifth cap after all rank5 changes. Thus the exact
density telescope is V3-V6=(V3-V4)+(V4-V5)+(V5-V6).

Above mu0, the second fallback has t2>=3/16 and the third fallback has
t2<3/16, so they are disjoint. Their third-tail approximation is kept.
All older higher-rank actions have t1<=mu0. Later rank6 changes either
add disjoint regions, upgrade from the current cap, or improve a bound
on the SAME difference. The lower-complement correction adds the missing
intermediate difference, not a second full difference. The four later
nonsingleton rank6 cells are counted only at the eighth-family stage.

There are exactly 30+15+13+29=87 adopted eighth domains:
 * 30 equal-parent domains with R6=L3=D(t2+t3), mask129={1,8};
 * 15 unequal-parent domains: thirteen winner9 and two winner50 cells,
   with masks138={2,4,8},184={4,5,6,8},134={2,3,8};
 * 13 ascending-pair minorant cells, t2+t3<mu0, B3=A(t2+t3);
 * 29 first-failure minorant cells, t2+t3>mu0, B3=A(t1), split as
   12+7+10 according to the first failed OLD D(t2+t3) history at3,5,7.
Failure of this sufficient minorant is not a first exit of the true tree.
Every actual history is below its B3<=L3 and is retained.

Their separations are structural. Unequal cells have L3<R6 and avoid the
first 30. Different old-plane assignments require a subset sum s<=e
versus s>=25/96-e/3, separated by 4(mu0-e)/3>0. Winner50 has a strict
mask9 comparison below R6, unlike winner9. The ascending and descending
pair groups have opposite sides of t2+t3=mu0. First-failure groups have
opposite inequalities at their earliest different history and avoid all
older domains requiring every D23 history. These proofs respectively
give all 105,78,406 within-family pair separations and all new-old ones.
The seed duplicated as family cell11 is excluded from the new12 sum.

For each eighth cell the actual sixth cap satisfies
lambda3<=lambda6_current<=R6, while retained H7=S6+3t7<lambda3 implies
lambda6_current-S8>2t7-t8>t8. Saved M>R6 and M-S8<3t8 give

 1<(lambda6_current-S8)/t8<=(R6-S8)/t8<(M-S8)/t8<3.

All eight full squares S_(j-1)+2t_j<M and ALL unselected complement
squares s_I+sum_(i<j,i notin I)t_i+2t_j<M are retained. Later complement
squares equal full squares. Hence M changes only later odd exits and
the exact ordinary upper residual identity yields the saving density

 4(M-R6)/[(R6-S8)(M-S8)*prod_(i=1)^8 t_i].                (1)

Using R6 instead of the actual smaller parent reduces the saving.
There is no Chen quarter or ordering factorial in (1). All 87 domains
have t8>53/13824. Jensen, log moments, curvature and mixed-transverse
improvements replace bounds on the same integrals, only by new minus
old. They do not activate another cap or supply another copy of a gain.

All these operations preserve prefix closure and ONLY ODD first exits.
```

Higher-rank overlaps are handled by the exact successive-tree telescope;
only domains acting at the same stage need disjoint interiors. An unused
or boundary upgrade inherits the prior policy; no even node is deleted.

3. Exact pointwise and literal raw-density identities

For a finite allowed prime set Q and 0<=h(p)<=1, the actual raw upper
coefficient lambda_T(d)=mu(d)*1_(d admitted) satisfies

 sum_(d|Q)lambda_T(d)h(d)
 =prod_(p|Q)(1-h(p))
  +sum_(odd first exits P|Q) h(P)*prod_(p|Q,p<p_last)(1-h(p)). (2)

Partition all excluded divisors by their unique first excluded prefix;
the remaining smaller primes sum to the displayed Euler product. The
minus sign of each odd prefix gives (2). At h(p)=1 on primes dividing
any integer n this gives the rough-indicator majorant, also at n=0.

For ordinary residual trees put g_T(p)=1_(p>2,p not dividing T)/(p-1).
The normalized odd/even exit sums E_(j,T) are nonnegative and satisfy
0<=E_(j,T)<=E_(j,0): a surviving prefix has exactly the multiplier
prod_(p_last<=p<y,p|T)(1-g_0(p))<=1, and other prefixes disappear.
For fixed rank j on a positive compact exponent range R=T^beta,
the preceding squares give A_i^2<R*A_(i-1), A_i<R^(1-2^(-i));
failure of the j-th cube forces
p_j>=T^[beta_min/(3*2^(j-1))]. Removing target primes at fixed rank
therefore costs o(1) uniformly in T. PNT gives the usual finite-rank
integrals a_j(s), with s=log R/log y.

Use the compact convergence of the positive exit series (Suzuki
Proposition 11.8, dimension1,beta2) and the opposite raw one-sided
bounds for fixed g_0 (Suzuki Lemma 14.7). First fix a sufficiently large
finite rank, then let T grow. The finite partial sums and E_(j,T)<=
E_(j,0) squeeze the actual ordinary coefficients to

 V_T^+(R,y)/V_T(y)=F(s)+o(1) (s>1),
 V_T^-(R,y)/V_T(y)=f(s)+o(1) (s>=2),   (3)

uniformly on the needed positive compacta. This does not assume PP or
a density for some other weight. For 1<s<=3, the exact upper first-cube
identity V_T^+(R,y)=V_T^+(R,R^(1/3)), and the Euler ratio3/s, give
F(s)=2exp(gamma)/s. This covers the entire range used in (1).

Mertens and the exact target Euler-product ratio give
V_T(T^u)=(2exp(-gamma)*C_T)/(u log T)*(1+o(1)), uniformly in even T.
Thus Fbar=F/(2exp(gamma)), fbar=f/(2exp(gamma)) give density coefficients
4Fbar(s)/u and 4fbar(s)/u. Exact signed subtree differences and (3),
not subtraction of two one-sided count bounds, evaluate every raw gain.

Finally for the SAME legal history projected to primes >=w=T^omega,
(2) shows

 V_T(w)*sum_(rough d)lambda_T(d)g_T(d)
 <=sum_(full d)lambda_T(d)g_T(d).   (4)

Only positive exits whose last prime is >=w survive on the left.
Equality of rough and full densities is neither true nor needed.

4. Finite realization fits the EXISTING allowances

The coarse third rule lambda_B=Gamma_B-2e on boxes of width chi<e/10
satisfies 0<=L_endpoint-lambda_B<=4e+6chi uniformly, including ties.
This follows from 9/2-Lipschitz branches and the nonleading-pair gate
surplus 3/256. The physical caps remain lambda_B throughout this proof.

Near a W gate, use ONLY an auxiliary support envelope: if inf_B A>=
lambda_B+e/2, take ell_aux=lambda_B+e/4 and the singleton licence;
otherwise sup_B A<lambda_B+11e/20, W>=Gamma_B, and take
ell_aux=lambda_B+e. The latter lies >=9e/20 above the gate and e below
W. The second initialization square has independent reserve3/256.
No density cap, coefficient, old integral or history is changed by this
factorization cover. Requiring ell_aux=lambda_B would be erroneous.

With u=rho0/2, the omitted third-activation tail has the common bound

 T3(h)<=18*h/(u-h)^2*[log(rho0/h)^2+2log(rho0/h)+2] ->0.   (5)

Indeed Fbar-fbar<=9/s^2 for s>=3, lambda_B-t1-t2>=u, and the
ordered reciprocal-prime triangle has mass log(rho0/t3)^2/2. This
controls the noncompact finite-activation limit independently of the
numerical split at t3=.01. All higher action cutoffs are positive and
their kernels are bounded after proper-wall trimming. A proper affine
wall has strip volume O(delta); genuine identities and inherit regions
are kept. Freeze ancestors on their own prefix labels before children.
Do not remove child-wall projections, which can have full parent measure.

Given the positive density allowance u_d in (6), allocate strictly
smaller fractions to the finitely many pieces. Choose h using (5), then
coarse buffers and widths so compact errors fit their fractions; choose
the finite higher caps in causal order. This gives an actual full raw
density <=(E+u_d+o(1))*C_T/log T. The entire 87-domain union shares these
ancestor choices. There is no minimum over an untrimmed equality wall.

The physical compiler has its separate allowance u_c. At even rough
rank take the outer endpoint tests, and at odd rank the inner tests,
always retaining every coarse historical decision. Then c_T(d)>=
lambda_T(d) coefficientwise. Attach a single small-prime beta+ to c>0
and beta- to c<0, where B-<=I_w<=B+. For every integer n,
c(d)B_sign(n)>=c(d)I_w(n). Thus the actual coefficient
rho_T(bd)=c_T(d)*beta_sign(b), with unique small/rough decomposition,
satisfies |rho_T|<=1, squarefree support, (bd,T)=1, and

 R_T(n):=sum_(q|n)rho_T(q)>=1_((n,P_T(T^rho0))=1)>=0.   (P)

The beta-density and tagged-cube boxing errors relative to V_T(T^rho0)
are O(exp(-c/eta)*omega^(-2)+C_policy*tau*omega^(-5)). The constant
C_policy depends on the frozen coarse policy, not the later atom count.
Together with (4), this controls the density of THIS rho from above.

Choose source zeta,sigma below the positive coarse reserves, then
ell_beta=nu0*eta, omega=nu0*eta^2 small enough for support, (5) and
beta error. Fix rough rank J0>theta_*/omega+2; then choose tau so its
rank inflation and compiler error fit u_c. Only afterwards choose the
actual fixed moment order K_mom, kappa<=min(sigma/200,1/(2K_mom)) and
0<kappa0<kappa/[4(1+kappa)], also within all old reserves. Retaining
the now unnecessary former long-branch tau^(4J0) order is harmless.
With support inflation below1/1000, theta_*=599/1000 is available;
shrink kappa0 so theta_*/(1-kappa0)<3/5. All choices precede T.

5. The current exact numerical endpoint, once

The old 87-domain endpoint was
 E_old=3045563858313172390552018158815166574109 63/
       4504130709723514489583681088560037888000 0.
Its raw construction is precisely the policy in section2. The only
subsequent upper improvement is the same original third-tail bound:
 T_old=1028753216356588863/2^80,
 T_new=319683857488126290909150567/2^111,

```
 Delta=T_old-T_new=188954685246505442944226165 7/2^111.
It occurs with coefficient+1 in E_old. No compact leaf or action changes.
The new bound follows by integrating the complete 0<t3<.01 tail using
D(s)<=D(10)*(10/s)^2 and the saved outward D(10) enclosure. It is not
a new removal of that domain or a change to the activation tail (5).

Both old upper allowances are retained and each gains Delta/8. Thus

 E=E_old-Delta
  =997970257671091123485061980781684786490735 9959/
   14759135509622012279467806190993532151398400 00,
 u_d=u_c=14500796122704831221059342753012800078 4681/
         11807308407697609823574244952794825721118720000,
 E+u_d+u_c=39918955314804871987714689824694921587630224517/
          59036542038488049117871224763974128605593600 00,
 K-E-u_d-u_c=71229286727542131248758574547050908572431/
            73795677548110061397339030954967660756992000000 0>0. (6)

The last surplus is 9.652230197507868...e-8. Exact fraction arithmetic
recomputed all four identities from the current and old assemblies.
These quantities are upper-density units; no Chen factor1/4 enters.
The current paired Y6/Y11 deletion changes lower costs only and leaves
every quantity in (6) unchanged. Section4 can fit STRICTLY inside each
positive allowance. Therefore the very coefficient in (P) satisfies

 S_T:=sum_q rho_T(q)/phi(q)
 <=(E+u_d+u_c+o(1))*C_T/log T <=(K+o(1))*C_T/log T.       (7)

The remaining strict surplus pays every final asymptotic density error.
No numerical endpoint requires attainment by a zero-buffer finite tree.

6. The derived restricted PP applies to this very coefficient

The coefficient theorem constructs exact labelled tensor atoms of rho,
with the original repeated-label factorial used once. Each is regrouped
into three one-bounded convolutions using one endpoint slot partition
valid throughout its box. Beta attaches through the unique small/rough
decomposition. Squarefreeness remains in the physical rho and (7); its
removal solely for the analytic tensor comparison costs
O((T+T^theta_*)*T^(-omega+o(1))). The coefficient is not reselected after
seeing a progression discrepancy or a Type-II arithmetic tuple.

Rough ranks0,1,2 have raw support exponent19/48, gap5/48 below half;
small losses preserve classical BV. Every rank>=3 has the genuinely
active third cutoff p3>=T^omega, a selected or repaired W licence and
all full/complement squares. Numerical t3=.01 is not an activation
criterion. Above-mu fallbacks have v(t1)>3t1 and their extra initial
squares <3t1. Every higher action retains the same descendant-square
closure proved in section2. The auxiliary envelope in section4 licenses
factorization without increasing lambda.

After the smooth complete-product trim, charts have
c*T^(1-kappa0)<=x<=C*T and c0*x<=T<=x^(1+kappa). The moving residue is
the full T and x the complete product, never a residual quotient.
All current four monomial inequalities have fixed positive reserves.
The proved medium theorem with the fixed Gevrey majorant, actual rough
SW, and direct Brun one/two-long alternatives therefore yield

 sum_q rho_T(q){sum_(m=T mod q)W_T(m)-X_W^(q)/phi(q)}
   =O_(A,W)(T/log(T)^A),                                  (8)

for every fixed A and every required smooth positive tuple family,
where X_W^(q)=sum_((m,q)=1)W_T(m) is the EXACT coprime mean. The finite
atom scalar mass and polylog chart counts are paid by requesting a
```

```
stronger logarithmic saving. There are only finitely many requested
families; one fixed realization and one eventual threshold suffice.
Equations (P),(7),(8) are precisely (U)(i),(ii),(iii).

7. Positive counts, the coprime mean and uniform negligible terms

For any prime p, squarefreeness and |rho|<=1 give
sum_(q:p|q)|rho_T(q)|/phi(q)<=1/(p-1)*sum_(e<=T^theta_*/p)1/phi(e)
<<log T/(p-1). Since W>=0 and there are at most k original prime slots,

 |sum_q rho_T(q)(X_W^(q)-X_W)/phi(q)|
 <=sum_tuples W(tuple)*sum_(p in tuple)O(log T/(p-1))
 <<X_W*T^(-a)*log T=o(H_T).                                  (9)

There is no T^theta_* factor. Repeated primes merely repeat a harmless
union-bound term. We need only C_T>=C_2>0, not an upper bound on C_T.

For primes T-m>=T^rho0, (P) majorizes the prime indicator. Smaller prime
remainders contribute T^(rho0+o(1)): each m has only fixed-order divisor
tuple multiplicity. If complete-product scales are needed, tuples with
m<2T^(1-kappa0) have total absolute progression contribution at most
sum_m tau_k(m)tau(|T-m|)=T^(1-kappa0+o(1)), with m!=T, and their
exact-coprime mean costs only another O(log T). Thus no tau(0) occurs.
For large T all original primes in these families are odd, so product=T
is impossible for even T; (P) also covers zero without needing that fact.

Positive majorization BEFORE distribution, followed by (8),(9),(7), gives

 sum_(m<T,T-m prime) W_T(m)
 <=sum_m W_T(m)R_T(T-m)+o(H_T)
 <=X_W*S_T+o(H_T).                                          (10)

Outward weights may include m>=T; this is harmless because R_T is
nonnegative for every integer, including negative ones. For A>2,
T/log^A T=o(H_T). Multiply (7) by the nonnegative X_W and take limsup
in (10): if limsup X_W*log T/T<=J, the result is <=KJ. Errors remain
absolute when J=0; no division by a family mass is used.

8. Sharp geometric restrictions and the required downstream families

On positive exponent compacta use t_i=log_T p_i (i<k),
t_k^*=1-sum_(i<k)t_i and y=product p_i/T. PNT in the last prime,
followed by reciprocal-prime summation, gives limiting unsifted measure

 dmu_k=dt_1...dt_(k-1)dy/[t_1...t_(k-1)t_k^*].              (11)

First restrict y>=eta; the remaining mass is O_a(eta*T/log T) by the
prime upper bound. Proper monomial walls are affine null walls in t;
pure complete-product walls must instead be treated in y. Identities
are kept. For fixed epsilon choose a smooth positive OUTER majorant
with excess limiting mass <=epsilon and all primes still >=T^a/2.
One global partition of unity repeats no mass. Apply (10), let T grow,
then eta and epsilon decrease. This proves (U*) on the sharp family.
One need not keep rho identical for different smoothing precisions:
at EACH fixed precision the SAME rho occurs in (P),(7),(8),(10).
No signed sharp-domain distribution statement is being inferred.

For ordered bad semiprimes r<=q, rq<T, r>q^b, fixed1/2<b<1, the trimmed
domain has positive prime cutoff. With beta=b/(1+b), (11) gives
J_b=int_0^1 dy*int_beta^(1/2)dt/[t(1-t)]=log(1/b). Therefore

 limsup_(even T) B_b(T)/H_T<=K*log(1/b).                    (12)

There is no second ordering factor or singular-series factor in J_b.
```

The equal-prime diagonal is O(sqrt T); it may be retained or restored.
All fixed positive switched leaves in the finite248/430 graph are also
covered: their positive cutoff and finite prime-slot bound are supplied
by the typed coefficient theorem. Their reciprocal normalizations do
not multiply (U*) by another factorial. The current U11 complement has
at most eleven original primes since12a=1.0008>1, a=417/5000. Its17772
mask cells give only finitely many proper geometric walls. Hence it is
also an instance of (U*), with its later parent-majorant allocation a
separate lower-count calculation. No H/h gain is needed to prove K.

9. Completion boundary and classical input inventory

The established classical library needed here is finite:
PNT and Mertens; the dimension-one fundamental lemma for the physical
beta compiler; ordinary cubic Rosser positive-exit convergence and raw
one-sided density bounds (Suzuki Prop.11.8 and Lemma14.7, with the
normalization in (3)); and the classical input library of restricted PP:
prime/Mobius SW, below-half BV and its fixed-family convolution form,
primitive multiplicative large sieve, standard divisor means, Shiu for
the preliminary nontrivial progressions, Fourier/Poisson, complete
Kloosterman Weil, Maynard I Lemma13.2, the stated normalized Kuznetsov
and regular spectral sieve, ABL3.1/3.2 in the coprime arithmetic-argument
case with continuous/oldform normalization, Kim--Sarnak theta=7/32,
and Pascadi A/J in their matching common-interval/common-phase form.
The q=1 shifted moments, W initialization, tensor identity and direct
Brun long alternatives are project derivations, not added conjectures.

The numerical input is the finite chain identified in sections2 and5,
not any adaptive or infinite optimization, and not an assumption about
the true prime count. Its complete certificate coverage is separately
audited. This report rechecked the source equations, actual causal tree,
same-rho signs, literal-density reduction, finite parameter order, PP
class matching, positive approximation, and all current scalar budget
identities; it did not rerun the large multidimensional quadratures.

Conclusion: the upper hypothesis (U) of the current downstream theorem
is a derived result at K=6.7617368 from the established analytic and
numerical lemmas just listed. There is no residual unspecified upper
distribution, same-weight, support-cover, density, smoothing or uniformity
assumption. Completion of the separate lower Chen theorem (C), the full
certificate coverage and the final combined theorem are the parallel
audit tasks. Nothing here restores unrestricted Lichtman or external
Li1.9728, proves an effective threshold, or asserts publication priority.
The open manuscript, earlier artifacts, state, formalization, GitHub and
external communications remain unchanged.

G The Chen lower construction with paired deletion

Source: global-chen-lower-completion-20261010.txt.

```
The current own Chen lower theorem: a completed modular derivation
10 October 2026. New proof only; the main manuscript and old artifacts are unchanged.

THEOREM AND THE PRECISE INPUTS

For an even integer T, let
 C_T=prod_{p>2}(1-1/(p-1)^2) prod_{p|T,p>2}(p-1)/(p-2),
 H_T=C_T T/log(T)^2,
 D12(T)=#{P<T: P prime, T-P>1, Omega(T-P)<=2}.
Omega counts prime factors with multiplicity. Then the current construction gives

 liminf_{T->infinity, T even} D12(T)/H_T
 >= 537979576980192301547849603498537562994346796 1 /
    27429525099786446047937783694182318080000000000
 > 12258223/6250000 = 1.96131568.                          (C)

Here are the separately established lemmas used in this proof; none is (C):

 P: the restricted signed prime-product distribution theorem, inserted for
    the ACTUAL finite coefficients by distribution-coefficient-completion-
    20261010.txt. Its analytic proof uses the current Gevrey and direct-Brun
    updates. In the lower proof its k=1 case acts on the original prime P;
    it does not turn an upper product estimate into a prime lower estimate.
 U: the positive product upper theorem with K=8452171/1250000, for every
    fixed finite nonnegative labelled geometric family with fixed positive
    prime exponents and bounded rank. Its normalized upper cost is K times
    the unsifted mass. U is proved before any H/h feedback or Chen lower c.
 D: literal ordinary raw Rosser density equalities for g_T(p)=
    1_{p>2,p does not divide T}/(p-1), on fixed positive compact parameter
    sets: V_T^+(T^beta,T^t)/V_T(T^t)=F(beta/t)+o(1) for beta/t>1,
    and V_T^-/V_T=f(beta/t)+o(1) for beta/t>=2. See section 2 below.
 N: the certified integral/finite-table lemma for the specifically selected
    functional in section 6, with its exact rational endpoint. This is an
    explicit collection of integral inequalities, not a numerical PASS
    used to infer an analytic statement. Its primitive/cover validation
    has the scope recorded in current-numerical-proof-obligations-20261010.

This note proves the implication from those lemmas to (C), including the
pointwise inequality, the actual lower coefficients, the finite feedback,
the current deletion and the common all-even quantifiers. P and U are
independent modules being audited separately; this is not a new verification
of their spectral proofs or of every quadrature. In particular no external
Li 1.9728 or unrestricted Lichtman moving-residue statement is a premise.

1. THE POINTWISE COUNT AND THE CURRENT PAIRED MASK

Put a=417/5000, h=19/125, d=59/250 and mu=25/128. For positive n<=T write
 Q_b(y;n)=1_{b|n} 1_{P^-(n/b)>=y}, with P^-(1)=infinity. Distinct prefix
primes have the following ranges; all the Q below have argument n:

 Y1=Q_1(T^a), Y2=Q_1(T^h);
 Y3=sum_{T^a<=p<T^(1/3)} Q_p(T^a),
 Y4=sum_{T^a<=p<T^d} Q_p(T^a);
 Y5=sum_{T^a<=p1<p2<T^h} Q_(p1p2)(T^a),
 Y6=sum_{T^a<=p1<T^h<=p2<T^d} Q_(p1p2)(T^a);
 Y7=sum_{T^d<=p1<p2<(T/p1)^(1/2)} Q_(p1p2)(p2),
 Y8=sum_{T^a<=p1<T^(1/3)<=p2<(T/p1)^(1/2)} Q_(p1p2)(p2),
 Y9=sum_{T^h<=p1<T^d<=p2<(T/p1)^(1/2)}
       Q_(p1p2)((T/(p1p2))^(1/2));
 Y10=sum_{T^a<=p1<p2<p3<p4<T^h} Q_(p1p2p3p4)(p2),
```

```
 Y11=sum_{T^a<=p1<p2<p3<T^h<=p4<T^d} Q_(p1p2p3p4)(p2).

The FULL last range is essential. Define
 W=3Y1+Y2-Y3-Y4+Y5+Y6-2Y7-Y8-Y9-Y10-Y11.
For a squarefree T^a-rough n, let r,s,t be the numbers of its prime
factors below T^h,T^d,T^(1/3). The first six counts are
 (1,1_{r=0},t,s,binom(r,2),r(s-r)).
When r>=2, the first two selected factors in Y10 or Y11 must be the two
least factors of n, since the quotient is sifted at the second selected
factor. Thus Y10=binom(r-2,2), Y11=(r-2)(s-r), and the first six weighted
terms minus these two terms equal s-t<=0. All remaining subtractions help.

For k=Omega(n)>=3, the two least factors satisfy q1*q2^2<n<=T. Consequently
Y7=1_{s=0}, Y8=1_{t=1}, and t>=1. For r<=1 the remaining upper bounds are
  r=1: 2-t-1_{t=1};
  r=0,s>=2: 4-s-t;
  r=s=0: 2-t-1_{t=1};
  r=0,s=1,t>=3: 3-t;
  r=0,s=1,t<=2: 3-t-1-1_{t=1}.
In the last line k>=4 would give n>=T^(h+d+2/3)>T; hence k=3 and
q1*q2*q3^2>=T^(h+d+2/3)>T, so the pair(q1,q2) contributes to Y9.
Every displayed bound is <=0. For k<=2 direct substitution gives W<=4;
the unit has W=4. Thus W<=4*1_{Omega(n)<=2} for squarefree n. If n is
not T^a-rough every Yj vanishes; for Y9 its quotient threshold exceeds
T^((1-d)/4)>T^a, which checks the only nonimmediate case.

Let E0 be the adopted 6091 whole-zero pair512 rectangles, and E1 the
disjoint selected 11481 rectangles in positive-pair-deletion-certificate-
20261010.json. This is a fixed set E=E0 union E1, in coordinates
(t,u)=(log_T p2,log_T p1), a<=u<h<=t<d. Write Y6(E) for its pair terms
and Y11(E) for the four-prefix terms whose LAST TWO selected primes
(p3,p4) have (log_T p4,log_T p3) in E. Define
 W_E=W-Y6(E)+Y11(E).
For squarefree n, a contributing Y11 tuple has forced first primes q1,q2;
sending it to (p3,p4) is injective into Y6(E). Its residual remains
T^a-rough. Therefore Y11(E)<=Y6(E) and W_E<=W<=4*1_{Omega(n)<=2}.

The prime-factor bound floor(1/a)=11 gives |W_E|<=1016 on rough n. There
are at most 2T^(1-a) integers with a repeated prime >=T^a. Summing over
P<T, prime and coprime to T, and excluding the unit loses at most
2032T^(1-a)+4=o(H_T). Restricting to these P only decreases D12. It follows
that 4D12 is at least the sum of W_E(T-P), up to this uniform error. This
also proves that the modified weight has the same eleven signs, with
Y6 and Y11 replaced by their complements. It does not assume Wu's
printed contracted-Y11 inequality.

2. WHY THE ORDINARY LOWER START IS A PRIME COUNT LOWER BOUND

Start at w=a or h with the ordinary lower tree of cap nu=249/448, using
a small root buffer. Causal third, fourth and fifth cap decisions depend
only on their own prefix and keep all ancestor decisions. A usual fourth
node must pass its cube at its ACTUAL third cap before its cap is changed.
Every first exclusion has even rank. For any finite prime universe and
0<=q(p)<=1 the exact first-exit identity is
 sum_d mu(d)1_{d retained}q(d)
 =prod_p(1-q(p))-sum_{even first exits e}q(e)prod_{p<p_last(e)}(1-q(p)). (2.1)
It proves a pointwise lower weight by q(p)=1_{p|n}; it also evaluates its
raw density by q=g_T. No pointwise monotonicity of cap changes is needed.

The restored fourth own-cube actions also satisfy(2.1): on an old first
exit select a new cap before that fourth cube, check every new support
gate, and require S3+3t4<lambda4. The attached ordinary lower residual has
R>p4^2, so its first square never excludes an odd full rank. Its exact
raw gain is g_T(P4)V_T^-(T^lambda4/P4,p4), old contribution zero. Current
```

```
restored slabs lie outside old E4 and the projection of E5 and receive no
new fifth gain. Alternative full covers are maximized/replaced, not added.

For a presieve z0=T^omega, let R_raw be this same tree restricted to
rough divisor primes. The positive even exits in(2.1) give exactly
 V_T(z0)R_raw - V_full
 =sum_{full even exits with last prime<z0}g_T(e)V_T(p_last(e))>=0.     (2.2)
Round positive even coefficients inward and negative odd coefficients
outward on one fine partition, retaining ancestry, to obtain
c(d)<=lambda_raw(d). Attach the lower beta polynomial B- to c(d)>0
and the upper polynomial B+ to c(d)<0. With B-<=I0<=B+ and I0>=0,
 sum_{q|n}rho_T(q)<=I0(n)sum_{rough d|n}c(d)
                   <=1_{(n,P_T(T^w))=1}.                          (2.3)
This does not require B- to be nonnegative. Unique small/rough division
keeps |rho_T|<=1. Equation(2.2), rather than c<=lambda alone, gives
 sum_q rho_T(q)g_T(q)>=V_full-small compiler density error.         (2.4)
The beta/fine-box errors tend to zero in the fixed-policy parameter order.

P is applied to this SAME rho_T, including the original marked/residual
factorials, fixed common beta and endpoint slot partition. It gives
 sum_{(q,T)=1}rho_T(q)[pi(T;q,T)-pi(T)/phi(q)]
 =O_A(T/log(T)^A).                                                 (2.5)
The exact-coprime mean differs here by only O(log(T)^2); all moduli are
coprime to T. The short-prime and square-collision terms have fixed power
savings. Thus(2.3)--(2.5) turn the raw density into the required lower
estimate for the original prime sequence. Historical invocations of a
supremum in unrestricted L23 are replaced by P, not retained as inputs.

D itself is genuinely a literal-density lemma. At any fixed rank, the
normalized positive exit for g_T is the corresponding g_0=1_{p>2}/(p-1)
exit times prod_{p_last<=p<y,p|T}(1-g_0(p)), or is deleted. It lies between
zero and the g_0 exit. Previous squares force a rank-j exit's last prime
>=T^(beta_min/(3*2^(j-1))). Thus its difference from the fixed g_0 integral
is uniformly o(1). Suzuki's beta=2 positive-exit convergence and one-sided
bound for the SAME raw weight give the opposite full-rank bounds and a
two-sided sandwich, first fixed rank then T. The upper range 1<s<=3 also
follows from the exact support identity at y=R^(1/3). This proves D,
including lower s=2, without subtracting two unrelated one-sided sieves.
The primary hypotheses and normalization are in rosser-raw-density-uniform-
audit-20261010.txt; its classical inputs are stated there explicitly.

At a retained prefix all residual primes are smaller than its least
prime; g_T(Pd)=g_T(P)g_T(d). Using D, its exact third/fourth/fifth raw
subtree differences are respectively upper-old minus upper-new,
lower-new minus lower-old, and upper-old minus upper-new. They yield the
saved I3/I4/I5 integrals of ONE causal tree. Prime harmonic summation and
V_T(T^t)~2e^(-gamma)C_T/(t log T) give precisely the factor 4 used below.

3. FINITE H/h CONSTRUCTION ON ACTUAL MARKED FAMILIES

Put A(s)=sF(s)/(2e^gamma), a_bar(s)=sf(s)/(2e^gamma). A=1 on[1,3],
a_bar=0 for s<=2. On a nonnegative finite marked prime-box family sigma(B)
use one total Q=T^lambda, residual L=lambda-log_T B>0, and
 Theta_Q=4 li(T) sum_B sigma(B) C(BT)/[phi(B)log(Q/B)].
An H(s) lower defect means Phi_s<=(A(s)-H(s)+o(1))Theta_Q; h(s)
means Phi_s>=(a_bar(s)+h(s)+o(1))Theta_Q. These are averaged estimates,
not estimates for each B and not assertions for a signed outer sigma.
The signed ordinary modulus atoms used to prove them are separate.

Enhanced states preserve every full square, every selected-remaining
square, their W branch and physical ceiling chi; L/s<=chi. Classical
states use a complete level below half. The physical cap lambda used
in density is never replaced by its possibly larger analytic support
cover ell_aux. The coefficient theorem P proves the representation on
```

that cover for the same rho, including all ordinary descendants.

Every ordinary child keeps Q fixed. For a singleton Lu,
L'=L(1-u), alpha'=1+(alpha-1)/(1-u); for a pair Lu,Lv,
L'=L(1-u-v), alpha'=1+(alpha-1)/(1-u-v), with alpha=(1-log_T B)/L>=2.
New full-square tests are 2u<1, or 2v<1 and v+2u<1. Once every selected
old slot has occurred, each new selected-remaining test equals its full
square. New primes remain below the old ceiling. In particular alpha
increases; it is never reset to 2 in a switched mass calculation.

The actual four-round seed gives H>=2327/100000 on[1,11/5]. Cutoff
monotonicity proves its upper extension below 2. The finite transitions are:
 BH: H_new(s)=integral_{max(2,s-1)}^5 h(w)dw/w,
     h_new(s)=integral_{s-1}^5 H(w)dw/w, s>=2;
 W1: the ordinary one-step base plus H(s0) and one half of its singleton
     lower-child defect integral; the positive switched cost is K I/8;
 W2: the repaired ordinary/switching base plus
   [4H(s0)+H(phi1)+sum_{three singleton domains} integral h(rho)du/[u(1-u)]
    +sum_{four good pair domains} integral H(rho)du dv/[uv(1-u-v)]]/5.

For W2 use a_i=1/phi_i, a0=1/s0. The pair domains/cutoffs are
 D5+: a0<u<v<a2, v+2u<1, cutoff La0;
 D6: (a0,a1)x(a2,a3), cutoff La0;
 D7: a0<u<v<a1, cutoff Lu;
 D8: (a0,a1)x(a1,a2), cutoff Lu.
Their arguments are s0(1-u-v) for 5,6 and (1-u-v)/u for 7,8. The
actual bank gates 2v<1, v+2u<1 make 1<rho<=s0-2<=3. Below 2 use
proved smaller-cutoff upper comparison. The excluded part of D5 receives
only a positive product cost, no ordinary H feedback. W2 terms 9--21 are
positive labelled families bounded by U; unit surfaces in 20,21 remain
in their masses. The valid uniform envelope is
omega(x)<=max(1/x,70893/125000) for x>=1, with the conservative extension
below 1. Thus the bases are valid for every actual alpha>=2.

These formulas follow by inserting the bounds, with their signs, into
the one-step or 21-term pointwise inequality at the SAME Q; child/parent
normalization yields exactly the displayed positive measures. All child
estimates on a right side come from a preceding finite generation. Take
a finite maximum of alternative defects, never their sum; use closed-cell
minima on crossing boxes. This is an induction on real positive counts.

There are 70 W1 and 36 W2 actions. The construction reaches 248 and then
430 total rounds after the seed; generation labels are not summed.
Their old K anchors are all >= the current K, so U licenses every retained
base. The 430 recost uses its proved net slopes, with old variable-kernel
savings already deducted. It is used ONLY in U3/U4. No infinite fixed
point is invoked: after at most 440 elementary transitions residuals are
>=ell*101^(-440), and every new prime exponent is >=this number/6. All
positive leaves therefore have a fixed cutoff in U's quantified class.

4. THE ELEVEN ROOT OCCURRENCES AND CAUSAL FINITE REALIZATION

Let v(t)=min((1+t)/2,57/64-3t/2), e=57/224, Lambda=153/256. The normalized
root lower functional is
 [3L1+L2+L5+L6-U3-U4-2U7-U8-U9-U10-U11]/4. (4.1)

L1 is the ordinary causal lower start at a, without an empty-prefix h
insertion. L2 is now the causal start at h. In making that change the
entire former H248 least-prime integral was removed. More explicitly the
change from w*=25/224 to w has Chen increment
 [B(w)-B(w*)+I(w)]/4-integral_{w*}^w 2H248((1-t)/(2t))dt/[t(1-t)],
 B(w)=4a_bar((249/448)/w)/(249/448)-8a_bar((1+w)/(2w)).
This follows from the exact least-prime identity and a_bar'(q)=A(q-1)/(q-1);
at w=h there is no surviving old H term to add again.

U3/U4 use H430 at s=max(2,L/a), L=v(t)-t on t<e, and the U3 tail
uses the limit of level 1/2 from below. The upper physical cutoff is
min(a,L/2), so its clamp s=2 is a real positive-measure state and is kept.
L5/L6 use h248 on their admitted pair domains, decreasing t>u, at
 L=V-t-u, V=max(v(t),v(t+u),Lambda-3a/8) for t<mu, and V=v(t) otherwise.
Only L/a>=2 contributes. All their squares, stable ceilings and alpha>=2
are checked by 2a>h and V<=(1+t+u)/2. Below 2 the lower option is zero.
U7/U8/U9 retain their exact positive labelled product masses and use U.
U9 multiplicities are not replaced by an indicator. Their unit products
have fixed exponent below 1 and hence negligible mass.

U10/U11 use on one common positive four-prefix partition the minimum of
the switched kernel K omega((1-S4)/z)/z and the eligible ordinary upper
kernel 4(1-H248(max(2,L/w)))/L, decreasing x>y>z>w. Every ordinary option
passes ALL full and selected squares and retains ceiling w; its argument
is <3. Saved parent refinements replace their old parents; overlapping
options are not added. The U11 family is subsequently restricted in section 5.

All ideal ordinary starts are realizable simultaneously. With
d0=249/448-3h>0, the third/fourth/fifth small-last-prime tails are
O(epsilon*(1+log^(r-1)(h/epsilon))). Ordinary Fbar/fbar differences cancel
the last-coordinate inverse-square singularity. On the remaining compact
domain freeze each cap at its own prefix, retest child history at that
actual cap, and preserve every inherit label. Proper affine walls have
vanishing mass; open equality/inherit regions are not discarded. Literal
raw subtree differences converge together by bounded convergence. Current
restoration slabs are separately licensed old first exits as in section 2.

The same nested finite refinement realizes the table graph. All kernels
have fixed positive denominators. One may bound each transition's positive
operator mass by 12 and the root coefficient norm by 7/2. Thus a local
oriented error delta propagates at most delta*(12^441-1)/11 before the
root normalization. Choose delta first within the final prescribed budget;
there are then finitely many queries. Zero-mass pieces are controlled by
absolute errors, not division by Theta. This proves common finite
realization; finite depth alone would not discretize continuous children.

5. THE DELETION IMPROVES THE SAVED FUNCTIONAL IN THE RIGHT DIRECTION

On the nonenhanced pair512 rectangle t in[mu,d], u in[a,h], put
R=57/64-(5/2)t-u. On each B=[tl,tr]x[ul,ur] let Rmax=R(tl,ul),
smax=Rmax/a and M_B=log(tr/tl)log(ur/ul). E0 has smax<2 throughout,
so its saved L6 lower floors are zero. The EXACT selected E1 has 227
crossing cells with zero saved floors by the source guards, and 11254
wholly positive cells with 2<=smin<=smax<=9/4.

On[2,9/4], F_lower(s)=log(s-1)/s is increasing. The saved ordinary
cell floor, including its alternative concave chords, is no greater than
(4/a)F_lower(smax)M_B. If H_B is the minimum h248 over the exact closed
argument support, the saved defect floor is <=4H_B M_B/Rmax: its directed
support contains the exact support and its denominator is Rmax. Hence the
discarded SAVED lower bound, in Chen-quarter units, is at most
 sum_{E1 positive B} M_B[log(smax-1)+H_B]/Rmax <= .002372094. (5.1)
This is deliberately not an upper bound for the actual removed Y6 count.

The same selected parents are all switched in the CURRENT n48 U11 cover;
none meets the priority40000 replacement. For E1 there are 61956 such
parents, and E0's corresponding 34405 parent check is also retained.
For a touched parent B let Z_B=int_B dmu4/z, K_anchor=6.7619 and g_B its
saved saving. Its source proof gives
A_B=K_anchor C_omega Z_B-g_B>=K_anchor omega_B Z_B. The frozen K feedback
has 0<=s_B<=A_B/K_anchor, so at current K,
A'_B=A_B-(K_anchor-K)s_B>=(K/K_anchor)A_B.

Thus the EXPLICIT density A'_B/(Z_B z) majorizes K omega(actual)/z on B,
preserving its old saving. All current omitted pairs satisfy a full-square
obstruction to every ordinary cap<=Lambda; the independent exact witness
intersection additionally verifies that no actual competing parent is used.

Throughout the U11 domain (1-S4)/z>=77/38>2 and omega>=1/2. Apply U to
the complementary positive family on a common refinement. Restricting this
majorant removes at least K I_E/8 in Chen units, where
 I_E=int_{(t,u) in E} int_{a<w<z<u}dt du dz dw/(t u z^2 w).
This subtracts majorant MASS, not an upper bound for an actual count.
For a rectangle B its inner integration is exactly
 log(tr/tl)*[log(ur/ul)/a+(log(ur/a)+2)/ur-(log(ul/a)+2)/ul].
N proves the safe E0 gross .002095161 (already adopted) and E1 gross
.003799514. Subtracting(5.1), the additional E1 raw improvement is
 G1=.001427420=71371/50000000. The two sets are disjoint. The retained
L6 is its old sum of author512 cell floors minus 4*.002372094; the
enhanced half is unchanged. No global maximum is given an unjustified
cell decomposition. Future K slopes remove GROSS/K, not net/K; the
present theorem keeps K fixed and needs no future optimization.

The mask introduces only finitely many proper exponent walls. Positive
smoothing, fixed refinement and U on the complement make their error
arbitrarily small. All prime exponents remain bounded below by a here.

6. THE EXACT NUMERICAL LEMMA, ALLOWANCES AND ALL-EVEN LIMIT

N concerns (4.1) with precisely the choices in sections 2--5: the current
causal starts and restored cover; H430 only in U3/U4; h248/ordinary H248
elsewhere; the retained five positive-cost K feedback; current n48/priority
minimum cover; and the paired E0 union E1 crop. Its certified RAW lower
endpoint is
 R=134557512953000373957168143230118885731275473 7 /
   685738127494661151198444592354557952000000000.   (6.1)
Its provenance is positive-pair-deletion-improvement-assembly-20261010.json.
It equals the preceding zero-mask raw endpoint plus G1, with no old table,
restoration, tail or E0 gain added a second time. N's requirements are the
one-sided integral inequalities, legal domains and finite recurrence
arrays; a digest or a successful process exit alone would not prove it.

The retained total lower density/finite-approximation allowance and lower
compiler/smoothing allowance are EACH
 B=250474831809194280822969421937979930755098 7 /
   548590501995728920958755673883646361600000000 0.   (6.2)
This is the old allowance plus G1/8, separately for each role. They include
the previous costs; they are not to be added to the old allowances again.
Sections 2--5 realize one finite lower proof within these strictly positive
allowances. The prime asymptotic errors are o(H_T) after every fixed choice.
Equations(2.3), (2.5), the finite table inequalities, (4.1) and section 1
therefore give
 D12(T)/H_T >= R-2B+o(1)   (6.3)
for every sufficiently large even T. Exact rational subtraction yields
 R-2B=537979576980192301547849603498537562994346796 1 /
       27429525099786446047937783694182318080000000000,
 R-2B-12258223/6250000
      =24854508869490553526005651070383185 21 /
       27429525099786446047937783694182318080000000000 >0.   (6.4)
This proves(C), and also the eventual inequality D12(T)>=1.96131568 H_T.

The quantifier order is essential but now explicit. Choose the finite
query graph, compact/tail accuracy, causal caps and support reserves;
then common beta/rough and fine-label parameters; then actual finite
divisor orders, the current permitted kappa (or a smaller one), and the
analytic logarithmic accuracy. Only then let T increase through ALL even
integers. Each distribution remainder is absolute, not an average in T;

```
all finite constants and thresholds can be maximized. C_T>=C_2>0 makes
fixed power savings and O(T/log(T)^A), A>2, negligible relative to H_T.
No effective threshold, almost-all substitute, or policy depending on T
is used. The strict number in(6.4) absorbs the remaining o(1).

CONCLUSION AND AUDIT BOUNDARY

The own lower theorem (C) is discharged by this modular proof once P, U,
D and the explicit certified functional N are inserted. It is not left
as an assumed Chen lower estimate. D, the pointwise weight, lower compiler,
finite typed induction, actual root insertion, paired restriction and
finite all-even realization have concrete derivations above. No additional
lower-side inequality was found missing or invalid on this selected scope.
Verification of P/U's own complete proofs and N's complete numerical lemma
belongs to their separate modules; this report does not silently label
that separate work as done. Neither formalization nor outside endorsement
is an extra mathematical hypothesis. This file does not settle frontier
priority, change the selected b, alter the main TeX, or authorize publication.
```

H The sharp-prime endpoint and lower interfaces

Source: final-lower-cross-review-20261010.txt.

```
Independent upper-author cross-review of the current Chen lower completion
10 October 2026. New proof addendum and source review; earlier files unchanged.

Result

I found one omitted justification in global-chen-lower-completion-20261010.txt:
equation(2.5) uses a SHARP original prime endpoint, whereas the current PP
theorem starts with a smooth original weight. Its permitted sharp individual
HB intervals do not themselves prove that original-prime assertion. Section1
below repairs the step explicitly, with arbitrary logarithmic saving and no
change of coefficient, K, c, density allowance or numerical query.

The exact-coprime O(log^2 T) correction, the finite W1/W2 transition norm12,
the simultaneous continuous-child approximation, the lower projection signs,
the complete displaced L2 correction and the saved-floor paired loss all
check out with the qualifications/proofs below. After the endpoint lemma is
inserted, no unmatched obligation remains among these audited lower interfaces.
This review uses the separately proved PP and U and the explicit selected
numerical inequality lemmas; it does not redo their spectral or quadrature
proofs. It is independent of the author of the lower-completion report.

1. The missing sharp-singleton lemma, now proved

Let the ACTUAL coefficient rho_T have |rho_T(q)|<=C, squarefree support
q<=T^theta, theta<1, (q,T)=1, with C fixed before T. Suppose the current
smooth-singleton PP holds, with arbitrary O_A(T/log^A T) error for its
allowed polylogarithmic Euler-derivative norms. For even T>2 it implies

 sum_q rho_T(q){pi(T;q,T)-pi(T)/phi(q)}=O_A(T/log^A T).      (S1)

Here pi(T) may count p<T or p<=T, which coincide for these T. This is
exactly(2.5), and it also applies to the bounded marked scalar classes.

Fix desired A. Put delta=log^(-B)T, with fixed B>A+3. Smooth the original
prime endpoint in [T-delta*T,T+delta*T], using a fixed smooth transition
rescaled by delta. Its derivative of order j in p/T is O_j(delta^(-j)),
a FIXED logarithmic power. Multiply by the already allowed smooth lower
trim eta(p/T^(1-kappa0)), which equals0 below T^(1-kappa0) and1 above
2T^(1-kappa0). The resulting W_T is supported on primes of a fixed positive
power, product O(T), and has the precise polylog derivative norms allowed
by distribution-prime-product-completion section1, lines25--32. Smooth
PP therefore applies to W_T with this same rho. Choose its local log/SW
accuracy after B and the fixed derivative orders, as in that proof's
sections3 and10. No parameter of the physical finite sieve changes.

The upper boundary error can be bounded AFTER summing all moduli, without
postulating a signed sharp-domain PP theorem:

 sum_(prime p,0<|p-T|<=delta*T) sum_(q|T-p)|rho_T(q)|
 <=C sum_(prime p,0<|p-T|<=delta*T)tau(|T-p|)
 <=2C sum_(1<=n<=ceil(delta*T))tau(n)
 <<delta*T*log T+log T.                                  (S2)

The elementary last estimate follows from sum_(n<=X)tau(n)=
sum_(d<=X)floor(X/d)<=X(1+log X). The excluded n=0 means p=T; it never
occurs because T>2 is even. Thus no tau(0), extra T^theta or signed
boundary-positivity argument is hidden in(S2).

The exact-coprime mean boundary is at most
 C*# {integers p in the strip}*sum_(q<=T^theta)1/phi(q)
 <<delta*T*log T+log T.                                  (S3)
```

One can ignore primality for this bound. The elementary reciprocal-phi sum is O(log T), obtained by n/phi(n)=sum_(d|n)mu(d)^2/phi(d) and the convergent sum of mu(d)^2/[d phi(d)]. With B>A+3, (S2),(S3) are O_A(T/log^A T).

The small-prime trim costs at most
C sum_(p<2T^(1-kappa0))tau(T-p)=T^(1-kappa0+o(1)),
and its exact mean at most O(T^(1-kappa0)log T). Here T-p is nonzero and <=T, so the usual pointwise divisor bound suffices. These fixed power savings are smaller than every prescribed logarithmic error. Removing both smoothing changes thus gives the sharp discrepancy with mean pi(T)-omega(q): all q<T, so every prime divisor of q lies below T.

Finally squarefreeness and |rho|<=C give

|sum_q rho_T(q)*omega(q)/phi(q)|
<=C sum_(prime l<=T^theta) 1/(l-1)
*sum_(e<=T^theta/l)1/phi(e)
<<log T*sum_(l<=T^theta)1/(l-1)<<log^2 T. (S4)

For the last crude bound, summing over all integers rather than primes already suffices. Equations(S2)--(S4) prove(S1). Thus the O(log^2 T) statement in C is correct, but the passage to its sharp pi needed this separate endpoint argument. A positive fixed-tolerance smoothing alone would not have justified it for a signed lower weight.

The lower pointwise count sometimes restricts original p to (p,T)=1. There are O(log T) omitted primes dividing T, and their total compiled divisor contribution is T^o(1), since p<T and |rho|<=C. This harmless power-small adjustment does not change (S1) or its normalization.

2. Lower projection and beta signs: checked as actual inequalities

own-lower-compiler-chain-audit-20261009.tex, section4, lines146--193, proves
V_T(z0)R_rough-V_full
=sum_(full EVEN exits,last prime<z0)g_T(P)V_T(p_last)>=0. (L1)
The finite tree's first exits are even. Restricting it to rough primes removes negative exit contributions, so the direction is indeed >=. It is different from the upper projection direction used in my U proof.

Positive even rough coefficients are rounded inward, negative odd ones outward, yielding c<=lambda coefficientwise. With B-<=I0<=B+, attaching B- to c>0 and B+ to c<0 gives, term by term,
c(d)B_sign(n)<=c(d)I0(n).
Since I0>=0, summing produces the actual rough-indicator minorant. The beta-minus polynomial need not be nonnegative. Unique small/rough division keeps the physical coefficient bounded; later atom decompositions do not replace it. Tagged-prime boxing bounds the ABSOLUTE density discrepancy, so (L1) gives sum rho/phi>=V_full-small error, as required. Coefficient order alone would not prove that density direction, but C includes (L1).

Combining this pointwise minorant with (S1) yields the original-prime lower count. PNT, literal raw density and V_T(T^t)~2exp(-gamma)C_T/(t log T) give the factor4 on Fbar/fbar exactly once. This is not an application of the positive-product upper K to obtain a lower prime count.

3. Positive transition norm and continuous children

I checked the actual positive measures and rational bank gates in own-chen-table-insertion-chain-20261009.tex, lines227--353, against lower-joint-typed-density-bridge-20261010.txt, sections2--4.
The child/parent normalization is du/[u(1-u)] for one new prime and du dv/[uv(1-u-v)] for two; its derivation keeps Q fixed. Alpha becomes 1+(alpha-1)/(1-u) or1+(alpha-1)/(1-u-v), so it never resets to2.

```
The following uniform operator masses are valid:
 BH <=integral_1^5 dw/w=log5<2;
 W1 <=1+(1/2)integral_(1/5)^(1/2)du/[u(1-u)]=1+log4/2<2;
 W2 <=1+(3/5)log4+(4/5)*125*(3/10)^2
     <1+6/5+9=56/5<12.                                  (T1)
For every good W2 pair domain, the bank gates give1/5<=u,v<=1/2 and
1-u-v>=u>=1/5, so its kernel<=125. The whole containing square has
area(3/10)^2; there are four domains. Singleton intervals are inside
[1/5,1/2]. Unchanged-prefix weights(4+1)/5 sum to1. The bad part of
Gamma5 and Gamma9--21 are fixed positive COST bases, not extra feedback
edges. The existing variable-kernel saving is already in that base.
Hence the norm12 does not omit a feedback term or divide by5 twice.

Finite maxima/minima of oriented bounds are1-Lipschitz. If each primitive
query and each local geometric approximation has error<=delta in its
parent normalization, e_(j+1)<=delta+12e_j, e0<=delta, yields
e_j<=delta*(12^(j+1)-1)/11. The eleven outer absolute coefficients sum
to14 before the sole Chen quarter, giving root norm7/2. This proves the
coarse amplification asserted in C, without numerical stability claims.

There is also a genuine finite approximation argument, beyond finite
depth alone. The 36 bank tuples have min s=101/50 and s0<5; BH retains
L/2, W1 retains >=7L/12 and each W2 child >=L/101. With depth<=440,
L_min=ell*101^(-440)>0; appended prime exponents are >=L_min/6.
Thus denominators and all switched positive cutoffs are fixed positive.

Refine physical appended exponents recursively. On each already chosen
positive parent, choose child approximation error within delta times
that parent's fixed normalized mass. Finite piecewise-rational cap and
table-knot equations have null proper walls on each nonidentical branch;
identical branches are retained. Structural clamps s=2 and an unchanged
inherit cap may have positive measure and are kept with the conservative
closed-cell value. Ancestor caps remain functions only of their own prefix.
Only the child's walls are trimmed, not their possibly full parent
projections. Ordinary compact integrands have bounded denominators;
positive switched leaves use the proved positive outer approximation.
By finite depth this gives a finite set of geometric queries, with total
error controlled by(T1). For its finitely many positive-mass vertices,
Theta_v/H_T tends to a positive constant; absolute o(H_T) PP/U errors
therefore become o(Theta_v) after T grows. Zero-mass pieces stay absolute.
This order avoids assuming a uniform error over an undiscretized continuum
or dividing by a normalization which tends to zero.

4. The entire old L2 H248 term is removed

The exact least-prime identity, not a difference of unrelated bounds, is
S(A,T^h)=S(A,T^w)-sum_(T^w<=p<T^h)S(A_p,p). The derivative identity
 d/dt a_bar((1+t)/(2t))=-A((1-t)/(2t))/[t(1-t)]
in own-chen-table-insertion-chain lines459--500 gives

 L2_new(w)=B(w)+8a_bar((1+h)/(2h))+I(w)
       +8 integral_w^h H248((1-t)/(2t))dt/[t(1-t)].       (Q1)

After the Chen quarter, the change from w*=25/224 is
[B(w)-B(w*)+I(w)]/4
 -2 integral_(w*)^w H248((1-t)/(2t))dt/[t(1-t)].          (Q2)
At w=h the residual positive integral in(Q1) is empty. It is the ENTIRE
248 correction of the old saved L2, not a184/seed part. I checked the
actual code: certify_sharp_184_baseline.py lines131--137 includes
df.l2_correction(), while its Defects(bank) loads the chosen bank's
H array. certify_combined_chen_functional.py lines90--97 computes that
correction with weight8 times the exact logit mass on [w*,h]. Dividing
once by4 gives precisely the weight2 in(Q2). The use of table430 only
at U3/U4 does not put any old L2 correction back.
```

5. The paired loss and retained negative majorant

The selected author512 L6 estimate is a sum of its pair_cell floors:
I checked sharp-248-pair-refinement-512.json and found chosen L6 equals
its recomputed512 total, not the alternative128 total. On the nonenhanced
half its sole residual is R=57/64-(5/2)t-u. In certify_sharp_184_baseline.py
lines50--96, the ordinary floor is the maximum of its constant and
eligible concave-chord lower floors; every chord requires pl>=2a.
The defect is floor(4*hlookup*mass/Rmax). SavedDefects.lookup returns0
when the outward lower argument is below2 and otherwise takes the
minimum over ALL closed cells meeting the directed interval.

Consequently crossing cells smin<2 have exactly zero saved floor: the
constant is0, the sole affine chord is ineligible, and hlookup is0.
I separately solved the rational grid equation s=2 and found no grid
endpoint equal to2. On positive selected cells 2<smin<=smax<=9/4,
fbar(s)=log(s-1)/s is increasing, so each legal ordinary floor is at
most(4/a)*fbar(smax)*M_B. The directed argument interval contains the
exact one; its h minimum is <=the exact closed-support minimum H_B.
Thus saved_floor/4<=M_B[log(smax-1)+H_B]/Rmax, exactly(5.1).
This bounds the discarded SAVED lower proof; it is not an upper bound
for the actual Y6 count. It applies to a max of cell estimates because
each candidate individually has that upper bound.

For the negative side, certify_sharp_248_four_min.py lines137--154 gives
g_B=max(0,floor(K0*C*Z_B)-ceil(K0*omega_B*Z_B)) at method0.
When g_B>0, A_B=K0*C*Z_B-g_B>=K0*omega_B*Z_B, a density-majorant
statement. I read the actual 1,190,700-row witness: among the34405 E0
intersecting parents, EVERY floor is positive (minimum integer floor
184409917917). E1's61956 positive-floor parent check is separately bound
in the current foundation report. Thus the stronger omega_B form used
by C has no zero-floor exception on either selected mask. In general
g_B=0 would instead require the baseline bound C>=omega(actual).

The fixed feedback proves0<=s_B<=A_B/K0, so
A'_B=A_B-(K0-K)s_B>=(K/K0)A_B. Therefore A'_B/(Z_B z)>=
K*omega(actual)/z. Apply U on the COMPLEMENT and restrict this actual
parent majorant. Since omega>=1/2 at the actual arguments>=77/38,
the removable Chen-quarter mass is at least K*I_E/8. No upper count is
subtracted as a lower count. The disjoint E0/E1 sets, unchanged priority
parents and proper positive smoothing are the required geometry inputs.
Gross removal, saved-floor loss and future slope remain distinct.

6. Exact checks, interpretation and conclusion

Using the current selected rational fields, I recomputed
R-B_density-B_compiler and its difference from c=12258223/6250000:
24854508869490553526005651070383185 21/
27429525099786446047937783694182318080000000000>0.
No budget is subtracted twice by the endpoint lemma: its error is an
arbitrary logarithmic o(H_T), and all physical finite coefficients stay
unchanged. All moment/profile choices remain fixed before taking every
sufficiently large EVEN T; shrinking logarithmic smoothing is part of
the permitted analytic profile, not a T-adaptive sieve policy.

With section1 appended, C's use of sharp original-prime PP is justified.
The five other requested interfaces are verified by their actual equations
and code, as above. There is no remaining gap on this cross-review's
scope. This is not an independent new computation of numerical lemma N
or of PP's spectral input; their separate completion records remain the
proper proofs of those modules. No main manuscript, old report, research
state, formalization, GitHub object or external message is changed.

I The selected finite numerical inequalities

Source: `global-selected-numeric-completion-20261010.txt`.

```
Selected finite numerical completion map and shorter certificate spine
10 October 2026. New audit; no old artifact, state or main source is changed.

Conclusion and exact scope

The selected finite numerical premises for K=6.7617368,c=1.96131568 have
enclosure proofs and complete output/selection coverage in the current files.
The accompanying obligation map covers ALL numerical components actually
entering those two selected inequalities, rather than a representative sample
of fourth/eighth integrals. No unmet selected finite numerical item was found.

This is a conclusion about finite numerical inequalities, given the specified
analytic functions and density functionals. It is not a proof that those
functionals equal the physical sieve densities, that the finite H/h actions
are analytically admissible, or that the required signed distribution theorem
holds. Those are separate substantive theorem requirements. Nor does this
audit invent a requirement for a new independent implementation of every
general-purpose routine. A proved algorithm and a complete corresponding
run suffice; separate implementations are useful corroboration, not axioms.

The new contribution is more than assembling old success labels:
 (i) a direct four-component upper expression minus22 final selected savings
     reproduces the current exact raw endpoint, without replaying historical
     safe constants or counting alternative refinements;
 (ii) a direct eleven-term lower expression with the six current I3/I4/I5
      values and the current restoration reconstructs a bound slightly ABOVE
      the conservative selected raw endpoint;
 (iii) one fresh430-round simultaneous map, beginning only with the original
       certified seed, reproduces ALL400 selected430 entries exactly. No
       intermediate248/312 generation arrays are inputs to this replay.

What is freshly run versus read

The new script imports only the Python standard library. It recomputes direct
rational ledgers, endpoint directions, reserves, count margins, seed rounds,
and all430 table transitions. It checks actual full-run count fields and
bound-file identities. It does NOT rerun large quadratures or every simplex
floor. Those are supported by their existing complete result records together
with the enclosure and partition algorithms identified in the obligation map.

Source examination here covered the directed arithmetic/series arguments,
ordinary interpolation and primitive interface, representative generic exact
polytope verifier, pulling triangulation and weighted moments, the selected
builders' complete finite sums, and the current selection/ownership formulas.
In particular the exact polytope algorithm does not accept a floating hull:
positive normal balance plus full rank proves boundedness; all active bases
of the selected facet system are solved exactly; every vertex satisfies every
original row; the intersection face lattice is reconstructed; reverse pulling
triangulates the convex polytope. This is the common algorithm for the entire
selected affine-cell family. Its complete output is needed separately for
every selected cell. The map names every such result family.

The obsolete bounded report and inventory were used to locate evidence, not
as blanket proofs of completeness. In particular all later replacements were
followed before treating an old shared-backend or partial-run statement as a
live gap. Every direct source binding used by the new report is rechecked.
Hashes identify the inspected evidence; they do not establish its mathematics.

1. The exact upper numerical spine
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```
Let U0=4/Theta+I1+I3_compact+I3_tail, with Theta=299/500, rho=19/96
and compact split z=1/100. The selected tail is the newer full-tail upper
bound, not the old tail plus a saving on a different domain. Then subtract:

 1 replacement fourth, all3699127 selected boxes;
 2 replacement old fifth,292633 parents and18728512 children;
 3 disjoint new fifth,1763420 boxes;
 4 replacement old sixth,632104 cubes;
 5 coarse above-mu rank2 fallback (not its larger fine alternative);
 6 above-mu rank3 fallback,4096 cells;
 7 final log-moment total on the original eleven new sixth cells;
 8 the two further descending33 sixth cells;
 9 ascending34 actual-parent difference;
10 descending34 actual-parent difference;
11 lower clip window;
12 lower descending34 window;
13 the SELECTED local cap corrections, not their larger all-replayed total;
14 four nonsingleton sixth domains;
15 radial maximum on the original eighth seed;
16 fourteen further ascending eighth domains;
17 fifteen descending eighth domains;
18 final log total on the fifteen unequal-parent eighth domains;
19 final log total on the thirteen ascending-third eighth domains;
20 final log total on the twentynine history eighth domains;
21 seed curvature increment over its already adopted log bound;
22 seed mixed increment over its already adopted curvature maximum.

Their source fields and exact fractions are in selected_upper_direct_savings.
Direct subtraction gives exactly
 E=9979702576710911234850619807816847864907359959/
   14759135509622012279467806190993532151398400000.
This agrees with the selected raw upper, not merely its rounded display.

All22 terms are lower bounds for positive improvements. Their selected cover
may be a proper subset of possible profitable regions: proving the omitted
gain nonnegative is sufficient. This is different from negative-cost upper
integration, for which the whole retained domain must be covered. Same-node
upgrades use the actual current cap, or a proved upper comparison that makes
the improvement smaller. Old/new sixth overlap with eighth cylinders is
handled by telescoping the actual cap differences. Within the87 eighth cells
the separate30+15+13+29 partition and pattern/half/history separators prevent
duplicate eighth integration. Navigation exclusions need not prove that the
chosen positive subsets exhaust every possible improvement.

Primitive coverage for U0 is also complete. I1 has a full positive series with
an explicit tail. Compact I3 has a complete separately reconstructed covering
partition whose upper endpoint is at most the selected upper. Boundary boxes
are kept with conservative upper mass. The noncompact tail uses
D(s)<=D(10)(10/s)^2 and is integrated over the ENTIRE z<1/100 region. It is
not a sampled small-z cutoff. Ordinary nodes and arbitrary rational queries
have the numerator-chord, shape and curvature proof; shared Q160 logs have
explicit series remainders. The old two-million heap need not be the selected
proof when the complete alternate integral covers the same upper allowance.

The two upper error reserves are reconstructed from the one post-fallback
reference and one eighth-of-each-subsequent-saving increment. They are equal,
positive and retained. The exact inequality is E+u_density+u_compiler<K,
with the already selected remaining slack about9.6522301975e-8. No Chen quarter
is introduced in these upper-density units.

2. A table proof from the seed directly to the final arrays

The seed numerical certificate has an ordinary lower value>.020249>.02 and
a positive feedback coefficient>.13. Consequently four exact applications of
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J -> .020249+.13 J, beginning withJ=.02, give
 .023273777353 > 2327/100000.
This establishes the input seed by finite arithmetic. Its typed-state
admissibility remains the separately proved analytic seed theorem.

Take the final106 action constants at K_anchor=6.761767:70 W1,3 originalW2,
and33 expandedW2. Their ordinary base, positive switched mass, variable-kernel
gain, feedback weights, target lists and anchor corrections all have the
complete evidence listed in T2--T4. The subtraction of the kernel slope is
kept in W2 recalibration. No action is silently evaluated at a different K.
The actual selected K=6.7617368 implies this weaker positive-product anchor.

On the200 closed cells with endpoints1+i/50 initialize
 H_i=floor(2^80*2327/100000) if the right endpoint<=11/5, otherwise0;
 h_i=0.
Apply430 times the final simultaneous positive action map. Every right side
reads only the preceding H,h arrays. Positive products are individually
rounded down, W2 feedback is divided by5, and the previous value is retained
by a maximum. Buchstab tails use the right endpoint of the target cell;
lower h remains zero below2. Thus each step is a valid finite lower-bound
propagation once the action identities and typed closure are available.

Fresh exact result:
 round248: all400 selected248 entries dominated;350 strictly;
 round312: all400 selected312 entries dominated;350 strictly;
 round430: all400 selected430 entries EQUAL, including every zero cell.

No selected intermediate array is used to generate the new arrays. They are
read only at the three comparison points. The final array hash equals the
original selected arm's hash. This provides a short numerical proof of the
selected arrays without an obligation to walk every historical24->56->120
->184->248->312 continuation. It is still a FINITE430-round proof, not an
infinite fixed-point assertion. No new larger table or gain is adopted.

Complete primitive counts:70 W1 bases,10 J queries,200 Buchstab weights,
1858 W1 weights;36 W2 bases,216 components,468 closed masses; original3 pair
feedback has2400 polygon queries/628 positive weights plus424 singleton
weights; expanded33 has12624 pieces/5224 pair weights/3626 singleton weights.
The variable-kernel covers have559560 and6089150 boxes. The original3 pair
backend gap has its later centered-series proof/output. The shared lower
ordinary profiles have all23199750 node intervals checked by the separate
chord construction, plus general query/interpolation proofs.

3. A direct lower expression with no obsolete patch

Use the current lower endpoints L1,L2 from the248 baseline, selected pair512
L5,L6 lower endpoints, baseline U3,U4 upper endpoints, selected new1024 U7,U8,
U9 upper endpoints, and the complete n48 U10+U11 upper cost. Define

 C11=[3L1+L2+L5+L6-U3-U4-2U7-U8-U9-(U10+U11)]/4.

This formula uses neither the old n32 minimum nor its patch/logmean gain.
Its exact value exceeds the conservative baseline-after-refinements by
1/241785163922925834941235 2. The n48 bound covers all1190700 ordered rows,
including tied-bin simplices; the count is C(51,4)+48*C(50,3). Its ordinary
and switched alternatives enter as one maximum saving. Every one of40000
priority parents is replaced by a maximum with its complete child sum;
all516779 children are covered. The fixed K slope excludes these parents.

Let B be the independent ordinary-start-change lower endpoint and Hrem the
whole removed H248 upper endpoint. Put

 C_at_K0=C11+priority_gain
   +[B-Hrem+3I3(a)+I3(h)+3I4(a)+I4(h)+3I5(a)+I5(h)+restoration(h)]/4.
```

```
All six I values are the current selected values. Specifically:
 - I3: full M1600 result limit0, both cutoff domains, actual selected totals
   covered by the independent arrangement method;
 - I4(h): the smaller independent M800 total, plus disjoint75609 bands;
   I4(a): complete6545-cell/420-positive-parent all-mask evidence;
 - I5(h): all-new158251 floors, excluding160 obsolete pre-Jensen parents,
   plus all23662 slab/tied additions; I5(a):679 full integrals;
 - restoration: selected slanted98668-box/789344-layer total. It replaces
   the old whole-box restoration and lies on the old rejected-node region.

Numerical omitted histories/boundary strips are permitted only for these
nonnegative lower gains. Tied boxes are not assigned a fictitious factorial
symmetry for an asymmetric integrand: the sorted actual target is symmetric,
and a valid full-box bound is obtained before division by the exact factorial.
Inner slices, log moments, ordinary primitives and all used argument ranges
have the explicit enclosure proofs identified in L4--L7.

Write A for the proved fixed five-cost slope, T312 for only the248->312
U3/U4 difference, and T430 for only the312->430 difference. The complete
current direct raw lower is

 C_at_K0 + A*(6.7619-K) + T312 + T430
             + .002095161 + .003799514 - .002372094.

The resulting exact rational is slightly above the stored selected raw.
The companion JSON records the exact surplus. It is not selected as a new
improvement; keeping the smaller historical raw value is conservative. This
identity exhausts every currently used numerical gain in c. It does not add
any old narrow seed contribution, repeated table generation, old n32 patch,
fifth alternative, or change of the U10/U11 table profile.

4. Paired masks, reserves and transfer

The6091 zero-floor and11481 new positive-net pair512 cells are disjoint.
Their actual whole-parent cover, including priority exclusions, is recorded
completely. Every new gross lower exceeds.003799514; the summed discarded
saved lower-estimator upper is below.002372094. These are leading functional
bounds. The loss is not an upper estimate for the actual removed Y6 count.

The full n48 ownership/remainder certificate supplies the needed restrictable
majorant after K feedback. It checks all1011131 retained switched rows. Its
identity A_B/K0>=s_B gives
 A_B-(K0-K)s_B >= (K/K0)A_B >= K*omega_B*Z_B.
The paired change therefore removes a majorant's positive mass on a fixed
subdomain; it never changes an upper bound for a removed count into a lower
bound. The geometric paired injection itself is outside this numerical audit.

Each new lower reserve retains the original table312 allowance and adds
[A*(6.7618-K)+T430+.002095161+.003799514-.002372094]/8.
Both reserves agree exactly with the currently selected fields. After BOTH
reserves the conservative current raw exceeds1.96131568 by the selected
positive rational slack about9.0612246399e-10. The larger direct reassembly
is not needed to obtain this strict inequality.

Future sensitivity is A-(.002095161+.003799514)/K. The positive-floor loss
is not K-linear, so subtracting the net gain/K would be incorrect. Fresh
192-term rational Mercator enclosures also give the two final count surpluses
about7.1260773775e-8 and3.4146633105e-8, respectively. Float displays decide
none of these signs.

5. What this resolves and what it leaves to the global proof

The finite obligation list P1,U1--U13,T1--T5,L1--L9,S1 is the requested
```

requirement-level numerical map. For each entry it supplies the mathematical enclosure/geometry method and the actual complete corresponding output. Every raw upper term, lower term, selected difference, table primitive, fixed mask, positive floor charge, and reserve appears in the direct ledgers. Full selected coverage does not mean exhaustive search of possible improvements.

There is no concrete failed numerical primitive, uncomputed selected term, unpaid numeric deficit, uncovered negative-cost region, or duplicated selected gain left by this map. This result should replace the earlier statement that only a bounded representative numerical audit is available. It should NOT be presented as a new simultaneous rerun or formal verification of all historical computations, or as a certificate for upstream analytic requirements.

The global theorem still needs its mathematical identification of these exact functionals: ordinary/Buchstab identities, raw same-weight density, W1/W2 mass and state derivations, compatible finite coefficient policies and PP, Chen pointwise weight and paired injection, finite approximation before the all-even limit, and final normalization. The companion map names these as outside this numerical subtask, not as asserted open gaps in the other agents' proofs.

No current constant, selected mask, table entry, budget, analytic policy, main source, formalization status, GitHub repository or external message is changed.

J The ordered bad-semiprime mass and final transfer

Source: current-downstream-proof-spine-20261010.txt.

```
Current downstream proof spine: fixed-b theorem and current all-even bounds
10 October 2026. Standalone implication; the current upstream inputs remain explicit.

Theorem. Put K=8452171/1250000=6.7617368 and c=12258223/6250000=1.96131568.
For even N>2 let C_2=prod_(p>2)(1-1/(p-1)^2),
 C_N=C_2 prod_(p|N,p>2)(p-1)/(p-2), H_N=C_N*N/(log N)^2.
Here p,q are prime and r is prime except where r=1 is expressly allowed;
Omega counts factors with multiplicity.
Define D12(N)=#{p<N: N-p>1, Omega(N-p)<=2}; define
 D_b(N)=#{p<N: N-p=r*q, q prime, r=1 or prime, r<=q^b},
 B_b(N)=#{p<N: N-p=r*q, q^b<r<=q}.
Each set counts p once. Assume the following TWO analytic conclusions.

(C) There is e_C(N)>=0 tending to zero along ALL even N such that
 D12(N)>=(c-e_C(N))*H_N.
The intended current source is the retained paired Chen weight
 W_E=W_full-Y6(E)+Y11(last pair in E), E=E0 disjoint-union E1,
with 6091 old zero-floor cells and 11481 new pair512 cells. In the full weight,
 W_full=3Y1+Y2+Y5+Y6-Y3-Y4-2Y7-Y8-Y9-Y10-Y11.
Its single global divisor is four. Hypothesis (C) requires the actual count
inequality and the lower estimate for this weight, not merely a scalar ledger.

(U) For each fixed positive smooth outer approximation required below,
one finite construction gives the SAME real coefficients rho_N(d) in (i)-(iii):
 (i) support is squarefree d|P_N(N^r0), d<=N^theta, |rho_N(d)|<=M,
     where P_N(z)=prod_(ell<z,ell prime,ell does not divide N)ell,
     M is fixed, theta<1 and 0<r0<1; for every integer n,
     A_N(n):=sum_(d|n)rho_N(d)>=1_((n,P_N(N^r0))=1)>=0;
 (ii) S_N:=sum_d rho_N(d)/phi(d)<=(K+e_U(N))*C_N/log N,
      with e_U(N)>=0 tending to zero;
 (iii) writing X_W=sum_(r,q)W_N(r,q), X_W^(d)=sum_((rq,d)=1)W_N(r,q),
       sum_d rho_N(d)[sum_(rq=N mod d)W_N(r,q)-X_W^(d)/phi(d)]
       =O_(A,W)(N/log^A N), for arbitrary fixed A>2 after summing charts.
This is required only for the actual licensed finite smooth positive families:
r,q>=N^a/2 for fixed a>0, rq=O(N), with bounded normalized derivatives.
If (iii) is available only on complete-product scales near N, use the fixed
cutoff rq>=N^(1-kappa0) below. All constants are uniform over even N.
The current high-level licence, stronger than theta<1, is an UPSTREAM input;
this theorem does not infer (iii) from support or allow arbitrary restrictions.

For every fixed 1/2<b<1 the hypotheses give
 liminf_(even N) D_b(N)/H_N>=c-K*log(1/b).
Thus every fixed coefficient strictly below this number is attained eventually.
In particular, for every sufficiently large even N, simultaneously,
 D_(74821909/10^8)(N)>=10^-5 H_N and D_(74821809/10^8)(N)>=10^-6 H_N.
Also D_(9/10)(N)>=1.24889 H_N>1.248 H_N. All sets are nonempty.
The smaller b is the endpoint improvement; the .9 bound is a count comparison
at a common exponent. No effective threshold or literature priority is asserted.

Proof. First C_2>0 since sum_(p>2)(p-1)^(-2)<infinity; each extra factor
in C_N is >=1. Hence C_N>=C_2 and H_N tends to infinity uniformly along
even integers. In particular N/log^A N=o(H_N) for A>2, as is N^(1-delta+o(1))
for every fixed delta>0. No uniform upper bound on C_N is needed.

For completeness the usual sieve normalization agrees with this H_N.
Set V_N(z)=prod_(2<p<z,p does not divide N)(1-1/(p-1)) and
M(z)=prod_(p<z)(1-1/p). Direct multiplication of local factors gives
 V_N(z)/(2*M(z)*C_N)
 =prod_(p>=z)(1-1/(p-1)^2)^(-1) prod_(p|N,p>=z)(1-1/(p-1)).
The absolute logarithm is <=(1+log N/log z)/(z-2): bound the first tail by
```

```
sum_(j>=ceil(z))1/[j(j-2)]<=1/(z-2), and count at most log N/log z target
prime divisors in the second. Mertens, with z=N^u and fixed u>0, now gives
 V_N(N^u)=(2*exp(-gamma)*C_N)/(u*log N)*(1+o(1))
uniformly in N. Thus ordinary densities normalize to 4*fbar(s)/u and
4*Fbar(s)/u, with fbar=f/(2*exp(gamma)), Fbar=F/(2*exp(gamma)). The Chen
quarter is applied once to its functional; it introduces no factor in B_b.

We next derive the positive-product consequence of (U), with absolute errors.
For any prime ell, squarefree support and the elementary bound
sum_(e<=Y)1/phi(e)=O(log(2Y)) give
 sum_(d:ell|d)|rho_N(d)|/phi(d)=O(log N/(ell-1)).
Consequently, since W_N>=0, a union bound over its two prime slots gives
 |sum_d rho_N(d)*(X_W^(d)-X_W)/phi(d)|
 <=sum_(r,q)W_N(r,q) sum_(ell in {r,q})O(log N/(ell-1))
 =O(X_W*N^(-a)*log N)=o(H_N), provided X_W=O(N/log N).
The argument includes r=q and has no N^theta factor. It does not require
r,q to avoid divisors of N; coprimality (d,N)=1 is already built into (i).

At primes N-rq>=N^r0, (i) dominates the prime indicator. The ordered count
with 0<N-rq<N^r0 is O(N^r0), because each remainder has at most one ordered
prime factorization. Positive majorization BEFORE (iii), followed by the
preceding exact-coprime correction, therefore proves
 count(domain)<=O(N^r0)+sum_(r,q)W_N(r,q)*A_N(N-rq)
              <=X_W*S_N+o(H_N).
Outward smoothing is safe even outside rq<N by the nonnegativity in (i).
The value at n=0 is a finite divisor sum; with positive-power prime cutoffs
the actual prime products are odd and cannot equal even N in any case.

If complete-product scales are needed, delete rq<2*N^(1-kappa0) first.
Its absolute divisor-sum contribution is bounded by
 O_W(sum_(m<2*N^(1-kappa0))tau_2(m)*M*tau(N-m))=N^(1-kappa0+o(1));
here m<N eventually, so tau(0) is never used. Its coprime mean has the same
power saving after an O(log N) factor. The smooth cutoff has bounded
multiplicative derivatives. On retained charts x is the COMPLETE product
scale, c1*N^(1-kappa0)<=x<=C1*N; no residual quotient replaces x.
For the current moment substitution choose the fixed moment order K_mom,
then 0<kappa<=min(kappa_old,1/(2*K_mom)), and
0<kappa0<min(kappa0_old,kappa/[4*(1+kappa)],1/10). This gives
c0*x<=N<=x^(1+kappa) eventually for the stipulated moving-residue range.
All such choices precede the N limit; no old equality kappa=sigma/200 is needed.

If limsup X_W*log N/N<=J_W, the last positive bound and (ii) give
 limsup count(domain)/H_N<=K*J_W.
Indeed multiply the positive upper bound K+e_U by X_W>=0 and use its
absolute normalized error. No division by J_W, X_W, or a boundary mass occurs:
the conclusion remains valid when J_W=0 or is arbitrarily small.

Now fix 1/2<b<1 and beta=b/(1+b). The actual bad products below
2*N^(1-kappa0) are at most 2*N^(1-kappa0), hence negligible. Otherwise
r>q^b implies r>(rq)^beta, so both primes exceed N^a for some fixed a>0.
Use t=log r/log N and y=rq/N. For fixed eta<=y<=C, PNT in q=Ny/r gives
 (N/(r*log N))*dy/[1-t+log y/log N].
Reciprocal-prime partial summation in r then gives limiting pair mass
 (N/log N)*dt*dy/[t*(1-t)]. The sum of PNT errors is uniform, because
the reciprocal-prime sum on fixed positive exponent ranges is bounded.
The elementary prime upper bound gives O_a(eta*N/log N) for y<eta.
Thus N tends to infinity first and eta then tends to zero.

The exact bad inequalities are
 t<=1/2+log y/(2*log N), t>beta+b*log y/((1+b)*log N), 0<y<1.
The limiting ordered mass is consequently
 J_b=int_0^1 dy int_beta^(1/2) dt/[t*(1-t)]
    =[log(t/(1-t))]_beta^(1/2)=log(1/b).
The r<=q ordering counts each composite once; there is no second ordering
```

factor and no singular-series factor in J_b. Smooth the total-product wall
in y and the remaining walls in t. For each epsilon>0 fixed first, positive
outer approximations have J_W<=J_b+epsilon; their normalized derivatives
are bounded for that precision. One partition of unity repeats no mass.
Use (U) for this fixed family, let N grow, and then epsilon decrease. We get
 limsup_(even N) B_b(N)/H_N<=K*log(1/b).

Unique factorization proves the EXACT identity D_b=D12-B_b. A prime remainder
is represented by r=1 and is always good. A square q^2 has r=q>q^b and is
bad; if a squarefree upper argument omits it, add its O(sqrt N) count first.
The unit N-p=1 is excluded in D12; a convention including it costs at most
one. Zero remainder is excluded by p<N. Small p was paid above, separately
from the small-product tail. Rough squareful exceptions in a Chen-weight
derivation cost O(N^(1-a+o(1))) for fixed a>0 and belong in (C)'s error.
All these costs vanish after division by H_N. Therefore (C) gives
 liminf_(even N) D_b(N)/H_N>=c-K*log(1/b).

Here is a fresh exact scalar certificate, independent of the old integrals.
For each of the three rational b, set x=(1-b)/b, so 0<x<1. Put
 L=sum_(j=1)^256 (-1)^(j+1)*x^j/j, U=L+x^257/257.
Integrating the finite geometric identity for 1/(1+t) proves
 0<log(1+x)-L=int_0^x t^256/(1+t)dt<x^257/257;
hence L<log(1/b)<U, with every displayed endpoint rational. Exact arithmetic
gives the following STRICT lower bounds (the JSON stores the recipe/results):
 b=74821909/10^8: c-K*log(1/b)>1007126077377507/10^20>10^-5;
 b=74821809/10^8: c-K*log(1/b)> 103414663310529/10^20>10^-6.
 b=9/10: c-K*log(10/9)>124889560400949969098/10^20>1.24889>1.248.
The first two surpluses exceed 7126077377507/10^20 and 3414663310529/10^20;
the third exceeds 560400949969098/10^20 above its stated count 1.24889.
For each b choose the fixed smoothing and realization errors smaller than
one quarter of its surplus, then let all remaining errors become smaller
than another quarter. The uniform hypotheses give an eventual bound for
EVERY even N. Take the maximum of the three thresholds. This proves the theorem.

Exact completion boundary. This proof establishes the downstream implication.
To discharge (U), the current K must come from one actual nonnegative upper
weight, its literal raw-density bound, and its signed compiled PP on precisely
the licensed finite families, with the corrected auxiliary support cover and
the narrowed moment parameters. The numerical endpoint plus two retained
upper allowances is strictly below K; this arithmetic does not prove them.
To discharge (C), retain the full-Y11 pointwise proof and the paired injection,
the actual lower compiler and literal residual-density equalities, licensed
table430 only at U3/U4 and table248 elsewhere, the complete old L2 H removal,
and the current restricted U11 parent majorants and discarded L6 floor losses.
All retained numerical inequalities and both finite/compilation loss budgets
must apply together: their lower endpoint exceeds c by only 9.0612246398e-10.
No current input is certified by an old PASS label, a density number alone,
an unrestricted distribution theorem, or this final subtraction. No complete
upstream replay, external validation, frontier-priority or whole-theorem claim
is made. The main manuscript, all old files, state and external services are untouched.

K The adopted same-mask refinement at threshold 7/3

Source: corpus-renewal-paired-certificate-v2-20261010.txt.

```
Same-mask paired certificate v2: improved Buchstab threshold7/3
10 October 2026. New files; the12/5 certificate and old state stay frozen.

1. Replacement, not accumulation of versions

This version replaces the prospective12/5 refinement by the stronger7/3
refinement. They are alternative estimates of the SAME new mass, and must
never both be added. The starting point remains the adopted E0/E1 assembly,
with K=8452171/1250000 and the same6091+11481=17572 pair512 rectangles.
The old assembly's17772 total field was a transcription error, not a mask
or numerical-sum discrepancy. No new cell, parent or table entry is used.

Keep the existing total gross .002095161+.003799514. Add ONLY the new safe
gross delta .000558583. Replace the old positive saved-estimator loss
.002372094 by .002334885. This supplies the raw increment

 .000558583 + (.002372094-.002334885) = .000595792 =37237/62500000.

The root must first match this arithmetic to its independent checks and
complete the final scalar assembly. This certificate alone does not adopt
new c,b, change the manuscript, or alter the research-state.

2. Elementary omega bound with the improved threshold

Claim: omega(s)>=11/20 for every s>=7/3.
First omega>=1/2 for every s>=1: it equals1/s on[1,2], and the delay
equation (s*omega(s))'=omega(s-1) proves the lower bound on successive
unit intervals from the value at2.

For7/3<=s<=3, the exact formula is omega(s)=(1+log(s-1))/s. For x>=1,
log x>=2(x-1)/(x+1), by the positive atanh expansion. Hence

 omega(s)>=(3s-4)/s^2.

After multiplying by20s^2, the difference from11/20 has numerator
60s-80-11s^2, a concave quadratic. Both endpoint differences are
positive, since27/49>11/20 and5/9>11/20. Thus the whole interval is
admitted, without using sampled monotonicity of omega.

For3<=s<=10/3, the global1/2 bound gives

 s*omega(s)>=1+log2+(s-3)/2 >=s/2+1/6,

using log2>=2/3. Therefore omega(s)>=1/2+1/(6s)>=11/20.
For s>=10/3 the derivative of s*omega(s)-(11/20)s is
omega(s-1)-11/20. The already proved interval starts at7/3, so induction
on consecutive unit intervals extends nonnegativity to infinity.
The numerical program checks the rational endpoints and independently
encloses log2>2/3. The certificate agent supplies a separate analytic review.

3. The inner fibre and its exact extra mass

The inherited parent majorant dominates
K*omega((1-t-u-z-w)/z)/z on a<w<z<u<h<t<d. For a pair rectangle
B=[tl,tr]x[ul,ur], now take

 Z=min(ul,3*(1-tr-ur)/13).

On a<w<z<Z, strict order gives z<u and w<z, and
```

```
 1-t-u-z-w >=1-tr-ur-2z >=(7/3)z.

Thus the majorant has an additional1/20 in omega on this whole inner
fibre, over the old1/2 already used. The additional gross mass in Chen
coefficient units, including its factor1/4, is at least

 (K/80)*log(tr/tl)*log(ur/ul)*[1/a-(1+log(Z/a))/Z].

The last bracket integrates log(z/a)/z^2 on[a,Z]. It has no extra
permutation factor. All remaining fibres retain their old bound.
The complete new gross is enclosed at

 E0: .0001931809125035256...
 E1: .00036540308173821915...
 sum: .000558583942417448... > .000558583.

These are ADDITIONAL masses only. The old gross gains and their rounding
surpluses remain inside the baseline. It is the existing explicit positive
majorant that is restricted, never a reversed upper estimate on actual
removed arithmetic counts. The same E0/E1 masks imply the exact same
saved method0 parents and no intersection with priority40000.

4. Replayed floor loss and feedback

The literal author512 evaluator, mesh16384 and h248 table are executed
on every deleted cell. All6091 E0 cells and all227 E1 crossing cells
return zero. The E1 sum is the Q80 integer11290806845175652020270.
After division by4*Q80, the exact discarded saved lower estimate is

 5645403422587826010135/241785163922925834941235 2
 =.0023348841306025784... <.002334885.

The root's independently authored upper bound can also be compared to
.002334885 without reproducing the original DDE mesh. The enhanced L6
half remains unchanged, and the retained lower bound becomes
old_L6-4*.002334885. This replacement concerns saved floors, not true
positive counts. The selected old whole L6 is verified to be the author512
sum, so it has this cell decomposition.

The correct new future K-feedback slope is

 A_new=A_old-.000558583/K=A_old-558583/6761736800.

The loss recovery is independent of K. Using the net gain divided by K
would give the wrong slope. Keep every old allowance and, conservatively,
add .000595792/8 to each lower finite-density and compiler allowance.
The resulting increment after allowances is3*.000595792/4=.000446844.
Prospective endpoints and the exact slope appear in JSON; selecting c,b
still requires the root's strict rational-log transfer calculation.

5. Numerical evidence and frozen boundaries

The new interval engine is standalone Q160. Logarithms use96 positive
atanh terms at Q224 with tail9/(4*193*3^193), then outward rounding.
All17572 rows store the group, cell indices, exact Z, discarded Q80
integer, and both extra-gross endpoints. Every bound is rational; floats
serve only displayed summaries. Literal old floor replay is explicitly
declared and not called an independent implementation.

Both current parent covers, their witness/source dependencies, all old
evaluator source modules, chosen h248 table, author512 sum and prior
assembly are hash-bound. Source hashes are checked before and after.
No old file, main, state, formalization or GitHub action is modified.
The12/5 certificate is retained as an obsolete candidate, not another gain.
```

L The final cross-check of the same-mask refinement

Source: corpus-renewal-paired-final-review-20261010.txt.

```
Independent final reverse review of the same-mask v2 refinement
10 October 2026. New review files only; no state, manuscript or old evidence edits.

Result: PASS for this incremental calculation, subject to the retained analytic
interfaces already used by the baseline. This review does not independently
reprove all those interfaces or establish external frontier priority.

1. Literal source, sign and endpoint checks

Read certify_sharp_184_baseline.py, lines50-95, rather than relying on its
description. With enhance=False and t>=mu=25/128, the active tent is exactly
R=57/64-(5/2)t-u. The other tent exceeds it by2(t-mu)>=0. The source first
maximizes its ordinary constant/chord alternatives, then adds a separate
nonnegative h correction. It does not sum overlapping ordinary estimates.

For a box with residual endpoints r,R0 and logarithmic mass M, let MR be
the integral of R against dt/t du/u. On the audited E1 boxes, sh=R0/a<=9/4.
When sl=r/a>=2, the exact ordinary endpoints are
fl=log(sl-1)/sl and fh=log(sh-1)/sh. They dominate the historical sf lower
endpoints; see certify_extended_prefix.py, lines54-110. The two old ordinary
floors, after division by the Chen factor4, are bounded above by

 fl*M/a,
 [fl*(R0*M-MR)+fh*(MR-r*M)]/[a*(R0-r)].

The coefficient integrals are positive, so replacing lower endpoints by true
values has the required upper-bound direction. Taking the maximum preserves
that direction. All fixed-point .lo roundings in the historical evaluator
can only decrease its saved value.

The minimum h248 entry over closed cells meeting the exact [sl,sh] interval
is at least the minimum over the old outward-rounded lookup interval. Thus
this closed-cell minimum times M/R0 is an upper bound on the old defect
floor in Chen units. At a knot both adjoining cells must be included.
This matches the literal lookup at certify_combined_chen_functional.py:68.
All h248 entries were separately checked nonnegative. If the outward old
lookup drops below2 and returns0, the same upper bound remains valid.

For sl<2, the constant is0, every active chord is skipped and lookup is0.
There are exactly227 E1 crossing boxes. All6091 E0 boxes satisfy sh<=2
and also have exactly0 saved floor. No true positive arithmetic count has
been bounded above or set to0 by these observations: only saved estimators.

The root's independent-v2 decimal interval arithmetic, 320-term alternating
Mercator sums and next-term remainder were read in full. Range reduction
leaves logarithm arguments in[1,3/2], every division has positive interval
denominator, and all integer arithmetic rounds outward. The code imports
no project interval or sieve implementation. Its proposed proxy is therefore
an independent analytic upper bound on the saved historical floors, not a
claim to reconstruct the old ordinary mesh exactly.

2. Actual old ledger and cellwise comparison

The historical L6 program certify_sharp_248_pair_refinement.py:28-35 uses
the enhanced [h,mu] half and the nonenhanced [mu,d] half, each with512
divisions. The selected L6 integer365077417315690091410168 5 equals its
recomputed author512 sum, not a different branch of the historical max.
The selected ordinary mesh is16384. Therefore subtracting the recorded
nonenhanced cell floors has the required decomposition; the enhanced half
is unaffected.
```

```
Reconstructed both old masks independently: E0=6091, E1=11481, disjoint,
total17572. Historical total17772 is a scalar addition/transcription typo;
its masks and actual sums have17572 entries. No old file was corrected.
E1 mask SHA256:
696c5feacd6404b94438131e767817f0bd379fbd171d946068c873609948ec55

The review program compared every E1 root proxy against the corresponding
literal author replay integer and found all11481 upper comparisons valid.
It also compared all11481 E1 Q160 and decimal extra-mass intervals: each
pair intersects. It independently checked every17572 exact Z value and
geometry inequality, row membership, zero boxes and positive h entries.
All53 unique source/dependency bindings from the two new ledgers passed
fresh hash checks. The review does not relabel this row check as a fresh
execution of the old DDE mesh.

Literal discarded floor loss:
5645403422587826010135/24178516392292583494123 52
=.0023238841306025784...
Independent root upper proxy:
.00233488413635981310147442734879042104825250253461 47...
Both are strictly below .002334885, so replacement of the old paid loss
.002372094 recovers exactly .000037209 safely.

3. Analytic new gross bound

The v2 omega proof is valid. First omega(s)>=1/2 for all s>=1 by its
formula on[1,2] and the delay equation. For7/3<=s<=3,
log(s-1)>=2(s-2)/s gives omega(s)>=(3s-4)/s^2. The quadratic
60s-80-11s^2 is concave with positive endpoint values, equivalently
27/49>11/20 and5/9>11/20. For3<=s<=10/3,
s*omega(s)>=1+log2+(s-3)/2>=s/2+1/6, using log2>=2/3.
Hence omega(s)>=11/20 there. The derivative of
s*(omega(s)-11/20) is omega(s-1)-11/20; unit-interval induction now
extends the bound from10/3 to infinity. The review also certified
log2>2/3 by its own exact rational positive series.

For each same-mask pair rectangle, Z=min(ul,3*(1-tr-ur)/13) satisfies
Z>a and1-tr-ur-2Z>=(7/3)Z. On a<w<z<Z the actual Buchstab
argument is therefore at least7/3. The old1/2 mass and the new1/20
subfibre mass are simultaneous pointwise bounds, not alternative estimates
of the same component. Existing parent majorant domination by K*omega/z
and positive restriction are retained inputs; their inequality direction
supports adding this new mass. The exact primitive of log(z/a)/z^2 is
-(1+log(z/a))/z. Including the Chen factor1/4 gives the additional mass

 (K/80)*log(tr/tl)*log(ur/ul)*[1/a-(1+log(Z/a))/Z].

There is no further permutation or factor2. The independent lower endpoints
sum to .0005585839942417448..., strictly above adopted .000558583.
This v2 value replaces the prospective12/5 value; both must not be added.

4. Independent scalar calculation and strict transfer

Read the original current selected_result and matched its selected fields
against positive-pair-deletion-improvement-assembly-20261010.json. From
that baseline, without importing the root's scalar calculation:

 K=6.7617368 remains fixed.
 Raw gain=.000558583+.000037209=.000595792.
 Each of the two old lower-error budgets increases by gain/8=.000074474.
 After-budget gain=3*gain/4=.000446844.
 New raw c lower=1.962824630647271778912819571344383598829165747...
 Each new lower budget=.000531052870574657461473432230858583127094513...
```

```
 New after-budget c=1.961762524906122463989872706882666432574976721...
 Safe c, rounded downward to8 decimal places, is1.96176252.

For both transfer checks the review used64 exact rational atanh terms:
log(q)=2*sum_(j=0)^63 z^(2j+1)/(2j+1)+R, z=(q-1)/(q+1),
0<=R<=2*z^129/[129*(1-z^2)]. No floating-point logarithm supplies any
proof endpoint. Exact fractions and their complete error bounds are in JSON.

Primary b=.74816964, count=1e-5:
c-K*log(1/b)-count >1.14971251731193680737366369829298458e-8.
Previous8-decimal b=.74816963 has margin <-7.8879929490426287e-8.

Secondary b=.74816865, count=1e-6:
c-K*log(1/b)-count >6.41628535880799327954570752244619892e-8.
Previous8-decimal b=.74816864 has margin <-2.6214320665192871e-8.

Thus each selected b is the minimal8-decimal value allowed by this safe
c and its separately stated count. The secondary b cannot be presented
with the primary count. The next8-decimal c exceeds the certified
after-budget endpoint and cannot be rounded upward.

5. Feedback and frozen boundaries

Future raw slope is A_new=A_old-.000558583/K, namely
25892776044271626542645156082021325504082078 7/
66239724740624887279841241469783045025300480 0
=.39089498251479843393319472362565855173819442997...
The .000037209 saved-loss recovery is independent of K and is NOT included
in this slope subtraction. All old allowances are retained. The root may
now assemble the new selected scalar ledger, subject to the same retained
proof interfaces; this scoped report itself changes no selected state.

Main manuscript SHA256 remains
06ecf0c7a8d767115dd0ef595e52844d49486b4b98d5979300faf47f4e09ff78.
No formalization, GitHub, external message, new distribution theorem or
external priority claim is part of this review.
```